

EXHALE: User Manual

Scope. This manual describes how to set up, run, and analyze models with EXHALE, the in-house extension of the ATES (ATmospheric EScape) code to trace metals, metal-line cooling, a runtime opacity dispatcher, the Roche tidal potential, and excited-hydrogen / Ly α physics. It documents every input parameter (with a brief physical meaning), every output file (column by column, with units), the analysis tools, and a set of worked examples. The equations and algorithms as the code applies them are in EXHALE_physics_and_algorithms.pdf; the detailed derivations live in the companion notes (the dated update logs Update_EXHALE_stage0.pdf to Update_EXHALE_stage3.pdf, whose Markdown sources are md/Update_EXHALE_stage1.md to stage3.md, transmission_spectrum.tex, cooling_formulas.tex (the analytic cooling coefficients), EXHALE_BC_and_IC.tex (the boundary- and initial-condition reference), and wind_ae_solver.tex, the Wind-AE steady-wind solver used by IC mode: windae); this document is the practical user guide.

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1. Introduction

1.1 What the code does

EXHALE solves the time-dependent, spherically-symmetric (1-D) radiation–hydrodynamics of an irradiated planetary upper atmosphere and the hydrodynamic wind it drives (“atmospheric escape”). It integrates the Euler equations for mass, momentum, and energy of a single fluid,

$$\partial_t \rho + \frac{1}{r^2} \partial_r (r^2 \rho v) = 0, \quad \partial_t (\rho v) + \frac{1}{r^2} \partial_r (r^2 \rho v^2) + \partial_r p = -\rho \partial_r \Phi,$$

$$\partial_t E + \frac{1}{r^2} \partial_r (r^2 v (E + p)) = -\rho v \partial_r \Phi + \mathcal{H} - \mathcal{C},$$

to a steady state, where Φ is the gravitational (optionally Roche) potential (§1.3), $\mathcal{H}$ is the XUV photoionization heating and $\mathcal{C}$ the radiative cooling. Coupled to the hydrodynamics, at every step it solves the *ionization equilibrium* of H, He, and the trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) with a Newton (MINPACK) solver, including photoionization (of the inner shells of the metals too, with their Auger electrons), collisional ionization, radiative + dielectronic recombination, and charge exchange. The photoelectron energy budget carries secondary ionization (a fast photoelectron drives further H I / He I / H₂ ionizations instead of thermalizing fully) on the Shull & van Steenberg (1985) ionization budget with the Dalgarno, Yan & Liu (1999) heating fraction and molecular channels. The ionizing recombination radiation of H II, He II and He III is absorbed on the spot, so that the ground captures count as case A where the gas is thin and case B where it is thick, and the He I 584 Å resonance line is scattered by He I and partly converted into the 2¹S two-photon continuum (§8.). The He ↔ H charge exchange (He⁺ + H, its reverse, and the radiative He²⁺ + H capture) acts in every helium-bearing ionization system. When the metastable He 2³S triplet is included the code adds its explicit collisional-excitation cooling (the 10830 Å channel) and the He 2³S + H / H₂ Penning ionization that removes the metastable. The converged run yields the radial profiles of density, velocity, temperature, and the ionization state, plus the steady-state *mass-loss rate* $\dot{M}$. A post-processing pass adds advection to the ionization balance, and the Python tool EXHALE_transit.py turns a converged model into transit *transmission spectra* (He 10830, Lyα, Hα/Hβ, the metal resonance lines Mg II, Ca II, Na I, and the O I 1302/1304/1306 triplet).

1.2 The lower boundary, radii, and normalization

The computational domain starts at a **lower boundary** placed at the 1 μbar pressure level, which defines $r = 1$ and the base radius R_p (the input Planet radius). The base number density n_0 and temperature T_0 (the equilibrium temperature) are set there. That level is the lower *face* of the first cell, $r_{\text{edg}}(0) = 1$, in every grid type: the first cell center lies half a cell above it and the two lower ghost cells lie below it (§2.6). Internally the code is non-dimensional: lengths in R_p , velocities in $v_0 = \sqrt{k_B T_0 / \mu}$, densities in n_0 , etc. *All output files are written back in cgs* (see §4. for the exact normalization of each column). The key dimensionless number is the *Jeans / escape parameter* $b_0 = GM_p \mu / (k_B T_0 R_p)$, the depth of the gravitational well in units of the thermal energy; small b_0 means an easily-escaping (or tidally unbound) atmosphere.

1.3 The gravitational potential

The potential Φ in the momentum and energy equations is selected by `Domain` mode (Table 3). The code writes it in the dimensionless units introduced above (lengths in R_p , potential in $v_0^2 = k_B T_0 / \mu$), so it is fixed by just three numbers: the Jeans parameter $b_0 = GM_p \mu / (k_B T_0 R_p)$, the stellar-to-planet mass ratio $q = M_\star / M_p$, and the orbital separation $\tilde{a} = a / R_p$.

Spherical mode (Domain mode: Spherical). Only the planet's own gravity is kept, so the domain can extend far past the Hill radius:

$$\phi(r) = -\frac{b_0}{r}.$$

Roche mode (default). The full co-rotating Roche potential, evaluated along the star–planet (substellar) line $x = r$, adds the stellar tidal attraction and the centrifugal term of the synchronously-rotating frame:

$$\phi(r) = -\frac{b_0}{r} - \frac{b_0 q}{\tilde{a} - r} - \frac{b_0(1+q)}{2\tilde{a}^3} \left(\frac{q\tilde{a}}{1+q} - r \right)^2,$$

where the second term is the star's gravity at distance $\tilde{a} - r$ and the third is the centrifugal potential about the center of mass at $x_{\text{cm}} = q\tilde{a}/(1+q)$ with orbital frequency $\Omega^2 = G(M_p + M_\star)/a^3$ (synchronous rotation). The effective gravity used by the hydrodynamics is $g(r) = -d\phi/dr$, with

$$\frac{d\phi}{dr} = \frac{b_0}{r^2} - \frac{b_0 q}{(\tilde{a} - r)^2} + \frac{b_0(1+q)}{\tilde{a}^3} \left(\frac{q\tilde{a}}{1+q} - r \right).$$

The inner Lagrange point L_1 is the root of $d\phi/dr = 0$, where planetary gravity, the stellar tide, and the centrifugal force cancel; in Roche mode the outer boundary is placed at this L_1 radius by default, beyond which gas is no longer bound to the planet.

Roche mode with an extended outer boundary (Outer radius in Roche mode). By default Roche mode truncates at L_1 . Supplying an explicit `Outer radius [R_p]: > 1` *without* `Domain mode: Spherical` now keeps the full tidal Roche potential above *and* pushes the outer boundary out to the requested radius (e.g. $10 R_p$), reproducing the extended-domain treatment of Yan et al. (2022) / Huang et al. (2023), whose escape models retain the tidal force but integrate well past L_1 . This differs from `Domain mode: Spherical`, which also extends the domain but *drops* the tidal and centrifugal terms (pure $-b_0/r$). Two caveats apply. First, beyond L_1 the Roche potential is past its maximum, so the net force is outward and the 1-D radial flow there is a spherical approximation to what is physically an L_1 funnel: the same approximation Yan/Huang make. Second, the cold hydrostatic IC is invalid past L_1 ; use `IC mode: auto` (or `transonic`) so the solver seeds a transonic wind through the interior sonic point. The tidal singularity of the $-b_0 q/(\tilde{a} - r)$ term sits at the star ($r = \tilde{a} \gg 10$), so a $10 R_p$ domain is numerically safe.

Full 3-D form (transmission post-processing). The 3-D Roche-equipotential reconstruction in `EXHALE_transit.py` / `roche_recon.py` (§5.2) needs the potential off-axis as well. With the substellar direction $\hat{x}$ pointing toward the star and (y, z) the sky plane,

$$\phi(x, y, z) = -\frac{b_0}{\sqrt{x^2 + y^2 + z^2}} - \frac{b_0 q}{\sqrt{(\tilde{a} - x)^2 + y^2 + z^2}} - \frac{b_0(1+q)}{2\tilde{a}^3} \left[(x_{\text{cm}} - x)^2 + y^2 \right],$$

which reduces to the substellar $\phi(r)$ above on the $+x$ axis. The centrifugal term involves only the rotation-plane coordinates (x, y) , not z , so surfaces of constant ϕ are triaxial (most extended toward L_1 , least along the spin axis z).

1.4 Workflow

A model run consists of:

1. Prepare `input.inp` (planet/star parameters, numerics, physics switches), by hand or via the GUI; optionally `metals.inp` (trace metal abundances) and `opacity.inp` (opacity model).
2. Build and run `./EXHALE.x`; it reads the input files from the current directory and writes profiles to `output/`.
3. Inspect with `EXHALE_plots.py` (live profiles) and the example notebook; compute transmission spectra with `EXHALE_transit.py`.

2. Installation and running

2.1 Requirements

A Fortran compiler (gfortran by default; Intel ifort/ix supported), make, and (for the GUI and analysis) Python 3 with tkinter, numpy, scipy, matplotlib, and astropy (the last is used by `EXHALE_transit.py` for `astropy.convolution`).

2.2 Build

EXHALE uses an incremental Makefile; module dependencies are auto-generated, so only edited sources and their dependents recompile.

```
make                # build ./EXHALE.x with gfortran
make FC=ifort       # Intel ifort (or FC=ifx for the LLVM compiler)
make wind_ae_ic     # standalone Wind-AE IC generator -> ./wind_ae_ic.x
make check          # regression against the stored goldens
make clean          # remove .o/.mod (keeps EXHALE.x)
make distclean      # wipe build/ and EXHALE.x
```

`make check` rebuilds, re-runs each regression case single-threaded (`OMP_NUM_THREADS=1`) and compares output/`Hydro_ioniz.txt` and output/`Ion_species.txt`, and the `_adv` pair where the case writes one, against the stored reference outputs, *bitwise* first and otherwise within `REGRESSION_REL_TOL` (default 10^{-3} relative), the verdict line saying which. The standing rule it encodes: a refactor with no intended physics change must be byte-identical, and a new physics option must default to off. The harness and its reference outputs live in `backup/regression/`; the target reports that it found no harness, without failing, in a checkout that does not carry it. `make wind_ae_ic` builds the out-of-process IC generator described in §2.10.

Thread count. The cell sweeps (ionization equilibrium, radiation field, rate coefficients, carrier operator) are OpenMP loops over cells, each cell solved from its own inputs, and the only reductions in them are integer counts, maxima and wall-clock times, none of which enters the state. A run is therefore meant to be the same function of its input at every thread count, and `src/tests/thread_count_identity` checks it: 200 marching steps of the metals-on

regression case `wasp_full` at 1 and at 8 threads must write bitwise identical data lines. That suite found one data race, fixed on 2026-09-26: a variable of the metal sweep (the inner-shell Auger share of neutral photoionization) was shared between threads, so every metals-on run at more than one thread between 2026-09-25 and that fix depended on the thread count (`md/metal_sweep_openmp_race_20260926.md`). The regression matrix still runs single-threaded, which is why it could not see that race. The line order of the EXHALE_STAGE_CHANNELS diagnostic (Table 1) is the one output that depends on the thread count.

2.3 Running

The driver script `run_EXHALE.sh` (1) launches a Python/Tk GUI that writes `input.inp`, (2) builds, and (3) runs `./EXHALE.x`:

```
chmod +x run_EXHALE.sh
./run_EXHALE.sh          # gfortran (GUI -> build -> run)
./run_EXHALE.sh --ifort  # Intel ifort  (--ifx for ifx)
```

The GUI writes the legacy core block only: it offers Load from file, Power-law and Monochromatic as spectrum types, and every optional key of Table 3 (Spectrum type: Planck and the two stellar quantities it needs among them) is added to the written file by hand or inherited by copying an `examples/` configuration.

You may equally skip the GUI: place a hand-written `input.inp` (and optional `metals.inp/opacity.inp`) in a run directory, ensure an output/ subdirectory exists, and run `./EXHALE.x` there. Each modeled planet typically lives in its own subdirectory holding its `input.inp`, its `EXHALE.x`, and its `output*/` folders. Progress (iteration counter, the mass-flux spread du , and the time-derivative norm dtu) is printed to the terminal; the run stops when du or dtu falls below its convergence threshold.

2.4 Convergence control

The time loop stops when *any* of the following criteria is met (defaults in `src/modules/init/parameters.f90`; most are runtime-settable, Table 3):

Parameter	Default	Stop condition
<code>du_th</code>	1×10^{-3}	mass-flux spread $du < du_th$
<code>dtu_th</code>	1×10^{-8}	time-derivative ∞ -norm $dtu < dtu_th$
<code>stall_tol, N_stall</code>	10^{-6} , 2000	du on a plateau for <code>N_stall</code> steps
<code>count_max</code> (Max steps:)	10^6	hard iteration cap

du is the radial spread of the mass flux $F = \rho v r^2$ over the convergence window $r \geq \text{Escape radius}$ (cells $j_{\min} \dots N$),

$$du = \frac{\max_j F_j - \min_j F_j}{|\bar{F}|}, \quad \bar{F} = \frac{1}{N - j_{\min} + 1} \sum_{j=j_{\min}}^N F_j,$$

with F taken *signed*: no absolute value is applied before the extrema, so a window whose flux reverses (inflow at one radius and outflow at another) is measured as the far-from-steady state it is rather than read as flat, and the normalization is the window's own mean level, floored at 10^{-30} so that a vanishing mean gives a large number rather than an infinity.

A steady spherical wind carries one mass flux at every radius, which is what this measures. It is evaluated once per step by a single function (`mass_flux_spread` in `EXHALE_main.f90`) that the du stop, the arming of the du triggers, the PLM $\rightarrow$ WENO3 stage flip, the secondary-ionization flip and the JFNK hand-off all read, so every one of them speaks about the same number; the value at the final step is written to `run.log`. It is *not* a velocity change or a step-to-step variation: that is dtu .

The stall and `count_max` backstops guarantee termination even when the strict du/dtu thresholds are physically unreachable for a given setup (e.g. a transonic wind whose du floors at $\sim 1\%$). Each iteration prints `count`, `du`, `dtu` (and the windowed flux-level change when `Level_tol` is set), and the run reports which criterion stopped it. The optional `Force_start` forces the first 1000 iterations regardless (useful when starting near a fixed point).

Two of these triggers are *armed* before they can fire. A generated initial condition (cold hydrostatic, transonic, warm-seed, Wind-AE) is an analytic profile the marching loop has not touched, so its du measures the smoothness of the formula that built it, not relaxation of the wind, and it can start orders of magnitude below every threshold. The $du < du_{th}$ stop and the Newton hand-off (together with the secondary-ionization flip that shares its du comparison) therefore require du to have been seen at or above their own threshold first, so only a *descending* crossing stops the run; both arm immediately when the run started from `Load_IC? True`, whose state was relaxed by the run that wrote it. Arming is reported in the log. The PLM $\rightarrow$ WENO3 switch is deliberately not guarded. The rationale and history of these settings are in `Update_EXHALE_solver`.

Caveat: du can stop a run prematurely. du measures only the *spatial spread* of $\rho v r^2$ in the wind window; it is blind to a uniform drift of the flux level (the mean it normalizes by drifts with it), and on strongly irradiated planets it *oscillates*, dipping below 10^{-3} transiently long before the wind settles. A WASP-121b reference stopped this way was low by ~ 0.08 dex in $\dot{M}$ with the heating/cooling grossly out of balance (energy residual ~ 30). For production answers use the residual-based stop (`Resid_tol`) or the Newton finish (`Solver: Newton`, §2.7); details in `steady_solver_memo`.

Staged secondary ionization. With `Secondary_ionization: True` (the default) the photoelectron coupling is *not* applied from step 0. From a cold initial condition it would deepen the base cooling valley and amplify the base acoustic mode into a NaN runaway on high-gravity cases. Instead the wind first relaxes without the coupling; the moment that first relaxation converges (or stalls) the coupling switches on (the terminal prints `secondary ionization activated at step N`), the convergence logicals and flux-level history are reset, and stops are held for `N_stall` steps while the base adjustment wave forms, after which the run re-converges with the full physics. Expect roughly double the step count on high-gravity cases. `Secondary_ionization: False` disables the coupling entirely; `Immediate` applies it from step 0 (for A/B tests only; it reproduces the cold-start crash on high-gravity cases).

A consequence worth knowing before reading a short run. Staged is not a synonym for on. A run that never reaches its first convergence (one stopped at a fixed step count, or one left at a du plateau) never switches the coupling on, and its answer carries no secondary ionization at all. The startup report (`EXHALE_setup.out`) says which of the three states the run is in, and the terminal prints the activation line only when it happens.

2.5 Base grid resolution (Base grid [dr,cells])

The Mixed grid stacks cells uniform cells of size dr (in R_p) on the lower boundary and fills the rest of the domain with $N - \text{cells}$ geometrically stretched cells; the total cell count N is set by the optional key `Grid cells`: (default 500). The two numbers share one input line because they are not independent: their product is the radial extent of the uniform region ($0.01 R_p$ with the default 2.0×10^{-4} 50).

The quantity that matters physically is how many uniform cells span the base density scale height, H/dr with $H = kT/(\mu g)$. Where H/dr is of order a few cells the discretization supports a *stationary* $2dr$ entropy (contact) mode that nothing in the scheme removes: HLLC resolves a contact of zero speed exactly and so adds no dissipation to it, the gravity source is cell-local, the WENO3 pressure gradient sees only interface pressures, and there is no conduction or physical viscosity unless they are switched on. The measured dependence is an alternating $\ln \rho$ amplitude of $\sim 10^{-2}$ for $H/dr < 5$ and $\sim 10^{-4}$ for $H/dr > 100$. High-gravity planets have the least margin: at T_{eq} on the default grid HD 189733 b spans only ~ 27 cells per H , against ~ 60 – 100 for HD 209458 b, WASP-52 b and WASP-121 b, and a cool, shielded base shrinks H further. The value in effect is echoed in EXHALE_setup.out as `Basescale-heightresolution:H(T_eq)/dr`, with a warning below 10 cells.

Refine at fixed extent, i.e. divide dr and multiply `cells` by the same factor: `Base grid [dr,cells]: 5.0e-5 200` is the $4\times$ refinement. Two costs follow. The CFL time step is set by the smallest cell, so a $4\times$ finer base means $\sim 4\times$ more steps for the same physical time; and at fixed N the stretched region loses those cells and coarsens (for a $4.43 R_p$ domain, the cell size at $1.5 R_p$ grows from 6.1×10^{-3} to $1.3 \times 10^{-2} R_p$). Raising `Grid cells`: alongside the refinement keeps the upper stretch fixed instead: the pairs `1.0e-4 100/500`, `5.0e-5 200/658`, `2.5e-5 400/916` and `1.25e-5 800/1375` hold the stretch ratio at its default-grid value, at the price of proportionally more work in every sweep. In practice the $2\times$ step (`1.0e-4 100`) is the one to try first: it is where the mode goes, and $4\times$ is not reachable to convergence on HD 189733 b within the step cap (`Max steps`;) because its time step is $4\times$ smaller.

The exact default width, and inputs written before it was pinned. The default dr is the double 2.0×10^{-4} written in the source with a `d` exponent, so the value is the one a reader of the key would spell: an `input.inp` without the key and one stating `Base grid [dr,cells]: 2.0e-4 50` build the same grid to the bit, the list-directed read of 2.0×10^{-4} into a `real*8` being that double. This matters because the grid is part of a stored state: a restart is refused when the file's cell centers differ from the run's by more than 10^{-10} relative. An input written before 2026-09-19 was run on the default-`real` literal, the width $1.9999999494757503 \times 10^{-4}$, whose centers differ from the present grid's by 3.8×10^{-9} to 6.6×10^{-9} relative, which exceeds that tolerance. Every such input standing beside stored results carries that width explicitly as `Base grid [dr,cells]: 1.9999999494757503e-4 50` (`src/utls/pin_base_grid.py`), and each preserved tree whose inputs were not edited has a `GRID_DEFAULT_NOTE.md` at its root stating the same line. The width a run used is recorded at round-trip precision in the resolved-configuration record, and the refusal itself prints it with where it came from, the key or the default. That $2\times$ setting is what HD189733b/`input.inp` carries; the other three production planets run on the default.

2.6 The lower boundary: a characteristic condition at the face

The lower boundary is imposed at the first cell *face* $r_{\text{edg}}(0)$, not in a ghost cell. At that face the characteristic speeds are $v - c$, v and $v + c$, and a wave enters the domain when its speed is positive; in EXHALE the radius increases outward, so gas entering the base moves outward and inflow is $v > 0$. For the subsonic inflow a wind rests on, two characteristics enter and one leaves, so the lower atmosphere states *two* conditions and the interior supplies *one*.

The two the lower atmosphere states are its **pressure** and its **specific entropy** at the base level $r = 1$, the entropy carried as the isentrope through (p_0, T_0) at the base composition. **The face is the level:** every grid type (Uniform, Stretched, Mixed) places its first face exactly at $r_{\text{edg}}(0) = 1$ (define_grid.f90 sets the first ghost center to the mirror of the first cell center, $r(0) = 2 - r(1)$, and stops the run if the face is not 1 or a ghost cell reaches $r \leq 0$), so the reservoir's (p, s) is the state of the face side it owns, with no transport along any isentrope, and the contact between the reservoir and the domain does not move with the cell width. The one the interior supplies is the outgoing acoustic invariant of the first cell,

$$p_b - \rho_i c_i v_b = p_i - \rho_i c_i v_i, \quad (1)$$

where the subscript i is cell 1 carried to the face and c_i uses cell 1's own γ_{eff} . The closed-form invariant $v - 2c/(\gamma - 1)$ is deliberately not used: the caloric equation of state gives every cell its own $\gamma_{\text{eff}}(x_{\text{H}_2}, T)$, and that invariant assumes one constant γ .

With Well balanced: True (boundary model v5, face_at_level_discrete_equilibrium_v5) cell 1 is carried to the face by its own constant-density hydrostatic equilibrium, $p_i = p_1 + \rho_1[\Phi(r_1) - \Phi(r_{\text{edg}}(0))]$, $\rho_i = \rho_1$, the pressure the well-balanced reconstruction extrapolates cell 1 to at the same face, and the ghost cells continue that same discrete equilibrium below the level: each ghost's constant-density equilibrium meets the face state's pressure (ghost 0) or the ghost above it (ghost -1) at its upper face, at the reservoir-side temperature. The boundary and the reconstruction then share one discrete mechanical equilibrium (Käppeli & Mishra 2016, A&A 587, A94, section 2.1): a column at rest in it, meeting the reservoir's pressure at the level, carries no flux through the base face and no momentum defect in the base cells, to rounding, whatever its entropy stratification, and the face state carries the interior's own mass flux $\rho_1 v_1 r_1^2$ at the density of the face state. The reservoir still states the pressure and the entropy of the gas it supplies. What this removes is an $O[(\Delta r/H)^2]$ truncation difference between two second-order constructions: on a column built in the discrete equilibrium the previous (v4) boundary left a base flux 1.47×10^4 times a Mach 10^{-8} wind; with v5 the base face carries that wind to 2×10^{-3} (src/tests/grid_and_gates, base_reservoir_equilibrium and low_mach_face_flux). A seed mapped from another grid or model is not in this discrete equilibrium, and its lower layer is not at rest until it is projected onto it.

Without it (boundary model v4) cell 1 is carried to the face along its own hydrostatic isentrope, and the ghost cells are the volume averages of the reservoir's hydrostatic isentrope through the face state continued *below* the face, i.e. below the level. In both models the velocity carries a constant $\rho v r^2$. The ghosts are not copies of the face state: the two differ by half a cell of stratification, and that half cell is what the previous ghost-cell closure left as a resolution-independent momentum imbalance of 25% of the weight of cell 1.

The face at the level (boundary model v4, 2026-09-27). Until 2026-09-27 the level was a ghost cell *center* ($r(0) = 1$ on the Mixed grid, $r(1 - N_g) = 1$ on the other two), the face lay half a cell (Mixed) or one and a half cells above it, and

the reservoir was carried up to the face along its own hydrostatic isentrope. That put a slab of reservoir gas, whose thickness scaled with the cell width, between the level and the domain, and made the base pressure of the domain $p_0[1 - \Delta r / (2H_{\text{res}})]$, a first-order error in Δr : MEASURED, the molecular lower layer of the LHS 1140 b He/H = 1.8 state sat 2.2% / 1.15% low in density on 500 / 1000 cells (md/ew_grid_order_20260926.md, section 3). With the face at the level the pressure gap between the interior at the level and the reservoir of that state falls from 2.93 / 1.44 / 0.70% to 0.114 / 0.067 / 0.038% on 500 / 1000 / 2000 cells (md/Update_EXHALE_stage3.md section 29.11). Two first-order terms remain at the base and are recorded rather than changed: the Dirichlet He/H of the element-diffusion operator is imposed at the cell-1 *center*, half a cell above the level (binary_element_diffusion.f90), and the conduction bath at the ghost center $r(0)$ (viscous_conduction.f90; conduction is off by default).

States written before 2026-09-27 do not load. Every cell center moved (by half a cell on the Mixed grid, by one and a half on the other two), so the restart grid check (Grid cells:, Table 3) refuses every such state. One enters a run only as an *initialization seed*, mapped onto the new grid with

```
python3 src/utils/map_state_to_grid.py <old_dir> <target_grid_file> <out_dir> --ic
```

where <target_grid_file> is any Hydro_ioniz*.txt written on the target grid (a one-step cold run of the target input.inp gives one), and --ic names the outputs Hydro_ioniz_IC.txt / Ion_species_IC.txt. The mapped pair states mode=init t_phys=0 certified=F cert_reason=mapped_seed whatever the source said, because an interpolated state is neither the certified state nor the physical clock of the run that wrote it; the new run has to be solved again. Two further options of the tool change the composition on purpose: --reservoir El/H <value> multiplies every row of that element by one factor, keeping the shape of its profile, and --uniform (with --reservoir) sets the element to that ratio in every row instead, which is the seed of a run whose element does not separate (helium with He_diffusion off, for which load_IC refuses any row whose He/H is not the input's) made from a state solved with separation. The header of src/utils/map_state_to_grid.py states the checks the tool makes before it writes anything.

Where the flow reverses ($-c < v < 0$) only one characteristic enters, so the reservoir states the pressure alone and the entropy at the face is the interior's, advected out. That choice is discrete and has no width: a contact carrying different entropies on its two sides has direction-dependent traces, and a face density between them is an interpolation rather than a law of thermal contact. The side that owns the face is the side the level's mass flux comes from, read from the wind window $r \geq r_{\text{flux}}$ where that window carries one flux (its relative standard deviation at or below 2×10^{-3} and its face-mapped Mach number at or above 10^{-12}), and from the matched face velocity where it does not, a velocity below the rounding of the cancellation it is formed from counting as zero. At rest and at inflow the reservoir owns the level, the physical assumption being that the lower atmosphere below it is a radiatively controlled heat bath. Supersonic outflow prescribes nothing, and that transition alone carries a declared numerical width, a smoothstep of half width 10^{-8} in M_i about -1 , where the number of outgoing characteristics changes. Supersonic *inflow* would need three reservoir conditions and a (p, s) reservoir has two, so it is reported as a configuration outside the model rather than repaired.

The composition of the two lower ghost cells is part of the boundary model as well, and is what distinguishes model v3 from v2 in the state file's # boundary_model line. The ghost composition is not prescribed but solved by the ionization sweep, and the sweep is a map from the composition it is entered at to the one it returns, so under v2 two entry compositions could return ghosts 4.7×10^{-3} to 5.5×10^{-2} apart in their trace ions, both accepted by every test the

sweep applies, and the base continuity row of cell 1 read 8.335×10^{-8} in one and 2.923×10^{-9} in the other against the same tolerance. v3 states both halves of the fix: the ghost solve starts from the reservoir row at the first sweep and from the ghost the previous sweep returned after it, and the ghost it returns is held to a fixed point of that same map, the largest absolute change of any ghost species density in the cell's own units at most 10^{-11} and the ghost's heavy-particle plus electron count moving by at most 10^{-6} of itself. A boundary that cannot reach that in twenty applications stops the run by name with both moves. Nothing here is a key: the seed and what the fixed point reached are printed in the boundary report, and the diagnostics that replace the seed for a measurement are the EXHALE_GHOST_* variables of Table 1, all default off.

Nothing in this is selectable. There is no ghost velocity, pressure or temperature closure to choose, and the four keys that used to choose them (Valve eps, Hydrostatic base, Base ghost temperature, Base velocity) are refused at startup with a message naming the replacement, rather than ignored. The one key that remains is Base BC:, which states *where* the boundary is, not what it does.

2.7 The steady-state Newton solver (Solver: Newton)

The finite-volume steady state can be solved directly ($F(Y) = 0$ for the hydro state, with the ionization equilibrium eliminated locally inside the residual) by a preconditioned Jacobian-free Newton–Krylov method (banded LAPACK preconditioner, frozen WENO weights, diagonal scaling, non-monotone line search; the boundary's one branch of nonzero width, the supersonic-outflow transition, sits decades from any operating point, so nothing has to be smoothed for it). Each unknown is scaled by its own cell's state (ρ , $\rho(|v| + c_s)$, E), so that no cell sets another's scale; the momentum scale uses the fastest signal speed because ρv itself vanishes wherever the flow reverses. This reaches residual levels unattainable by marching at a small fraction of the cost: WASP-121b converges in 8 Newton iterations (~ 200 marching-steps of cost, vs. $> 35,000$ insufficient marching steps), HD189733b in 33 s where marching spent > 3 h without converging.

Two ways to use it:

- **Production:** add Solver: Newton to input.inp. The run marches (including the two-stage PLM→WENO3 when Reconstruction scheme: PLM+WENO3 is set) until the flux metric du (the radial spread of $\rho v r^2$) drops below the hand-off threshold newton_du_switch (default 10^{-2} ; the du -based *convergence* stops are suspended), then the Newton solve finishes to Resid tol (default 10^{-5}), and the standard outputs and post-processing follow. The hand-off point is adjustable through an optional third token on the key, Solver: Newton <du_switch> (e.g. Solver: Newton 5e-3 hands off later); a bare Solver: Newton keeps the 10^{-2} default.
- **From an existing state** (recommended for re-converging old runs): put the state in output/*_IC.txt, set Load IC? True and Resid tol: 1.0e-5, and run with the environment hook

```
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x
```

which solves from the loaded state, writes output/, and stops.

The warm start matters: Newton has a finite basin, so the hand-off must wait until the marching warm-up has flattened the wind. The default $du < 10^{-2}$ is calibrated to land inside the basin: handing off earlier (a larger newton_du_switch)

risks starting Newton outside it and stalling, whereas a cold-IC start (no warm-up at all) stalls outright. Earlier revisions keyed the hand-off on the residual itself ($\|R\| < 5 \times 10^{-2}$); the *du*-based key is equivalent in practice and consistent with the flux-based convergence decision (§2.4), while the residual remains the *target* of the JFNK finish. Convergence is linear (frozen-weight refresh and the chemistry noise floor). Planets with a breathing base (HD189733b) used to stop at a practical floor $\|R\| \sim 3 \times 10^{-4}$; the 2026-08-11 line-search and base-energy fixes removed it, and a warm re-convergence of HD189733b now reaches $\|R\| = 1.2 \times 10^{-5}$ in 11 outer iterations (re-measured 2026-08-15). The Newton finish is also *safe to fail*: the solver tracks the best iterate seen, aborts once the line search has failed to find any acceptable step in 12 consecutive outer iterations, and on failure the run prints JFNK failed, keeps the best iterate, and falls back to plain time-marching with the *du*-based stops re-armed: an unconverged Newton state is never accepted as the final answer. The associated diagnostic hooks (EXHALE_RESIDUAL, EXHALE_JAC_TEST, EXHALE_NEWTON_TEST, ...) are environment variables, collected in §2.8 (Table 1). Full development record and validation: `steady_solver_memo.pdf`. Figure 1 shows the whole route end-to-end on WASP-121b (Roche): the monitored residual during the marching warm-up (left; note that the *du*-based stop would have fired long before the residual is actually small) and the JFNK finish (right).

What certifies a converged state, and what the exit status says. Reaching `Resid tol` is not the same as being certified. At the marching stop, at the JFNK final and at a pseudo-transient declaration, one evaluator builds an inventory of every balance the configuration activates and returns, for each, a status (`not_applicable` / `evaluated` / `unavailable`), the row measure (the maximum over physical cells of $|res|/\max(\text{scale}, \text{floor})$), the volume average, the worst cell and the scale it was divided by. The tolerances are 3×10^{-12} for the mass row, 10^{-8} for momentum and 10^{-6} for energy, the level balances and the eliminated-species closures.

The species rows are gated by regime. An element row and a carrier row are judged against 10^{-5} for cells at $r \geq 1.20R_p$ only; below that radius the row is measured and printed and does *not* gate. The gate is anchored on the element operator's own stated floor and sits a decade above the discretization error a smooth manufactured column shows at the production spacing (1.5×10^{-6}); the layer is excluded because its element fluxes are not conserved there at all (their radial spread over $r < 1.10$ is 9.8 times the median), so no threshold means anything against them yet. The band $1.10 \leq r < 1.20$ is likewise reported and does not gate. A state that passes with species rows present reports that it is certified *in the wind*, and its state file records `cert_reason=certified_in_wind`. The two radii are properties of the flow and are constants of certification. `f90`, not input keys.

A run that DECLARED a stationary state and then failed certification exits with status 2; a run that ended on a cap makes no such claim, writes `certified=F` `cert_reason=no_stationary_claim` in its state files and exits 0. The regression harness never tests the status.

Transported species: the outer passes of the stationary route. A species the run moves by a transport step instead of a Newton row (the element fractions of `He_diffusion` always, the molecular carriers unless `Coupled carrier solve: True`) has a balance no hydrodynamic solve touches, so the stationary route alternates: solve the wind at fixed transported composition, relax the composition to its own steady state in that wind, solve again (`steady_wind_with_element_diffusion`, EXHALE_main.f90). Each such round is an *outer pass*; the cap is 20 passes (EXHALE_OUTER_PASSES overrides it), and a run with no transported species has one pass, the wind solve alone. The

element update is under-relaxed by a factor ω that starts at 0.5 (EXHALE_DIFF_OMEGA) with floor 0.125, and the carrier update is limited by a *movement bound*, the largest relative change of one cell's particle count $n_{\text{tot}} + n_e$ a pass may make, which starts at 10^{-2} (EXHALE_CARRIER_TRUST) with floor 10^{-3} and ceiling 0.5. The iteration ends when the joint certification passes, when a row is not finite, when the element update is refused, when three consecutive passes make no progress, or when the pass budget ends.

The progress control (revised 2026-09-27). A pass makes progress when the joint distance of the state falls, or, where that distance is carried by a row that is not hydrodynamic, when the composition's own residual falls. A pass whose carrier update ended on its movement bound, whose solve converged and whose joint distance rose by no more than a factor 1.1 is *not* judged by those two measures: the state moved by the whole bound, the relative measures fall together with their scales and cannot show it, and such a pass is neither counted toward the ending nor allowed to shorten a step. The two step lengths answer to the hydrodynamic solve that consumes the update: the movement bound doubles, up to 0.5, after such a pass; it halves when a bound-limited update raised the joint distance of a solved wind by more than 1.1, or when it preceded the first failed solve after a converged one, or when a pass without progress moved the carriers short of the bound on a solved wind. ω halves on a pass without progress whose wind was solved and on the first failed solve after a converged one, and doubles back toward its starting value after a converged solve on a pass with progress. A failed solve that follows another failed solve shortens neither. EXHALE_CARRIER_TRUST_HOLD freezes the bound at a stated value for every pass. What the revision measured, and what it leaves open (the starting bound for transported ionization stages; the overshoot of the H₂ row at the outer cells of the molecular He/H = 0.083 continuation, which neither the old nor the new control certifies), is in md/Update_EXHALE_stage3.md section 29.7.

A restartable state after every pass (2026-09-28; published as generations since 2026-09-29). Every outer pass after which the iteration goes on publishes the state the next pass starts from as one generation output/pass_state/<state_id>/, named by output/pass_state/current, so a run stopped from outside (a wall-clock limit, a kill) keeps its last completed pass; §4.3 describes the files and the restart from them.

2.8 Environment-variable hooks (diagnostics and expert runs)

A small set of *environment variables* expose diagnostic and expert entry points without touching input.inp (Table 1). They are read once at startup; when unset they have strictly zero effect, so normal runs are unaffected. All of them are diagnostics or validation aids, apart from EXHALE_ROOT, which the lower-atmosphere pre-step needs: the others used in a production workflow are the EXHALE_PTC family, which implements the “Newton from an existing state” recipe of §2.7 (and examples/04_newton_from_state/), and EXHALE_CARRIER_TRUST / EXHALE_DIFF_OMEGA, which set the starting step lengths of the stationary outer passes and are how a run continued from a pass snapshot takes over the step lengths it had reached (§4.3). Typical usage:

```
EXHALE_RESIDUAL=1 ./EXHALE.x # rank a "converged" state
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x # re-converge it
```

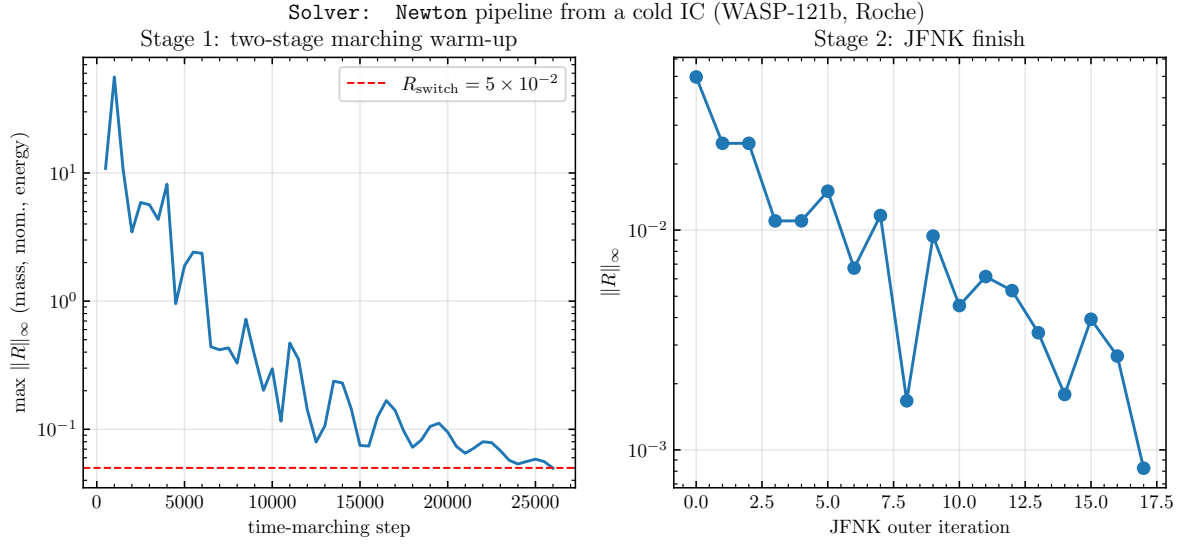


Figure 1: The Solver: Newton route from a cold IC (WASP-121b, Roche). Left: maximum steady residual $\max \|R\|_\infty$ over the mass, momentum, and energy equations during the two-stage (Reconstruction scheme: PLM+WENO3) marching warm-up; the dashed line is the legacy residual hand-off threshold $R_{\text{switch}} = 5 \times 10^{-2}$ used when this figure was produced (the current default keys the hand-off on $du < 10^{-2}$ instead, at a comparable point on the warm-up). Right: residual vs. JFNK outer iteration during the Newton finish, down to Resid tol (10^{-5}).

Table 1: Environment-variable hooks. All are read once at startup and have zero effect when unset. The four diagnostic tests (top block), EXHALE_PARSE_DUMP and EXHALE_PTC runs stop after printing/writing; they do not march.

Variable	Effect
EXHALE_DUMP_IC=1	After Load IC? True, write the restored state to output/Hydro_ioniz.txt and output/Ion_species.txt and stop, <i>before</i> the first ionization-equilibrium solve re-equilibrates the species fractions (verifies a snapshot/restart round-trip).
EXHALE_RESIDUAL=1	Evaluate the steady residual R of the loaded state (WENO3, boundary conditions applied first), print the norms for each equation (mass, momentum, energy), and stop. The criterion-independent way to rank candidate “converged” states (a premature-dip state shows $\ R\ _E \sim O(10)$ despite $du < 10^{-3}$).
EXHALE_NEWTON_TEST=1	Self-test of the Newton infrastructure (pack_U/unpack_U round-trip and the frozen residual against the EXHALE_RESIDUAL values); prints and stops.
EXHALE_JAC_TEST=1	Verify the colored-FD banded Jacobian against a directional finite difference $Jr \simeq [F(Y+\epsilon r) - F(Y)]/\epsilon$; a correct band layout matches to $\sim 10^{-6}$, a wrong one is $O(1)$ off. Prints and stops.

continued on the next page

Variable	Effect
EXHALE_PTC=1	Solve the <i>loaded</i> state directly with the steady solver (no marching), write output/, and stop. Requires Load IC? True; set Resid tol: 1.0e-5 (§2.7). Default engine: banded PTC.
EXHALE_PTC_JFNK=1	With EXHALE_PTC=1: use the full JFNK solver (GMRES on the matrix-free J) instead of the banded PTC, the recommended engine.
EXHALE_PTC_DTAU0= $\langle v \rangle$	With EXHALE_PTC=1: override the initial pseudo-time step $\Delta\tau_0$ (default: the CFL Δt of the loaded state). A large value (e.g. 1.0) probes the pure-Newton regime directly.
EXHALE_PTC_NFIX= $\langle n \rangle$	With EXHALE_PTC=1: anchor the first n physical cells (frozen-base experiment; solves the wind on top of a fixed base).
EXHALE_DIAG_BASE=1	Write output/base_diag.txt (r, n, v, T , heating, cooling of the first few cells over the first steps) to trace IC-startup transients at the base.
EXHALE_FORCE_HYBRD1=1	Force every ionization-equilibrium solve to use MINPACK hybrd1, skipping the analytic-Jacobian Newton: lets the same binary produce both solvers' outputs for an A/B validation (Update_EXHALE_solver.pdf §6).
EXHALE_DIFFUSION_CHECK=1	Write the face-by-face elemental fluxes of every element-diffusion step to output/element_flux_profile.txt (§4.), and the composition profile of each direct-steady outer pass to output/diffusion_pass_profiles.txt. With a lower-atmosphere profile in use the flux file is written anyway.
EXHALE_DIFF_OMEGA= $\langle v \rangle$	Set the starting under-relaxation factor ω of the element update of the stationary outer passes (default 0.5; the progress control halves it down to 0.125 and doubles it back up to this starting value, §2.7), so the undamped loop (1.0) can be compared with the damped one without a rebuild. A restart does not inherit the ω a run had reached: to continue from a pass snapshot (§4.3) at the value its # pass_snapshot line records, pass it here.
EXHALE_CARRIER_TRUST= $\langle v \rangle$	Set the starting movement bound of the carrier update of the stationary outer passes, the largest relative change of one cell's particle count one pass may make (default 10^{-2} ; the progress control moves it between 10^{-3} and 0.5, §2.7). A state that has to move far, such as a restart that turns Ionization transport on, can need a large start (MEASURED: the LHS 1140 b molecular He/H = 1.8 state with transported stages certified in 14 passes from 0.5 and not from 10^{-2}). Also the way to continue from a pass snapshot at the bound its # pass_snapshot line records, which a restart does not inherit.
EXHALE_OUTER_PASSES= $\langle n \rangle$	The pass budget of the stationary outer iteration (default 20).
EXHALE_PASS_STATE_STOP_AT= $\langle k \rangle[:\langle p \rangle]$	A measurement key: stop the run (exit status 3) right after step k of the publication of the pass state of pass p (of every pass when p is not given): $k = 1$ after the temporary directory is written, 2 after it is renamed onto its generation name, 3 after output/pass_state/current names it (§4.3). Unset: no effect.

continued on the next page

Variable	Effect
EXHALE_MAXSTEPS= $\langle n \rangle$	Deterministic step cap: leave the marching loop after n steps and write the outputs. Separate from the Max_steps: key (§2.4) and only ever earlier; it is what the regression harness uses to pin a relaxation-snapshot case.
EXHALE_PROFILE=1	Print a timing breakdown: the <code>ioniz_eq</code> share of the step every 500 steps, and the totals at the end of the run.
EXHALE_PARSE_DUMP=1	Write every value <code>input_read</code> derived from <code>input.inp</code> (plus any <code>base.inp</code> override) to <code>parse_dump.txt</code> and stop, before any grid/IC/output work. Feeds the parser regression corpus.
EXHALE_ROOT= $\langle \text{path} \rangle$	Install directory for the lower-atmosphere pre-step generators (§3.7); required when <code>Lower_atmosphere:</code> is set on a relocated install. Not a diagnostic.
EXHALE_JFNK_MAXIT= $\langle n \rangle$	Cap a stationary solve at n outer iterations in place of the number its caller passed (the default is 3000 at all three call sites). It also caps the <i>run</i> : under this variable the run takes its FIRST stationary solve and refuses later entries by name, so that a bounded measurement is one solve of one state rather than the sum of several from different states. Nothing about the step depends on it.
EXHALE_INVENTORY_REPORT=1	Print the elemental inventory of every cell (what the carriers hold, what is locked, what the reservoir specifies, what the closure supplies, and the budget each side spends) at the carrier operator's entry.
EXHALE_ELEMENT_ASSERT= $\langle 1 2 \rangle$	Verify the element census after each write-back and report (1) or stop on (2) a breach; off by default, and the only check that sees a write-back that creates nuclei.
<i>Measurement options of the stationary solve. Each was implemented, measured against both species-row fixtures and left in the default the measurement supported; none changes a default run.</i>	
EXHALE_ELEMENT_CONSTRAINT_ROWS=0	Limit a species trial by the five coordinate-box corners instead of the shared element constraint rows (the state before 2026-09-09; the corners let two carriers of one element each spend the whole budget).
EXHALE_ELEMENT_WRITE_BACK	The element-conserving write-back of each Newton iterate.
EXHALE_TR_RESTART_STALL= $\langle n \rangle$	Restart the trust-region state after n outer iterations without improvement. Delivered disarmed: no reset measured helped either fixture.
EXHALE_KRYLOV_ON_THE_BALL=1	Steihaug–Toint truncation of the Krylov cycle at the trust-region radius. Much cheaper in products, worse on the carrier fixture; default off.
EXHALE_GM_REORTHO=1	Reorthogonalize the Arnoldi basis. The measured orthogonality loss is 10^{-13} to 3×10^{-12} , so it is not the limiter, and the option is worse on both fixtures; default off.
EXHALE_COLUMN_EQUIL=1	Equilibrate the columns of the model as well as of the factored band. It moves the trust-region coordinates and trials ascend; measured and not adopted.
EXHALE_CAUCHY_LEG_BY_IMAGE=1	Let the image-share test alone decide whether the approximate-gradient leg is dropped. Worse on both fixtures; default off.

continued on the next page

Variable	Effect
EXHALE_CERT_ANCHOR=1	Run and print the five anchors behind the species-row certification tolerances (reproducibility, closure seed, state perturbation, derivative, manufactured column). The default certification report is byte-identical without it.
EXHALE_ELEM_DIAG	The diagnostic line the species-row step prints at every trial. <i>The lower boundary's composition. All default off; with none of them set the seed is the boundary model's own (§2.6) and not one number of the run moves.</i>
EXHALE_GHOST_COMPOSITION_SEED=<s>	Enter the two lower ghost cells into the composition sweep at a stated composition instead of the model's seed. the_state_file_ghost_rows takes the ghost rows the restart pair carries, which the loader otherwise drops; the_previous_sweep_ghost takes the ghost the previous sweep of this run returned; a path takes a file of rows in the format EXHALE_GHOST_SEED_WRITE writes. A run with it set is entered at a composition the boundary model does not state, so its result is a measurement and never a certificate, and the boundary report says so.
EXHALE_GHOST_SEED_WRITE=<path>	Write the lower ghost composition as it was installed, in the format the seed key reads back, so that a perturbed seed is made from a measured one rather than assembled by hand.
EXHALE_GHOST_RECORD=1	Append one record per evaluation to output/ghost_record.txt: per lower ghost cell the thermodynamic row the boundary left there, the heavy-particle and electron counts of the returned composition, its molecular and ionization fractions, the charge and elemental budgets, the reaction residual the cell was accepted at, the residual of its imposed molecular partition and the pass the closure ended on; and, in the same record, the base cell's continuity row with the two face mass fluxes it is built from, their signed difference, the row scale and the row's rounding floor.
EXHALE_INTERIOR_COMPOSITION_HELD=1	Return the composition the sweep was handed on every cell above the two lower ghosts, and the solved composition on the ghosts themselves, so that a base row measured after the call carries the ghost's own refresh and nothing else. The heating and cooling returned are those of the composition the sweep solved on every cell, so a run with the key set is not a state anything may be certified on.
EXHALE_BOUNDARY_TRACE=1	Write the lower boundary at named points of an evaluation to output/boundary_trace.txt, so that the order in which the boundary and the composition are built can be read off a run instead of argued.
EXHALE_CARRIER_TRUST_HOLD=<v>	Hold the bound on how far one outer pass may move the molecular carriers at v for every pass, taking every change the progress control would make to it (growth and shortening alike) out. It separates a relaxation the bound throttles, which decays at a rate proportional to the bound, from a mode the alternation cannot damp, which decays at the same rate whatever the bound.

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Variable	Effect
EXHALE_STAGE_CHANNELS={list}	Up to eight grid cell indices, comma separated (107, 248). Every evaluation of the ionization stage rows at one of those cells appends one line to output/stage_channels.txt carrying the gas branch, the cell, a call counter, the kernel's own arguments and the channels, so the rows can be re-evaluated afterwards at the very state the run gave them. A line holds the state entries 1–29 of both gas branches (temperature, densities, photoionization, recombination and collision coefficients; 26, 27 and 29 the $\text{He} \leftrightarrow \text{H}$ charge-exchange coefficients of $\text{He}^0 + \text{H}^+$, $\text{He}^+ + \text{H}^0$ and $\text{He}^{2+} + \text{H}^0$, 28 the $2^3\text{S} \rightarrow 1^1\text{S}$ de-excitation q_{31g}), entries 30–59 of the molecular network in a molecular branch, then the 35 gross channels of the H II, He II and He III rows; the file's header names every entry and channel (ion_residual_core.f90). The rows run inside the cell sweep, so the order of the lines is deterministic only with one thread.

2.9 Metal cooling on/off comparison

Metal-line cooling is active *iff* a metals.inp is present (§3.); remove or rename it for a metals-off run. Because the metal abundances are read at run time, the executable is built once and the two cases differ only by whether metals.inp exists: no source edits or rebuild. To run the *same* planet both ways:

```
make                                # build once
# metals ON (solar C/N/O from metals.inp):
printf 'CI 2.69e-4\nNI 6.76e-5\nOI 4.90e-4\n' > metals.inp
./EXHALE.x ; mv output output_metals_on
# metals OFF (no metals.inp):
rm -f metals.inp
./EXHALE.x ; mv output output_metals_off
```

The notebooks for each planet metals_comparison*.ipynb read the two folders and plot the temperature, velocity / mass-flux, H and metal ionization, and heating/cooling differences (cf. §7.).

2.10 Wind-AE warm-start initial conditions

A 1-D steady-state Parker-wind solver (a Fortran port of *Wind-AE*; Murray-Clay et al. 2009 / Broome et al. 2025) is included under src/modules/wind_ae/ and can supply the initial condition for a hard-to-launch EXHALE run. There are two ways to use it; the underlying algorithm, ramp, and BC/IC are documented in docs/wind_ae_solver.tex.

In-process (recommended): IC mode: windae. Add the single line

```
IC mode: windae
```

to input.inp (with Load IC? False). On startup EXHALE maps the planet's parameters to Wind-AE, ramps a shipped seed solution to the planet, solves the steady wind, and writes the IC onto the EXHALE grid (output/*_IC.txt),

which is then loaded automatically: the whole input $\rightarrow$ Wind-AE solve $\rightarrow$ IC $\rightarrow$ run sequence is one `./run_EXHALE.sh` (one `EXHALE.x`) invocation. This is a *warm start*: EXHALE re-solves the ionization (and any metals) from the first step, so the Wind-AE IC need not be exactly self-consistent. Always finish with Solver: Newton for a quantitative $\dot{M}$ (§2.7). The shipped seed and spectrum live in `inputdata/windae_seed.csv` and `inputdata/windae_spectrum.inp`. The ramp runs in two stages, and the second one is what makes far-from-seed planets reachable. It first tries the *static-BC* ramp, which moves the system parameters (flux, gravity, star) while holding the seed's base boundary conditions fixed: fast, and enough for a seed-adjacent hot Jupiter such as HD 209458 b. If that stalls, the seed is reloaded and the ramp is retried with the *self-consistent-BC* continuation, which re-converges the base $R_{\min}/T/\rho$, the sonic-point column density, and the bolometric/molecular layer turn-off as the parameters move (`wae_continuation.f90`). That is the default path, and with it the ramp converges the far-from-seed HD 189733 b as well as the seed-adjacent HD 209458 b (`examples/11_windae_ic` and `examples/12_windae_ic_hd209`). Only if *both* stages fail does the run stop with a message advising IC mode: `auto`. What the Wind-AE IC does not fix is EXHALE's own HD 189733 b base-breathing instability, which limits the warm start there whatever the IC source.

Because a generated IC is an analytic profile that the marching loop has not touched, its du can start far below every threshold and would once have tripped a false convergence stop within a few steps. Both the du stop and the JFNK hand-off now require du to have been seen at or above their own threshold first, so only a descending crossing counts (§2.4). Measured on `examples/12_windae_ic_hd209` (2026-08-15): du at step 2 is 1.43×10^{-4} , neither trigger fires, and the two arm at steps 23 and 222 on the first ascending crossing. The false stop is structurally gone for this IC mode; an $\dot{M}$ read off a du stop still carries its usual path-dependent spread, so the Newton finish remains the prescription.

Standalone generator: `wind_ae_ic.x`. The same solver builds an IC out of process:

```
make wind_ae_ic      # builds ./wind_ae_ic.x (separate from EXHALE.x)
./wind_ae_ic.x <input.inp> <seed.csv> <spectrum.inp> \
                <IC_dump_grid> <outdir> [rhoscale]
```

It reads an EXHALE `input.inp`, ramps from the seed, and writes `<outdir>/{Hydro_ioniz,Ion_species}_IC.txt` on the grid given by an `IC_dump.txt` from a prior EXHALE run with the same domain settings. Run EXHALE on the result with Load IC? `True` + Solver: Newton. (The standalone tool is independent of the in-process path above; either can be used.)

3. Input files

3.1 `input.inp`: the main parameter file

`input.inp` is read *line by line in a fixed order* for the core block (Table 2), followed by an *optional keyword block* (Table 3) whose lines are matched by keyword and may be omitted (older files without them keep the default behavior). A few lines are *conditional*: the spectrum-property line depends on Spectrum type, and the energy-band / X-ray-luminosity lines depend on Use only EUV?.

Table 2: input.inp core block (fixed order). Conditional lines are marked (*).

Label	Value	Meaning
Planet name:	string	Label for the run (cosmetic).
Log10 lower boundary number density [cm ⁻³]:	real (log ₁₀)	log ₁₀ n_0 , the H number density at the base. <i>Optional since 2026-09-03</i> : a base.inp carrying p_base states the base level itself and n_0 follows from it. Give one or the other; if both are given they must agree to 1% or startup refuses the pair.
Planet radius [R_J]:	real (R_J)	Base radius R_p (the 1 μ bar level); the grid radius r is in units of this. The unit is the IAU 2015 nominal Jupiter <i>equatorial</i> radius, $R_J = 7.1492 \times 10^9$ cm (parameters.f90); the Python tools take the same value from RJ_CM of examples/exhale_io.py.
Planet mass [M_J]:	real (M_J)	Planet mass M_p (sets gravity / the Jeans parameter b_0). The unit is the IAU 2015 nominal $M_J = 1.8982 \times 10^{30}$ g; the stellar mass below is in $M_\odot = 1.98842 \times 10^{33}$ g.
Equilibrium temperature [K]:	real (K)	Base temperature T_0 .
Orbital distance [AU]:	real (AU)	Star–planet separation a (sets the XUV flux $\propto a^{-2}$ and the tidal/Roche potential).
Escape radius [R_p]:	real (R_p)	Inner radius of the constant-momentum convergence window; must lie <i>inside</i> the domain ($< r_{\max}$). For L1-truncated Roche runs use a value below the lobe.
He/H number ratio:	real	Helium abundance $n_{\text{He}}/n_{\text{H}}$; > 0 enables He chemistry.
2D approximate method:	see text	Day–night averaging of the XUV flux and $\dot{M}$: Rate/2 + Mdot/2 (dayside, $\xi = \frac{1}{2}$), Mdot/4 (full-sphere), Mdot, or alpha (then a trailing number sets the rate-correction a_τ).

Table 2, continued

Label	Value	Meaning
Parent star mass [M_sun]:	real ($M_{\odot}$)	Stellar mass $M_{\star}$ (tidal potential, Roche-lobe size).
Spectrum type:	Power-law Planck Load Monochromatic	The incident spectrum. One type builds every band of the photon grid, the sub-13.6eV band and the Balmer continuum included (§3.5); no band is filled from a type the input did not select.
(*) Power-law index: / Spectrum file: / Photon energy [eV]:	real / path / real	Property line of the chosen type: the power-law slope E^{PLind} , the numerical SED file, or the monochromatic energy. Planck has no property line: it is built from Stellar Teff [K] and Stellar radius [R_sun] of the optional block, and startup refuses Planck without both of them. With metals or the He 2 ³ S triplet, mind the required wavelength coverage: a Loaded table that does not reach the band an active absorber needs stops the run (§3.5).
Use only EUV? True/False:	logical	True: EUV band only. False: include X-rays (adds the band's third energy and an X-ray luminosity line).
(*) [E_low, E_mid, E_high] = [. . .]	reals (eV)	Band edges: two values (EUV only) or three (with X-rays).
(*) Log10 of X-ray luminosity [erg/s]:	real (log ₁₀)	Only if X-rays are included; log ₁₀ L_X .
Log10 of EUV luminosity [erg/s]:	real (log ₁₀)	log ₁₀ L_{EUV} (with a , sets the incident XUV flux).
Grid type:	Uniform Stretched Mixed	Radial grid spacing. Mixed (fine near the base, stretched outward) is the usual choice. Every type puts the first cell face at the base level, $r_{\text{edg}}(0) = 1$ (§2.6).

Table 2, continued

Label	Value	Meaning
Numerical flux:	HLLC ROE LLF	Riemann solver for the hydrodynamic fluxes (Num_Fluxes.f90). HLLC is the usual choice; LLF (local Lax–Friedrichs) is the most diffusive and is kept for solver-isolation tests. MEASURED 2026-09-17 (item L25): on a wind whose base is subsonic to $M \sim 10^{-7}$ the HLLC flux can fail to reach the stationary root on a 500-cell grid where it reaches it on 1000 cells, and ROE reaches it on 500; where both fluxes solve the same case they differ by -0.3 to -0.6 percent in $\dot{M}$ and by 6 to 8 percent of opposite sign in the first cell's T and ρ .
Low Mach velocity jump:	logical (def. False)	Scale the normal velocity jump of the ROE branch by $\min(\text{Ma}, 1)$ after Rieper (2011, J. Comput. Phys., 230, 5263); no effect under HLLC or LLF. Delivered off and NOT recommended on a near-hydrostatic column: the factor vanishes exactly where two neighboring velocities cancel in the Roe average, so it removes the damping of the odd-even velocity mode at the faces that mode lives on, and it was measured to prevent a solve that unmodified ROE completes. Rieper's premise, that the local and the global Mach number are comparable, is violated by eight decades on such a column.
Reconstruction scheme:	PLM WENO3 PLM+WENO3	Spatial reconstruction order and staging. PLM or WENO3 runs a single stage (only the first du_th value is used as the convergence threshold); PLM+WENO3 selects the two-stage mode (march in PLM to the first du_th value, then switch to WENO3 and converge at the second).

Table 2, continued

Label	Value	Meaning
Include He23S?:	logical	Include the metastable He 2 ³ S triplet (needed for the He 10830 line). <i>On by default</i> , and the only line of the core block that may be omitted: a file without it gets the triplet. False is the deliberate opt-out, used to compare against the atomic helium limit. Forced off when the gas carries no helium.
Load IC?:	logical	Restart from the previous run's output/*_IC.txt (warm start). What the restart <i>continues</i> is stated by Restart intent: below; the state files also carry the configuration they were produced under, and a restart under another composition, grid, constant set or option set is refused naming the field (md/input_schema.md appendix D.2), unless the differing options are named by Restart option change: below.
Do only PP:	logical	Skip the time loop and only run the post-processing on existing output. It still takes one time step before it stops, so it is not an identity on the state it reads, it reconstructs with whatever Reconstruction scheme: states rather than with the scheme the solution was reached under, and it writes the stepped state back over Hydro_ioniz.txt. For the advection-corrected products of a <i>solved</i> state use Restart intent: stationary evaluate instead, which takes no step at all (item L9, 2026-09-17).
Force start:	logical	Force the first 1000 iterations regardless of the convergence test (useful when the IC is close to a fixed point).

The constants these units mean. Every physical constant has one definition, in `src/modules/init/parameters.f90`, and the Python tools take the same numbers from `examples/exhale_io.py`. The ones a user input is quoted in, or that set the scale of a reported number, are

$$\begin{aligned} R_J &= 7.1492 \times 10^9 \text{ cm} & M_J &= 1.8982 \times 10^{30} \text{ g} & M_\odot &= 1.98842 \times 10^{33} \text{ g} \\ R_\odot &= 6.957 \times 10^{10} \text{ cm} & \text{AU} &= 1.495978707 \times 10^{13} \text{ cm} & G &= 6.67430 \times 10^{-8} \text{ cgs} \end{aligned}$$

R_J , M_J and $M_\odot$ are the IAU 2015 nominal values (Prša et al. 2016, AJ 152, 41, Table 1), R_J being the nominal *equatorial* radius, which is what a transit depth measures. The mass unit of the code is the hydrogen *atom*, $m_H = 1.67353284 \times 10^{-24}$ g, and one helium nucleus weighs $m_{\text{He}}/m_H = 3.9715$: the helium-4 atom mass $6.6464790722 \times 10^{-24}$ g divided by m_H , not the integer 4, so the mass density, the mean molecular weight and the base composition are the helium-4 gas and not a 0.7% heavier one. The ionization thresholds are the NIST ASD values, one definition each and every cross section turning on exactly there: $e_{\text{th}}(\text{H I}) = 13.598434599$, $e_{\text{th}}(\text{H}_2) = 15.425927$, $e_{\text{th}}(\text{He I}) = 24.587389$, $e_{\text{th}}(\text{He II}) = 54.417765$ and $e_{\text{th}}(\text{He } 2^3S) = 4.767775$ eV.

Table 3: `input.inp` optional keyword block (any line may be omitted; defaults restore the standard behavior).

Keyword	Value	Meaning
Domain mode:	Spherical	If present and Spherical: pure planetary potential $-b_0/r$, domain extended to Outer radius. Absent (default): full Roche (tidal) potential; outer boundary at the Hill/ L_1 radius unless Outer radius is given (§1.3).
Outer radius [R_p]:	real > 1 (R_p)	Outer boundary (e.g. 10–15 R_p). In Spherical mode it sets the pure $-b_0/r$ domain; in Roche mode (no Domain mode line) it extends the domain past L_1 while <i>keeping</i> the tidal potential (Yan/Huang-style; pair with IC mode: auto).

Table 3, continued

Keyword	Value	Meaning
Base grid [dr,cells]:	real (R_p), int (def. $2.0\text{e-}4$ 50)	Resolution of the uniform region of the Mixed grid: cells uniform cells of size dr stacked on the base, then N -cells stretched cells out to $r_{\max}$. The product $\text{dr} \times \text{cells}$ is the extent of the uniform region ($0.01 R_p$ by default), so refining the base at fixed extent means dividing dr and multiplying cells by the same factor ($5.0\text{e-}5$ 200 is the $4\times$ refinement). Requirement: dr must resolve the base density scale height $H = kT/(\mu g)$; below ~ 5 cells per H the scheme carries an undamped stationary $2dr$ entropy mode at the base (§2.5). Ignored by Grid type: Uniform and Stretched. At fixed Grid cells: a finer base leaves fewer cells for the stretched region; raise the two together to hold the upper stretch (§2.5).
Grid cells:	int ≥ 10 (def. 500)	Total number of computational cells N (ghost cells not counted). The default reproduces the value that used to be compiled in, so an input.inp without the key is unchanged. A restart (Load IC? True) must load an IC written on <i>this run's grid</i> : the row count is checked against $N + 2N_g$, and then the file's cell centers are compared with the ones define_grid built over the physical cells $1 \dots N$. A relative difference above 10^{-10} anywhere stops the run, naming the cell and both radii. The reader does not interpolate, so a state written under a different Grid type, Base grid or Outer radius, or by a binary older than 2026-09-27 (whose first face was not the base level), has to be mapped onto this grid with src/utils/map_state_to_grid.py --ic before it is handed back, and then loads as a seed and not as a continuation (§2.6; load_IC.f90 verify_restart_radii).

Table 3, continued

Keyword	Value	Meaning
Outer shells [r_face,cells]:	real (R_p), int (def. none)	Append cells cells beyond the outer face of the grid the keys above construct, the last of them ending on the face r_face, which must lie beyond $r_{\max}$; center spacings in geometric progression (define_grid.f90 append_outer_shells). The constructed cells, faces and volumes are unchanged bit for bit, so the key moves the outer boundary without moving the interior, where Outer radius rebuilds every cell. N becomes Grid cells + cells; both numbers are required and a missing or non-positive one stops the run.
Max steps:	int (def. 10^6)	Hard cap on marching iterations (count_max); the run reports that it stopped without converging when the cap is reached. Lower it to bound an exploratory run, raise it for a high-gravity case whose base time step makes 10^6 steps too short (§2.4).
Coronal cutoff width:	real (def. 0.1)	Roll-off width w of the coronal-excitation guard below the 10^3 K CHIANTI fit floor: the coronal part of every metal-line coefficient is multiplied by $\exp(-x^2)$ with $x = (T_{\text{floor}}/T - 1)/w$, so smaller w cuts the extrapolated coronal cooling off more sharply. Since the ground-term fine-structure floors were solved in statistical equilibrium (§8.) the results no longer depend on w : the base-cell balance temperature of all four paper planets is identical to every digit printed over $w = 0.02$ – 1.2 , against a 500–650 K spread before. The default is kept at 0.1, and the guard is kept because extrapolating a fit below its fitted range is the wrong thing to do, not as a tuning knob (md/coronal_cutoff_width.md, section 7).

Table 3, continued

Keyword	Value	Meaning
Base IR field:	True/False False)	(def. Let the infrared coolants of the molecular layer see the thermal radiation of the atmosphere below the base instead of emitting into vacuum. The lower atmosphere is taken to be black at those wavelengths and to radiate $B_\nu(T_0)$ over the sky fraction $1 - \sqrt{1 - (R_p/r)^2}$, and the cooling becomes the net rate, emission minus absorption of that field, so each band stops cooling at its own radiative-equilibrium temperature (576–642 K for the C I/OI fine-structure lines and 941 K for the H_3^+ bands, at $T_0 = 1140$ K and half-sky coverage). Applies to the eight ground-term fine-structure lines of C I, C II, N II, O I and to the H_3^+ infrared cooling; every other channel keeps the optically thin, no-incident-field limit. Off by default, and with no effect on an atomic run, where those channels carry no cooling (md/lower_atmosphere_coupling.md).

Table 3, continued

Keyword	Value	Meaning
Molecular IR bands:	True/False False)	(def. Add the infrared coolants an H ₂ atmosphere carries below the H ₂ → H front and the atomic channels do not: the H ₂ quadrupole and magnetic dipole line spectrum (Roueff et al. 2019, A&A 630, A58) and the H ₂ O and CO vibration–rotation bands (HITEMP, through the correlated- <i>k</i> coefficients distributed with Photochem). Each is computed in LTE, in the optically thin limit, as the <i>net</i> exchange $4\pi n_X \int \sigma_\nu(T) [B_\nu(T) - W B_\nu(T_0)] d\nu$ with the same diluted field Base IR field supplies, so each stops cooling at its own radiative-equilibrium temperature: 889, 918 and 984 K for H ₂ O, CO and H ₂ at $T_0 = 1140$ K and half-sky coverage. The closure keeps stimulated emission, so the net rate is identically zero for gas at T_0 with the whole sky black. H ₂ needs only Molecular chemistry; H ₂ O and CO exist only with Oxygen chemistry, and the key is inert on species the run does not carry. Used together with Base IR field: on its own the new bands emit into vacuum, which deepens the collapse of the molecular layer instead of holding it, and input_read warns. Writes three net columns (H2_IR, H2O_IR, CO_IR, negative where the band heats) and the Planck-mean optical depths of the H ₂ O and CO columns to output/Cooling_breakdown.txt, and a column-integrated infrared block to output/FUV_bands.txt. Cross sections tabulated 50–2000 K and clamped outside it (src/modules/lower_atmosphere/molecular_infrared_cooling.f90, cooling_data/molecular_infrared_bands.py).

Table 3, continued

Keyword	Value	Meaning
Molecular reaction heat:	True/False (def. True)	Deposit the energy the <i>collisional</i> reactions of the H_2/He network release into the gas. On by default; the key exists to turn it off. In an atomic gas the ionization energy a photon spends comes back as the Lyman photon of a radiative recombination and leaves, so it is correctly absent from the heating. In a molecular gas it does not: the cycle $H_2 + h\nu \rightarrow H_2^+ + e$ (the photon pays $I(H_2) = 15.43$ eV and the photoelectron keeps the rest, which the photoheating already deposits), $H_2^+ + H_2 \rightarrow H_3^+ + H$ (+1.70 eV), $H_3^+ + e \rightarrow H_2 + H$ (+9.25 eV, <i>dissociative</i>) returns to H_2 having turned one H_2 into $H + H$ and left $I(H_2) - D_0(H_2) = 10.95$ eV in the gas with no photon to carry it away. Measured at 81 percent of the total heating rate at $1.02r_{\text{base}}$ on the converged $He/H = 0.0793$ hot Uranus. The heat is built from one table of species enthalpies, so a closed chemical cycle releases exactly zero; the photon-driven reactions, the radiative recombinations, the collisional ionizations, the Penning channels and the H/He charge exchange are excluded because their energy is already in the ledger (the charge exchange in its own channel, <code>heat_charge_exchange</code>). Needs Molecular chemistry, so an atomic run is unaffected to the bit. Writes the column <code>heat_mol_chem</code> to <code>output/Heating_breakdown.txt</code> (<code>src/modules/lower_atmosphere/molecular_reaction_heat.f90</code> , <code>Update_EXHALE_stage1</code> Sec. 141).

Table 3, continued

Keyword	Value	Meaning
H2 double ionization:	chung80 yan_rho off (def. chung80)	Resolve $H_2 + h\nu \rightarrow H^+ + H^+ + 2e$ as a channel of its own, vertical threshold 51.4 eV (Yan, Sadeghpour & Dalgarno 1998), two protons an event where the single dissociative channel gives one. chung80: 20% of the H^+ -producing events at and above 80 eV, the number Chung et al. (1993) state outright, ramped linearly from the threshold (the ramp is not in the source); yan_rho: built from the double-to-single ionization ratio of Yan et al. (1998). At 110 eV chung80 gives 15% more double ionization than yan_rho. Running both is how that uncertainty is reported. off folds the channel back into the single dissociative one, which reproduces the code before the channel existed and is not a physical statement. Any other word stops the run (src/modules/functions/h2_photo_channels.f90).
H2 neutral dissociation:	True/False (def. True)	Resolve $H_2 + h\nu \rightarrow H + H$, the absorptions that make no ion, over the 33–41 eV window in which Chung et al. (1993) Table 1 measure a photoionization yield below unity (up to 7.4% of the absorptions at 37.5 eV); zero outside it, where their source assumes unit yield. False folds that share back into the ionizing channels, which reproduces the code before the channel existed.

Table 3, continued

Keyword	Value	Meaning
Stellar LW flux [erg/cm ² /s]:	real (def. 0)	Band-integrated stellar flux in the H₂ Lyman–Werner bands (912–1201 Å, i.e. 10.3–13.6 eV) <i>at the planet’s orbit</i>. Switches on H₂ photodissociation, $\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}$, in the molecular network. The rate is the incident band photon fluence $F_{\text{LW}}/\overline{h\nu}$ contracted with a dissociation cross section $\sigma_{\text{diss}}(N_{\text{H}_2}, T, n_{\text{H}})$ read from a level-resolved CLOUDY table (line self-shielding and the trapping of the fluorescent decay photons are both inside it) times $e^{-\tau}$ of the continuum of the same interval, and it is averaged over each cell between its two faces rather than taken at one of them (the cross section falls by four decades across the self-shielding transition). Plus 0.4 eV of heating per dissociation (Black & Dalgarno 1977). The band is a separate input because the photon grid need not carry it: the grid floor is the lowest ionization threshold of an <i>active</i> absorber (§3.5), so a molecular run with no metastable, no low-IP metal and no armed excited-H coupling has no grid point in 11.2–13.6 eV at all, and where the grid does reach that far, a power-law type gives only the XUV fit extrapolated downward rather than a stellar FUV spectrum. With Spectrum type: Load the key may be omitted: the band is then integrated from the spectrum file over 912–1201 Å (the exact integral of the piecewise-linear table, the flux interpolated at each clipped edge; one read of the file serves the LW, B3 and B4 bands; the file is at the planet already) and the setup report says so. A stated value always wins, and a stated 0 switches the band off even when the file could supply it (the comparison with a model that has no H₂ photodissociation). A file that does not reach both edges of the band stops the run and asks for the key: a positive field is never turned into zero silently. Without a spectrum file, 0 (default) = no photodissociation. The key is inert without Molecular chemistry: True. Writes output/Lyman_Werner.txt

Table 3, continued

Keyword	Value	Meaning
Stellar FUV B1 flux [erg/cm ² /s]:	<i>retired</i>	Band B1 was 1110–1201 Å, between the Lyman–Werner interval and Ly α . The H ₂ Lyman and Werner lines pump on both sides of 1110 Å at the 700–3200 K of a planetary base, so a band edge there normalized the H ₂ pumping per photon of a band narrower than the one the lines absorb from. On 2026-09-06 the band was merged into the Lyman–Werner interval, which is now 912–1201 Å, and this key was retired: a file that still carries it stops the run with a message asking for the two numbers added into Stellar LW flux.
Stellar FUV B3 flux [erg/cm ² /s]:	real (def. 0)	The same over 1231–1450 Å (band B3, branching 0.89/0.11/0.00). Band B2 is the Ly α line and comes from Stellar Lya flux. With Spectrum type: Load and Oxygen chemistry: True, a run that does not state this key gets the band integrated from the spectrum file over 1231–1450 Å (the exact piecewise-linear integral, as for Stellar LW flux; a file that does not reach the band stops the run), and the setup report says so; a stated value, zero included, always wins.
Stellar FUV B4 flux [erg/cm ² /s]:	real (def. 0)	The same over 1451–2304 Å (band B4, branching 1.00/0.00/0.00), integrated from the spectrum file under the same rule when not stated. <i>Read the caveat in §3.7:</i> the H ₂ O cross section falls three decades across this band, so a single band average misrepresents it by a factor of several for a red spectrum.

Table 3, continued

Keyword	Value	Meaning
Stellar Teff [K]:	real (K)	Stellar T_{eff} . Two consumers: it arms the excited-H model when both this <i>and</i> Stellar radius are > 0 , and it is the temperature of the photosphere under Spectrum type: Planck, which is refused without both. Arming the excited-H model makes $H(n=2)$ an absorber of the grid-floor rule, so the photon grid then reaches down to the Balmer head $e_{\text{th,H}}/4 = 3.400 \text{ eV} = 3647 \text{ \AA}$ (§3.5).
Stellar radius [R_sun]:	real ($M_{\odot}$ -radius)	Stellar radius: sets the dilution $(R_{\star}/a)^2$ of the Planck field and of the Balmer continuum, and in EXHALE_transit.py it normalizes every transit depth and caps the impact-parameter integration at $R_{\star}/R_p$. Supply it for any run whose spectrum will be synthesized: absent, the transit script falls back to a built-in radius that need not match the system, and the depths scale accordingly.
Jlya RT file:	path	Read an external $\bar{J}_{\text{Ly}\alpha}(r)$ profile to pump $H(n=2)$ (sets jlya_mode=1).
Jlya escape-prob:	True	Compute $\bar{J}_{\text{Ly}\alpha}$ in-line each step by escape-probability RT (jlya_mode=2).
Stellar Lya flux [erg/cm2/s]:	real	Incident stellar $\text{Ly}\alpha$ flux at the planet (the stellar source for jlya_mode=2).
Lya stellar halfwidth [km/s]:	real (km/s)	Stellar $\text{Ly}\alpha$ line half-width; controls how deep the trapped stellar beam penetrates (default 70).
Lya stellar boost [-]:	real	Bounded buildup cap for the trapped stellar $\text{Ly}\alpha$ beam (default 5).
Lya absorbing bottom:	logical (def. False)	Treat the lower boundary of the jlya_mode=2 closure as a pure absorber (Huang et al. 2017: the H_2 layer beneath the base destroys $\text{Ly}\alpha$ through accidental resonances): a downward wing-escape channel β_{bot} on the depth to the bottom joins the upward and Sobolev channels, lowering $\bar{J}_{\text{Ly}\alpha}$ (and n_2) in the bottom few scale heights only. No effect on jlya_mode 0/1.

Table 3, continued

Keyword	Value	Meaning
Deexc heat:	logical (def. True)	Collisional de-excitation of $H(n=2)$ returns 10.2eV to the electrons. It is not a double count of the H I collisional-excitation cooling, which is the one-way coronal sum over the CHIANTI 11.0.2 H I levels (the same $1s \rightarrow 2s, 2p$ rates the $n=2$ model uses) with no de-excitation term in it; most of the $n=2$ population is maintained by $Ly\alpha$ pumping, so the term is absorbed $Ly\alpha$ thermalized by a collision. Active only with the excited-H model; False reverts to the one-way ledger.
Transonic IC:	True	Use a transonic isothermal-wind initial condition (solved from the potential via the Bernoulli integral) instead of the hydrostatic IC; needed for deep-RLOF bases that do not launch from rest. Auto-enables Force start.
Newton solver:	logical (def. True)	Solve the ionization equilibrium with the analytic-Jacobian Newton method (MINPACK hybrd1 takes over where it fails); False reverts to the legacy hybrd1. Default is faster ($\sim 25\%$) and reproduces hybrd1 to $\sim 10^{-7}$.
Brent solver:	logical (def. True)	Solve the post-process energy equation with a bracketing (Brent) root-finder that selects the physical root; False reverts to legacy hybrd1 + a $2\times$ -band reject.
du_th [PLM,WENO3]:	two reals	Convergence thresholds (e.g. $0.5 \quad 1.0e-3$). Which are used is set by Reconstruction scheme: above. With PLM+WENO3 (two-stage) EXHALE marches in PLM until $du < \text{first value}$ (or a PLM stall), then switches to WENO3 and converges at $du < \text{second value}$; with single-stage PLM or WENO3 only the first value is used and the second is ignored. The PLM+WENO3 mode replaces the manual PLM $\rightarrow$ stop $\rightarrow$ Load IC+WENO3 restart of the official ATES recipe.

Table 3, continued

Keyword	Value	Meaning
Reconstruction continuation:	real $\Delta\lambda$ [, real τ] [, fixed adaptive] (def. $\Delta\lambda = 0$, i.e. the one-step hand-off)	Walk the PLM $\rightarrow$ WENO3 hand-off of a two-stage run along the homotopy $R_\lambda = (1 - \lambda)R_{\text{PLM}} + \lambda R_{\text{WENO3}}$ instead of changing the discrete operator in one step. $\Delta\lambda$ is the advance in λ per marching step (≤ 0 off; 1.0 reproduces the one-step switch, byte-identically). τ (def. 1.2) is the step control: λ advances while du stays within τ times its value at the start of the ramp, and a step that exceeds it halves $\Delta\lambda$ instead of advancing. fixed disables that control, adaptive is the default; the two optional fields are told apart by content, not position. Requires Reconstruction scheme: PLM+WENO3 with a non-empty PLM stage, otherwise a warning is printed and the key is ignored. Once λ reaches 1 the continuation disarms and the run continues in pure WENO3, so the JFNK finish and every acceptance gate see the production operator.
Stall [tol,N]:	real, int	Stall-detector override (relative du change and window; defaults 10^{-6} , 2000).
CFL:	real (def. 0.6)	CFL-number override.
Hot Parker IC:	real (K)	Warm-seed initial condition: stable hydrostatic density with a warm T /ionization/velocity overlay.

Table 3, continued

Keyword	Value	Meaning
IC mode:	word	Initial-condition family: cold (default), transonic, hot_parker (synonyms for the two keys above), auto, which picks automatically from the cold sonic-point topology of the actual potential: an interior sonic point (deep-RLOF via the Roche L1 topology, or low-gravity boil-off, $r_c \simeq b_0/2c_0^2$ inside the domain) selects the transonic IC, otherwise the cold hydrostatic IC; or windae, which builds the IC in-process with the ported Wind-AE steady-wind solver (§2.10). No tunable threshold for auto; the choice and the escape parameter b_0 are logged. Explicit Transonic IC/Hot Parker IC keys override auto. See, for windae, docs/wind_ae_solver.tex.
Level tol:	real	Mass-flux <i>level</i> -stability gate: a converged/stalled stop additionally requires the mean $ \rho v r^2 $ over the wind window to change by less than this (relative) across the last <code>N_stall</code> steps. Guards against stopping on a transient du dip while the flux level still drifts (≤ 0 = off, default; §2.7).
Flux spread tol:	real [real]	FLUX gate (§133): a steady solve is accepted only when the radial spread of $\rho v r^2$ over $r \geq r_{\text{flux}}$ is below this tolerance AND the residual is below <code>Resid tol</code> . Defaults 2×10^{-5} (<code>flux_spread_th_default</code>) and $r_{\text{flux}} = 1.2 R_p$; ≤ 0 disables it.

Table 3, continued

Keyword	Value	Meaning
Resid tol:	real	Residual-based convergence: stop when the steady residual falls below the value (monitored every 500 steps), <i>replacing</i> the <i>du</i> test (≤ 0 = off, default; the JFNK finish then uses the code's own 10^{-5}). The number compared against it is $\max_k \max_j R_{j,k} /s_{j,k}$, each row divided by a bound on the largest term that row itself contains: mass $\max(F r^2$ at the two faces)/ ΔV , momentum $\max(\Delta F_2 , S_2)$, energy $\max(\Delta F_3 , S_3 , \text{heat}, \text{cool})$, plus the operator-split viscous and conduction sources where those are on. The maximum is over cells, so one cell out of tolerance keeps the state out of tolerance; the volume-weighted norm is reported beside it and no longer decides acceptance. <i>du</i> alone is blind to the energy imbalance (§2.7); the scales and the reason for each are in <code>md/input_schema.md</code> K35.
Time stepping:	Local	Pseudo-time steps in each cell (steady-state acceleration; not time-accurate). Reaches the deep state fast but leaves residuals in each cell; pair with a Newton finish.
Energy solver:	Explicit	Revert to the original forward-Euler source update (solver-isolation tests).
Solver:	Newton [real]	Production steady-state solve: marching warm-up until the flux metric <i>du</i> (the radial spread of $\rho v r^2$) drops below the optional value <code>newton_du_switch</code> (default 10^{-2}), then the JFNK Newton finish to <code>Resid tol</code> (default 10^{-5}), then the standard final outputs and post-processing. See §2.7.

Table 3, continued

Keyword	Value	Meaning
Run mode:	init phys (def. init)	What this run is doing, stated rather than inferred. init: initialization or continuation. It claims no elapsed time and permits local pseudo-time, pseudo-transient continuation and a chemistry that is not at its root, all as numerical devices; the physical clock <code>t_phys</code> stays where it started and no output reports it. phys: physical integration, one global dt per step, the clock advanced only by a step accepted in full, the temporal error of the step sampled every 20 accepted steps, and the certification refusing a state whose chemistry has no root. Any other word is a fatal error stop 1. Refused: phys with Time stepping: Local, since cells advanced by different intervals do not form one trajectory. The mode sets the default of Restart intent: below, is written into the state file's metadata block as mode and read back on a restart (a phys run loading a file that says mode=phys without a finite non-negative <code>t_phys</code> is refused), and is echoed in EXHALE_setup.out together with whether it was given or defaulted. See <code>md/input_schema.md</code> K45.

Table 3, continued

Keyword	Value	Meaning
Restart intent:	trajectory relaxation stationary [evaluate equilibrate] (def. trajectory with Run mode: phys, else relaxation)	What a loaded state is continued as. trajectory: the physical clock continues from the t_phys the state file records. relaxation: march toward stationarity and then hand off to the solver, which is what every restart did before this key existed. stationary: rebuild the derived quantities of the loaded state without taking a step, measure its stationary residual and its certification <i>as loaded</i> , and enter the stationary solve at once; the second word evaluate writes the measured state back with its certification and stops, and equilibrate puts the loaded composition on its own fixed point before the measurement (a change of the state, reported when taken). This is the route that certifies a written state without the two CFL steps a marching hand-off takes first. Since 2026-09-17 (item L9) evaluate also names the three states it handles apart (the state as loaded, the work state whose pressure and temperature are derived from the loaded conserved state, and the advection-derived one) and writes every product a solved case is read for from the work state with no time update: the two state files, the heating and cooling breakdowns, the *_adv profiles and the mass-loss line. The *_adv files carry a # derived_from: provenance line rather than a # coupling: header, because they describe a composition the certification was not made on. Refused: the key with Load IC? False, trajectory with Run mode: init, stationary without Solver: Newton.

Table 3, continued

Keyword	Value	Meaning
Restart option change:	token list, separated by commas or blanks (def.: no option may differ)	Which switches of the state file's # options line are <i>allowed</i> to differ between the loaded state and this run. <code>carrier_newton</code> is not among the tokens that have to be named: it states the ROUTE and not the equation set, the rows, columns and tolerances being the same either way, so a difference in it is admissible with nothing named, the load says so and the change is recorded as a # <code>route_change</code> line (item L22 increment I3, 2026-09-17). The restart contract otherwise refuses any option difference, which forbids the ladder of settings this code is converged with: converge without an option, restart with it on, converge again. Naming a token (<code>sec_ion</code> , <code>He23S</code> , <code>mol_heat</code> , <code>base_ir</code> , <code>cond</code> , <code>interdiff_enth</code> , ...) permits exactly that token to differ; every other difference still refuses the load and names the token. The loaded state is then a solution of the <i>former</i> option set and a starting point for this one, so the change is written into the state this run produces as a # <code>option_change</code> line and inherited by the rungs that follow, and EXHALE_setup.out states it. Refused: an unknown token; a token that decides how many unknowns the state has (<code>metals</code> , <code>mol</code> , <code>oxychem</code> , <code>carrier</code>), whose change makes the rows of the file not the rows of this run and is a cold start and not a restart (<code>iontrans</code> is not one of them: the three ionization stages have a column in every state file, and the key changes whether that column is a local root or a transported partition, so it may be named); a word naming the grid, the reservoir or the constant set; the key with Load IC? False. See <code>md/input_schema.md</code> appendix D.2.

Table 3, continued

Keyword	Value	Meaning
Low-Mach damping:	real ε_4 [, real M_{th}] (def. -1, i.e. off; $M_{\text{th}} = 10^{-3}$)	Artificial dissipation of the $2\Delta r$ contact/entropy mode. The HLLC flux carries the middle wave at $S^* \simeq v$, so the dissipation it applies to the entropy family vanishes as the flow stagnates and that mode becomes marginally damped; where metal cooling stalls a shell (HD 209458 b with solar C/N/O, $r \simeq 1.02$, $M \sim 10^{-5}$) it grows to several times the local mean $ v $. This key adds a gated fourth-difference (Jameson, Schmidt & Turkel 1981) stress $\varepsilon_4 g(M) \rho_f \lambda_f (v_{j+2} - 3v_{j+1} + 3v_j - v_{j-1})$ to the numerical <i>momentum</i> flux, and its work $v_f D_p$ to the energy flux, with the gate $g = [\max(0, 1 - M^2/M_{\text{th}}^2)]^2$ exactly zero for $M \geq M_{\text{th}}$. It enters the numerical flux, so the marching loop and the JFNK steady residual see the <i>same</i> equation. Explicit stability requires $\varepsilon_4 < 1/(16\text{CFL})$ (0.10 at the default CFL); the classical range 1/64–1/32 works. $\varepsilon_4 \leq 0 =$ off, bit-identical. See <code>src/modules/flux/low_mach_dissipation.f90</code> .

Table 3, continued

Keyword	Value	Meaning
Well balanced:	logical (def. False)	Carry the <i>departure</i> from the cell's own local hydrostatic equilibrium through the reconstruction, the Riemann jumps and the pressure force, instead of the state itself. Inside cell j the equilibrium is the constant-density one through that cell's own density and pressure, $p_{\text{eq},j}(r) = p_j - \rho_j[\phi(r) - \phi(r_j)]$, so no thermal, entropy or compositional stratification is assumed; the pressure stencil acts on the departure of a neighboring cell's pressure from that equilibrium continued <i>through</i> the face between them (this cell's density up to the face, the neighbor's beyond it), which vanishes at a discrete equilibrium, and the momentum row is assembled from the face pressure measured against that equilibrium while the gravitational source term is zero. The equilibrium's flux difference and its source then cancel in the algebra rather than in floating point, and a discrete hydrostatic equilibrium is preserved to rounding (Käppeli & Mishra 2016, A&A 587, A94; the face-pressure-difference form of the source is Käppeli & Mishra 2014, J. Comput. Phys. 259, 199). Exact preservation also needs a numerical flux that resolves a stationary contact, so it holds with Numerical flux: ROE and HLLC but <i>not</i> with LLF, and not at a face the positivity repair has replaced with a first-order flux. It applies to both reconstructions and to the PLM $\rightarrow$ WENO3 continuation. Turning it on <i>moves every result</i>: the pressure/gravity pair is in every run, and the two discretizations are both second-order consistent with neither a subset of the other. It is stated in EXHALE_setup.out and EXHALE_resolved.out, and written as the option token wellbal in a state file's # options line, where Restart option change: may name it, since it does not decide how many unknowns a state has. See src/modules/states/Reconstruction.f90.

Table 3, continued

Keyword	Value	Meaning
ATES_photoionization_rate:	True	Revert the He I (1^1S) photoionization cross section to the legacy ATES two-term fit (default: Verner+1996). Does <i>not</i> affect the He 2^3S Penning rate, which is always the temperature-dependent form (§8.).
Legacy_HHe_rates:	logical (def. False)	Revert the H/He case-B recombination and collisional ionization to the legacy ATES fits (Hui & Gnedin 1997 recombination; Abel+1997 / Hui & Gnedin 1997 collisional ionization). Default uses Badnell (2023) RR (+ He II DR) minus the ground capture α_1 for case B, α_1 being the Milne relation on the ground-state photoionization cross section the radiative transfer absorbs with (for H I, He I and He II), and Voronov (1997) collisional ionization (§8.). The ground captures added back by H_rec_escape and He_rec_coupling are that Milne α_1 in either setting. The free-free Gaunt factor and its ion net-charge weighting are always the van Hoof et al. (2014) form, independent of this switch.

Table 3, continued

Keyword	Value	Meaning
Atomic rate set:	default Koskinen2022 (def. default)	Swap the four atomic H/He rate coefficients for the fits Koskinen et al. (2022) list as R1–R4 of their Table 1, for a like-for-like comparison with their Model A: recombination $4.0 \times 10^{-12}(300/T)^{0.64}$ for H^+ and $4.6 \times 10^{-12}(300/T)^{0.64}$ for He^+ (Storey & Hummer 1995), and collisional ionization $2.91 \times 10^{-8}U^{0.39}e^{-U}/(0.232+U)$ with $U = 13.6\text{eV}/kT$ for H and $1.75 \times 10^{-8}U^{0.35}e^{-U}/(0.180+U)$ with $U = 24.6\text{eV}/kT$ for He (Voronov 1997). The two collisional fits are the ones EXHALE already uses, so only the two recombination coefficients change value; the branch does pin the Voronov form against Legacy_HHe_rates. The default set is the physically preferred one here (case B where the Lyman continuum is optically thick, the escaping ground captures added where it is thin; H_rec_escape, He_rec_coupling), so the key exists to reproduce their choice and not to replace ours. It swaps <i>rate</i> coefficients only, and the recombination cooling follows them: every recombination cooling term is $k_B T \alpha (3/2 + d\ln\alpha/d\ln T)$ on the coefficient the balance uses, so the electrons removed and the energy removed come from one rate. With He_rec_coupling or H_rec_escape on, the Milne ground capture α_1 is added to Table 1’s coefficient, which has no such split, so the closest like-for-like run also sets He_rec_coupling: False and H_rec_escape: False. Any other word stops the run (§8.).

Table 3, continued

Keyword	Value	Meaning
Secondary_ionization:	False True Immediate (def. True, staged)	Secondary ionization by fast photoelectrons in the main loop: a photoelectron ejected with $E_0 > 30\text{eV}$ deposits only $f_{\text{heat}}(x, E_0)$ of its excess as heat and drives further HI / He I / H ₂ ionizations, where x is the free electron density over the H+He nuclei. The ionization budget is Shull & van Steenberg's (1985) with the energy dependence of Dalgarno, Yan & Liu (1999); the heating fraction, the H ₂ target and the molecular heat returns are Dalgarno et al.'s. The default is <i>staged</i> (§2.4): the coupling switches on only after the wind first converges without it, then the run re-converges. False restores full photoelectron thermalization, bit-identical to the legacy path; Immediate applies it from step 0 (A/B tests only).

Table 3, continued

Keyword	Value	Meaning
He_rec_coupling:	logical (def. True)	Absorb the ionizing recombination radiation of $\text{He II} \rightarrow \text{He I}$ and $\text{He III} \rightarrow \text{He II}$ on the spot (§8). He II: the ground captures (α_1, the Milne relation on the He I cross section) count as recombinations with the local escape weight y, the fraction of their 24.6 eV photons that He I does not re-absorb in the cell, so the He II coefficient is $\alpha_B + y\alpha_1$ (case A where the cell is thin, case B where it is thick); the photons that H I, H_2 and the metals take ionize those species; and the case-B captures leave through their own exits, the 584 Å line (scattered by He I and partly converted into the 2^1S two-photon continuum), the 19.8 eV 2^3S line and the two-photon continuum, shared by the network's rates (the H-ionizing photons of a case-B capture are $z = 0.966$ as $n_e \rightarrow 0$ and 0.660 at $n_e = 10^8 \text{ cm}^{-3}$, at 10^4 K in a cell without He I, as the physics probe <code>recombination_radiation_on_the_spot</code> measures them; He I scattering of the 584 Å line lowers z). He III: the ground capture, the direct capture into $n = 2$, He II Lyα and the $2s$ two-photon continuum (Nussbaumer & Schmutz 1984), with the $2s$–$2p$ mixing of He II by ion impact. All the photoelectron energy is heat (channel <code>heat_He_recomb</code>). With the He 2^3S network on, case B is split into triplet and singlet captures with Hummer & Storey (1998) Table 5 in both settings of the key. False = case B for both recombinations with the cascade photons lost.

Table 3, continued

Keyword	Value	Meaning
H_rec_escape:	logical (def. True)	The $\text{H II} \rightarrow \text{H I}$ ground captures whose 13.6 eV photons leave the cell or are taken by a metal ion count as recombinations: the H II coefficient is $\alpha_B + y_{\text{HI}} \alpha_1$, α_1 the Milne relation on the hydrogenic ground cross section and $y_{\text{HI}} = 1 - n_{\text{HI}} \sigma_{\text{HI}} \Delta r (1 - e^{-\tau}) / \tau$ the escape weight of the same construction as He II's (τ the cell's optical depth at 13.6 eV over H I and the metal ions), so case A holds in the thin outer wind and case B in the thick base. The recombination cooling uses the same coefficient at the frozen weight; the $n = 2$ sources of the H($n=2$) model stay case B. The closure is local: a photon that leaves the cell is dropped, so where the cells are neither thick nor thin the weight depends on the cell width. On by default because it corrects the case-B assumption where it does not hold; False = case B everywhere (e.g. for a like-for-like comparison with Koskinen et al. 2022). Written as use_h_rec_escape in the parse dump (EXHALE_PARSE_DUMP=1).

Table 3, continued

Keyword	Value	Meaning
He_H_charge_exchange:	logical (def. True)	The $\text{He} \leftrightarrow \text{H}$ charge exchange: $\text{He}^+ + \text{H}^0 \rightarrow \text{He}^0 + \text{H}^+$ as the sum of its radiative channel (Stancil, Lepp & Dalgarno 1998, from Zygelman et al. 1989) and its non-radiative one (the Maxwellian average of the $\text{H}^+ + \text{He}$ cross section of Loreau, Ryabchenko & Vaeck 2014, turned forward by microscopic reversibility; their calculation starts at 15 eV/u, and since 2026-09-26 the reverse cross section between the threshold $x_{\text{th}} = 13.655$ eV/u and 15 eV/u is taken as $\sigma_r(15) \sqrt{(x - x_{\text{th}})/(15 - x_{\text{th}})}$, i.e. $\sigma_f v$ constant at low forward energy, an assumption no calculation tests; the channel is 3.3% of the radiative one at 10^4 K and 5.9% at 10^3 K); $\text{He}^0 + \text{H}^+ \rightarrow \text{He}^+ + \text{H}^0$, the detailed-balance reverse of the non-radiative channel (endothermic by 10.99 eV); and the radiative capture $\text{He}^{2+} + \text{H}^0 \rightarrow \text{He}^+ + \text{H}^+ + h\nu$ (the Maxwellian average of the cross section of West, Lane & Cohen 1982, $1.6\text{--}2.0 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ at $10^3\text{--}3 \times 10^4$ K), which has no reverse. The non-radiative exchange heats the gas by 10.99 eV an event and its reverse takes that back; the radiative channels give their energy to the photon (channel heat_charge_exchange). Applied by dedicated routines in <i>every</i> helium-bearing ionization system, independent of cx_full (which gates the metal groups C and D). False leaves all three out.
He_diffusion:	logical (def. False)	Enable diffusive separation of helium: transport the helium <i>mass fraction</i> of the two-component H/He gas with advection + molecular-diffusion settling, so He/H falls with altitude. Off = He/H frozen at the input He/H number ratio (§3.6).

Table 3, continued

Keyword	Value	Meaning
He_Kzz:	real (cm ² s ⁻¹ , def. 0)	Eddy-diffusion coefficient used with He_diffusion: mixes the composition toward uniform below the homopause. Default 0 = pure molecular diffusion; an eddy term belongs to the atmosphere being modeled, so it is stated rather than inherited (Taylor+2025 use 10 ⁹).
He_ambipolar:	logical (def. True)	Ambipolar-corrected settling: in the ionized wind the polarization field lifts He ions, reducing settling. The field is computed from the local electron-pressure gradient, $eE = -kT \partial \ln(n_e T) / \partial r$; False = $eE = 0$, i.e. neutral settling $\Delta m = 3$.
He_alphaT:	real (def. 0)	Thermal-diffusion factor α_T added to the settling drift ($\alpha_T \partial \ln T / \partial r$); 0 = off.
He_metal_diffusion:	logical (def. False)	Also diffuse each trace metal against the hydrogen background with its own mass, mean charge and stage-resolved diffusion coefficient; heavier elements settle more. Off = metals frozen to H (§3.6).

Table 3, continued

Keyword	Value	Meaning
Interdiffusion enthalpy flux:	True/False True)	(def. Carry the interdiffusion enthalpy flux $q_d = \sum_s h_s J_s$ in the energy equation of the diffusing mixture (Cook 2009, Phys. Fluids 21, 055109, eqs. 11–13): the sensible enthalpy the helium (and, with He_metal_diffusion, each metal) carries as it moves against hydrogen, and the sensible enthalpy each transported molecular carrier (H ₂ ; OH, H ₂ O, CO with the oxygen chemistry) carries relative to the rest of its elements, in the marching energy update and in the stationary energy row alike (§3.6). It acts with He_diffusion: True and helium in the gas, or with Molecular carrier transport: True; otherwise the key is inert, bit for bit. A restart from a state written without the carrier term names the token carrier_enth on a Restart option change: line. On by default, because the term belongs to the energy equation of any mixture whose components move relative to one another ; False exists only to reproduce published escape models that omit it (Koskinen et al. 2013, 2022; Yelle 2004). MEASURED on the LHS 1140 b He_diffusion states: 13–26% of the local energy budget at 1.1–5 R_p (md/interdiffusion_enthalpy_flux_20260926.md). Written as the option token interdiff_enth of a state file’s # options line; a state written before the key existed reads as interdiff_enth=F, so restarting such a He_diffusion state under the default needs Restart option change: interdiff_enth.

Table 3, continued

Keyword	Value	Meaning
Molecular chemistry:	True/False False)	(def. Add H_2 , H_2^+ , H_3^+ and HeH^+ to the coupled ionization equilibrium, with the H_2 photoionization opacity and heating and the H_3^+ infrared cooling. Requires He/H number ratio > 0 . He_diffusion is allowed (since 2026-08-26): the element transport closes over the molecular carriers. Turning it on also turns <code>Molecular base</code> on (see there). A <code>metals.inp</code> may be present: the metal stages are then solved in the same system (§3.7).
Molecular base:	True/False False)	(def. Remove the H_2 -bound particles from the base particle count <code>ntot_bc</code> using the chemical-equilibrium $\text{H}_2/\text{H}/\text{He}$ partition, giving a lower base pressure and a heavier base μ with the species arrays still atomic (Tier 2a). Forced on by <code>Molecular chemistry: True</code> , because an atomic <code>ntot_bc</code> under a molecular base inflates the base temperature by the particle-count ratio and dissociates the layer the chemistry just built, so with the chemistry on the key is redundant (§3.7).

Table 3, continued

Keyword	Value	Meaning
Oxygen chemistry:	True/False False)	(def. Add OH, H ₂ O and CO to the same coupled system, with the FUV photolysis of H ₂ O and OH, so that the code computes its own base H ₂ /H partition instead of importing one. Requires Molecular chemistry: True, He/H > 0 and a non-zero oxygen abundance; refuses q_H2_base (it computes that quantity) and He_metal_diffusion (that option counts an element over its ion stages alone, and CO carries an oxygen <i>and</i> a carbon nucleus, so it cannot follow two element factors at once). He_diffusion alone is accepted. Supply the band fluxes with Stellar LW flux, Stellar FUV B3/B4 flux and Stellar Ly α flux; with Spectrum type: Load the three continuum bands (LW, B3, B4) are integrated from the spectrum file when their keys are absent, and only Ly α has to be stated. Without band fluxes the run returns the chemical equilibrium of the O/OH/H ₂ O family rather than a photochemical partition, and says so at startup. Writes output/Oxygen_chemistry.txt and output/FUV_bands.txt and appends OH H2O CO to Ion_species.txt (§3.7).

Table 3, continued

Keyword	Value	Meaning
Molecular carrier transport:	True/False (def. True whenever Oxygen chemistry is on)	Transport the molecular carriers H ₂ , OH, H ₂ O and CO vertically instead of closing them by a local steady state: one implicit backward-Euler diffusion-advection step per pass, solved together with the same chemistry rows the local equilibrium uses, with molecular diffusion from Blanc's law over the background carriers, the eddy coefficient K_{zz} of He_Kzz or of a lower-atmosphere profile, and settling. False restores the local steady state, which isolates the chemistry for testing and is not a model of a base: at a cool base the H ₂ chemical time and the flow time are comparable. With $K_{zz} = 0$ everywhere the transport is pure molecular diffusion, the wrong limit for a lower atmosphere, and the startup report says so. It acts only with Molecular chemistry: True, not with Oxygen chemistry: H ₂ is the first carrier and is transported with or without the oxygen cycle, which only adds OH, H ₂ O and CO to the set, and Ionization transport adds H ⁺ . With the molecular network off there are no carriers and the key changes nothing; input_read reports that at startup rather than refusing the run, the state produced being the correct atomic one (§3.7).
Oxygen transport:	<i>retired</i>	The former name of Molecular carrier transport, retired on 2026-09-02 because the operator transports H ₂ whether or not the oxygen cycle is on. A file that still carries it stops the run with a message naming the new key; it is not read as an alias.

Table 3, continued

Keyword	Value	Meaning
Coupled carrier solve:	False True On stall (def. False; a word the key has no meaning for stops the run rather than falling through to False)	Solve the transported balances as unknowns of the stationary system itself, $(\rho, \rho v, E, n(\text{H}_2)/n_0, \dots)$ per cell, instead of alternating a wind solve with a fixed-wind carrier relaxation. The alternation was measured not to converge: the two halves are coupled through the particle count, which sets the temperature, so a splitting that evaluates each at the other's old state gets the sign of the H_2 front's motion wrong. On a molecular configuration this key requires Molecular carrier transport: True, and the combination without it is refused at startup. With the carriers eliminated, $n(\text{H}_2)$ is not an unknown but whatever the local-equilibrium sweep returns for the composition it was seeded with, and in the shielded layer the sweep cannot return a content at all: the fast chemistry cycles $\text{H}_2 \rightarrow \text{H}_2^+ \rightarrow \text{H}_3^+ \rightarrow \text{H}_2$ without changing the number of H_2 nuclei, so the local balance rows fix the partition among the molecular species and leave their sum where the seed put it. The stationary residual is then not a function of its unknowns (measured seed dependence 3.2×10^5 times <code>Resid tol</code> per unit relative seed change, against 9.7×10^{-10} with the H_2 row carried), so it has no root. A run whose only transported balance is an element (<code>He_diffusion</code> in an atomic gas) is not affected, and neither is marching. The carrier densities of the coupled system are carried as $\ln n$, so that the positivity of a carrier is a property of the unknown space rather than a bound the step control has to enforce; the floor under $\ln n$ is 10^{-20} of the cell's own element budget, the fraction of its element below which the carrier row scale already calls a row converged (<code>EXHALE_CARRIER_LOG_UNKNOWN=0</code> restores the density unknown for measurement).

Table 3, continued

Keyword	Value	Meaning
Composition update holds:	energy pressure (def. energy)	Which thermodynamic variable the composition update of the stationary alternation (§2.7) keeps fixed beside ρ and v while the element and carrier relaxations change the composition at the fixed hydrodynamic state. energy: the conserved E , and p and T follow at the new composition. pressure: p , with $T = p/(n_{\text{tot}} + n_e)$ of the new composition and E rebuilt from its caloric equation of state, the kinetic part unchanged. A route through the iteration and not an equation set: a stationary state is stationary under either. Any other word stops the run.

Table 3, continued

Keyword	Value	Meaning
Ionization transport:	True/False (def. False)	Carry the hydrogen ionization state with the flow: H^+ becomes a fifth transported carrier of the same operator that carries H_2, and the ionization sweep is handed the transported fraction instead of solving the H/H^+ partition as a local root. The local partition is the right answer only where a parcel is ionized faster than it leaves its shell, $Pr/ v \gg 1$. On the Koskinen et al. (2022) Model A comparison that ratio is 0.15–0.35 above $1.5 r_{\text{base}}$, and the local root then gives $x(H^+) = 0.129/0.327/0.585$ at $2.0/2.4/3.0 r_{\text{base}}$ where integrating along the same flow gives 0.057/0.078/0.106 and Model A, which advects every species, has 0.020/0.055/0.080 (md/k22_electron_density_excess.md §7). The proton's molecular diffusion is set to zero, an ion in a neutral gas being ambipolar- or Coulomb-limited rather than hard-sphere and the term this option exists for being the bulk advection, while the eddy coefficient K_{zz} still applies, it being a bulk mixing coefficient. There is no base Dirichlet value, no handoff stating an ionization fraction, so the lower ghost cells stay with the sweep. It requires Molecular carrier transport: True and Molecular chemistry: True; each is a fatal refusal at startup. With Solver: Newton both stationary routes carry it: with Coupled carrier solve: True the stationary system carries the proton's row and the sweep is handed the Newton unknown; without it the route is the partitioned alternation, whose hydrodynamic solves hold the composition and whose sweeps are handed the transported proton fraction of that composition, the carrier relaxation between the solves transporting it (until 2026-09-12 this combination was refused, on the premise, true of the plain Newton finish and not of the alternation. Kwang-II Seon / Last updated: 2026-09-30 that the last sweep would put the proton back on its local root). Marching requires neither. The direct steady route (EXHALE_PTC=1) is refused

Table 3, continued

Keyword	Value	Meaning
Lower atmosphere:	none analytic vulcan [, real (R_f)] (def. none)	Lower-atmosphere pre-step run before the wind: generate the base composition over the 1-bar to 1 μ bar column, analytically (src/utls/run_lower.py) or by calling VULCAN (src/utls/vulcan_driver.py, hours on the first run, cached after), and hand it over through base.inp (Tier 3). The second token is the 1-bar (transit) radius and is required. An existing base.inp is used as is and nothing is regenerated; the generator needs EXHALE_ROOT set to the install directory (§3.7).
Lower column:	real (R_f)	Tier-1 consistency <i>report</i> only: integrate the isothermal- T_{eq} hypsometric column with the chemical-equilibrium $H_2/H/He$ partition from the given 1-bar radius up to 1 μ bar and print the derived base radius, base fractions and μ next to the input Planet radius. Nothing is overridden (§3.7).
Lower atmosphere profile:	path (def. none)	Take the lower atmosphere from a <i>profile</i> (the lower model's own solution tabulated over an interval of pressure) instead of the single-level scalars of base.inp. The named file supplies the base state at its matching pressure (T_0 , R_0 , p_{base} , the H_2 fraction), the elemental reservoirs (He/H and every element column it carries, overriding metals.inp), and the eddy coefficient $K_{zz}(r)$ as a profile, which makes He_Kzz inert. The profile also fixes the base <i>level</i> : n_0 follows from p_match_bar exactly as it does from p_base of base.inp, and a Log10 lower boundary number density line given beside a profile is refused unless the two agree to 1%. A missing file, a malformed table, or a base.inp beside it that still states a physics key stops the run (§3.8).

Table 3, continued

Keyword	Value	Meaning
Base BC:	density pressure [, real (μ bar)] (def. density)	What the lower boundary anchors. density (legacy) fixes the base number density at the n_0 of the core block. pressure (CETIMB-style) fixes the base pressure at the optional second token (default 1 μ bar) and <i>derives</i> n_0 from $n_0 k_B T_0 n_{\text{tot},bc} = p_{\text{base}}$; the reservoir of the characteristic boundary (§2.6) then holds that pressure and T_0 at the level, the lower face of cell 1, and ρ follows. It is a re-parameterization of the same reservoir, whose physical effect is a much less dense base.
Shapiro filter:	real ε [, int n] (def. -1, i.e. off; $n = 4$)	Apply a weak Shapiro (1970) 1–2–1 low-pass filter $u_j += (\varepsilon/4)(u_{j-1} - 2u_j + u_{j+1})$ to the conservative variables every n steps, damping the gravity-unbalanced sound-wave (base-breathing) mode the same way CETIMB does. $\varepsilon \leq 0 = \text{off}$, and off is the default on purpose: with the filter on, an HD209458b cold-IC sweep left the transonic-wind saddle for an infall state ($v < 0$ everywhere) regardless of the velocity BC, while filter-off relaxed to a clean outflow. Enable it only for a case that actually breathes. Unlike Low-Mach damping it touches only the marching state, not the numerical flux, so the JFNK residual does not see it.
Viscosity:	True/False, or real μ_0 [, real s] (def. False)	True adds the radial viscous force $(\nabla \cdot \tau)_r$ and its dissipation $q_\mu = \frac{4}{3}\mu(\partial_r v - v/r)^2$ with the calibrated $\mu(T) = \frac{4}{15}(m_H/k_B)\kappa(T)$ (monatomic Chapman–Enskog at Prandtl 2/3, tied to the Watson et al. 1981 conductivity): the molecular transport that damps the near-base momentum imbalance the inviscid HLLC scheme lacks. A numeric value is instead a diagnostic power law $\mu = \mu_0 T^s$ in code units ($s = 0.7$ by default), kept for sensitivity scans. Off = byte-identical to the inviscid code.

Table 3, continued

Keyword	Value	Meaning
Conduction:	logical (def. False)	Add heat conduction $r^{-2}\partial_r(r^2\kappa\partial_rT)$ with the conductivity of the mixture $\kappa = [n_e 1.2 \times 10^{-6} T^{5/2} + n_{\text{HI}} 379 T^{0.69} + n_{\text{HeI}} 299 T^{0.69} + n_{\text{H}_2} k_{\text{H}_2}(T)]/n \text{ erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-1}$ (Banks & Kockarts 1973 as Salz et al. 2015 and Sutton et al. 2015 state it; k_{H_2} an Eucken-form fit to Incropera et al. 2007 Table A.4, valid 200–2000 K; n all particles). Off = byte-identical.
Caloric EOS:	ladder monatomic (def. ladder)	The caloric equation of state. ladder: every atom, ion and electron stores $\frac{3}{2}kT$ and H_2 its rovibrational energy (the Roueff et al. 2019 ladder), so γ_{eff} depends on composition and T ; in an 80% H_2 layer that is not 5/3. monatomic: H_2 too stores $\frac{3}{2}kT$ and $\gamma = 5/3$ everywhere, kept as a <i>comparison</i> option for a published model whose energy equation is $u = c_v T$ with a monatomic c_v . In an atomic gas the two agree exactly. Any other word stops the run.
Photoelectron heating:	excess full real $f \in (0, 1]$ (def. excess)	What a photoionization deposits as heat. excess (physical): the electron energy, $h\nu - I$ for an outer-shell absorption and, for an inner-shell absorption by a metal ion, the photoelectron's $h\nu - E_{\text{th},s}$ plus the Auger electrons of the vacancy (Kaastra & Mewe 1993; §8.). full: the whole photon energy $h\nu$; a number: the fraction f of $h\nu$ (both keep the outer-shell form for the metals). The last two are <i>comparison</i> options. Koskinen et al. (2022) Figure 9 shows a stellar heating rate 2.5–3 times what $h\nu - I$ gives at their ionization rate and spectrum, i.e. between the two accountings, and a fraction reproduces their profile without claiming a physics; a run that uses one must say so. Anything else stops the run.

Table 3, continued

Keyword	Value	Meaning
Resid norm:	retired	Selected between a volume-weighted and a max-over-cells combination of the scaled residual rows. The acceptance norm is now the cellwise maximum and the volume-weighted number is reported beside it every time, so there is nothing left to select. The label is still matched and a file carrying it draws a warning saying so, rather than failing the unknown-key scan.
Wind-AE seed:	path, or grid auto	Which converged Wind-AE solution IC mode: windae ramps from; the default is the shipped inputdata/windae_seed.csv. grid/auto instead picks the nearest solution in inputdata/windae_grid/ to this planet's (M_p, R_p, F_{XUV}), which is a shorter and more robust ramp than always starting from the single fiducial seed (§2.10).
Wind-AE seed out:	path (def. none)	Write the converged Wind-AE relaxation solution to this file as a reusable seed, e.g. to grow inputdata/windae_grid/. Omitted = nothing is written.

3.2 metals.inp: trace-metal abundances

metals.inp is *optional*: if absent, the run is metals-off (H/He only). If present, it lists the total elemental abundance n_X/n_H (summed over ionization stages; the equilibrium solver distributes each element over its stages) for any of C, N, O, Mg, Si, Ca, Na, K, S, Fe. Blank lines and # comments are ignored; any non-zero abundance turns metals on. Each quantity may be stated once; a second line for it stops the run. Six control keys are also accepted: pp_metals (metal treatment in the advection post-process: 0=metal-free, 1=frozen at the converged equilibrium (default), 2=re-solve the metal stages inside the post-process iteration, at fixed element totals), cx_full (1 adds groups C (metal+He/He⁺) and D (metal+metal) of the Huang et al. 2023 Table 4 reaction list to the metal+H/H⁺ group A that is always active; it does not gate the He $\leftrightarrow$ H reactions, which have their own input.inp key He_H_charge_exchange, nor the two group E electron captures below), cno_cool (C/N/O line-cooling source: 1=closed-form fits to CHIANTI v11, including NI/NII cooling (default); 0=legacy AIOLOS analytic fits, which have no N cooling and deviate by 40–70% from CHIANTI for O I/O II in the wind-launch region, see cooling_data/cno_cooling_comparison.ipynb), and eos_metals (bulk-gas metal policy: 1=metals contribute their mass to the gas density and base density, their

electrons to the free-electron density and the base pressure correction, and their nuclei to the total particle density, a $\sim 1\%$ mass effect at solar abundances, consistent with the charge balance inside the ionization solver (default); θ =legacy trace approximation, H/He-only gas budget), and the two group E scales `cx_02p_H` and `cx_N2p_H` (default 1 each = the published rate). Huang+2023 Table 4 has no doubly ionized oxygen or nitrogen row, so re-sourcing the charge-exchange network to that paper dropped $O^{2+} + H^0 \rightarrow O^+ + H^+$ and $N^{2+} + H^0 \rightarrow N^+ + H^+$. EXHALE carries both separately at the rates of one calculation, Barragán et al. (2006, ApJ 636, 544; their Table 1 k_1 for oxygen, the calculation CHIANTI v11 and Cloudy both adopt, and their Table 2 k_4 for nitrogen, interpolated in log-log and held at the ends of the calculated 10^2 – 10^5 K). `cx_02p_H<scale>` and `cx_N2p_H<scale>` rescale them, with θ leaving the row out exactly and other values serving as bounding experiments. They are deliberately *not* controlled by `cx_full`, which means “all of Table 4”. Measured on HD 209458 b, carrying the oxygen reaction suppresses the O III fraction by 4.7 decades at $1.05 R_p$, 3.7 at 1.1 and 2.7 at 1.2, and raises O I by 3% below $1.2 R_p$ and 52% near 2 – $3 R_p$, while $\dot{M}$ moves by 0.01 dex (§103 and §107 of Update_EXHALE_stage1; measured 2026-08-30, before the charge-exchange corrections of 2026-09-24 to 09-26). The nitrogen capture ($9.70 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ at 10^4 K) beats N III recombination wherever $n(\text{HI})/n_e$ exceeds a few 10^{-3} ; MEASURED on the wasp_full regression case (2026-09-26, md/charge_exchange_detailed_balance_20260924.md section 12.6), N III/N falls from 0.444 to 0.0029 at $1.3 R_p$ and $\log_{10} \dot{M}$ moves by -3×10^{-5} dex. Neither reverse is carried: each is endothermic by more than 16 eV (21.5 eV for oxygen, 16.0 eV for nitrogen), and the nitrogen product $N^+(2s2p^3 D^o)$ decays by the 1084 Å multiplet, which leaves the cell, so that event heats the gas by 4.57 eV of its 16.00 eV defect. Solar reference values (Asplund+2009):

```
# metals.inp -- solar n_X/n_H (Asplund+2009)
C 2.69e-4 N 6.76e-5 O 4.90e-4
Mg 3.98e-5 Si 3.24e-5 Ca 2.19e-6
Na 1.74e-6 K 1.07e-7 S 1.32e-5 Fe 3.16e-5
pp_metals 1
```

3.3 opacity.inp: opacity model (optional)

`opacity.inp` selects the bound-free opacity model via `OPACITY_MODEL = A|C|P|T`: **A** analytic (default, the standard ATES cross sections); **C** constant (threshold σ times the factors for each species `OPA_CONST_HI/HEI/HEII/HEIIR`); **P** pressure-broadened (Robinson & Catling $f(p) = 1 + a(p/p_0)^n$ via `OPA_PB_FACTOR=a`, `OPA_PB_EXPONENT=n`, `OPA_PB_PIVOT=p_0`); **T** tabulated (two-column .opa files per species, energy [eV] vs. σ [10^{-18} cm^2], given by `OPA_FILE_HI` etc.). Missing keys keep their defaults, so an absent `opacity.inp` is equivalent to model A.

3.4 Excited hydrogen and the Ly α pumping field

The non-LTE $H(n=2)$ model (the metastable $2s$ + resonance $2p$ levels in the level balance of Christie et al. 2013, with the rate coefficients listed in Table 7: $A(2s \rightarrow 1s)$ of Drake 1986, the CHIANTI $1s \rightarrow 2s, 2p$ collision rates, the $2s$ – $2p$ mixing by electrons of Seaton 1955 and by H^+ , He^+ and He^{2+} of Pengelly & Seaton 1964, and the case-B coefficient of the balance split with the $2s$ fraction of Pengelly 1964) feeds two things: in EXHALE.x the $H(n=2)$ photoionization adds a Balmer proton source and photoelectric heating to the energy/ionization balance; in EXHALE_transit.py the

$n=2$ population sets the $H\alpha/H\beta$ transmission. The model is *off by default*; it switches on only when both `Stellar Teff` and `Stellar radius` are given, so the standard run is unchanged. Those two keys *arm* the coupling; the Balmer continuum that ionizes and heats $n=2$ is then the run's own spectrum over $[e_{\text{th,HI}}/4, 13.598]$ eV (the power law for a Power-law run, the photospheric blackbody for Planck, the loaded table for Load), integrated against the hydrogenic $\sigma_2(E) = \sigma_2(e_{\text{th,HI}}/4)(e_{\text{th,HI}}/4E)^3$, and not a blackbody of its own (§3.5). Arming the coupling also lowers the photon-grid floor to 3.400 eV, so a Loaded table that stops above 3647 Å stops the run.

The $2s/2p$ populations are driven by the **Ly α mean intensity** $\bar{J}_{\text{Ly}\alpha}(r)$, which pumps $1s \leftrightarrow 2p$. How $\bar{J}_{\text{Ly}\alpha}$ is obtained is selected by three mutually-exclusive options:

(mode 0) Parameterized: the default with no external input. If neither `Jlya RT file` nor `Jlya escape-prob` is present, $\bar{J}_{\text{Ly}\alpha}$ is the Huang et al. (2017, Eq. 6) estimate computed internally from the run's own state,

$$\bar{J}_{\text{Ly}\alpha}(r) = \frac{0.1 F_{\text{LyC}}}{\Delta\nu_D(r)} \frac{1}{1 + \tau_{\text{Ly}\alpha}(r)},$$

where $\Delta\nu_D = \nu_0 \sqrt{2k_B T / \mu} / c$ is the local Doppler width, $\tau_{\text{Ly}\alpha}(r)$ is the *top-down* line-center Ly α optical depth integrated from the outside in through the HI column, and

$$F_{\text{LyC}} = \xi \frac{10^{L_{\text{EUV}}}}{4\pi a^2} \left(1 - e^{-\sigma_{\text{LyC}} N_{\text{HI}}}\right)$$

is the deposited Lyman-continuum flux (the incident EUV flux times the dayside factor ξ from 2D approximate method and the absorbed fraction set by the neutral-H column N_{HI}). This needs *no* external file: it is a self-contained, approximate “staging” field, adequate to get the coupling and signs right but not a true radiative-transfer solution.

(mode 1) External RT file: `Jlya RT file`: <path>. Read a tabulated $\bar{J}_{\text{Ly}\alpha}(r)$ profile (two columns, r/R_p and $\bar{J}$ in cgs) produced by a real Ly α radiative-transfer calculation (e.g. a Monte Carlo run); it is interpolated onto the grid and used directly.

(mode 2) In-line escape probability: `Jlya escape-prob`: True. Compute $\bar{J}_{\text{Ly}\alpha}$ every step from the current state by escape-probability radiative transfer: a trapped *internal* field (recombination + collisional sources closed by the escape probability β) plus a trapped *stellar beam* set by `Stellar Lya flux` (with the penetration controlled by `Lya stellar halfwidth` and the buildup capped by `Lya stellar boost`). $\beta = 1/(1 + \langle N \rangle)$ with $\langle N \rangle$ the mean number of scatterings before escape of the static plane-parallel damping-wing slab (Neufeld 1990, ApJ 350, 216, eq. 3.27 at zero continuum destruction; for a mid-plane source, Harrington 1973, MNRAS 162, 43, eq. 40, $\langle N \rangle = 0.909316\tau$), so β goes as $1/\tau$ in the wings and tends to one when the column thins. By default the lower boundary reflects, which makes the emitting plane a symmetry plane; `Lya absorbing bottom`: True instead puts the planet-ward face of that slab at the bottom of the domain, where the H₂ layer beneath the base absorbs Ly α (a pure absorber, as in the Huang et al. 2017 Monte Carlo), which shortens the random walk and lowers $\bar{J}_{\text{Ly}\alpha}$ in the bottom few scale heights only. This is the most physical of the three but still an approximation; see Update_EXHALE_stage1 §12 for the full formulae.

EXHALE_transit.py rebuilds the $H\alpha/H\beta$ $n=2$ population from the profiles with the solver's level balance and the solver's rate coefficients, each transcribed as its own function of exhale_transit_lib.py (`n2_populations` agrees

with the Fortran to 3.1×10^{-13} relative, MEASURED 2026-09-25, md/stale_texts_20260925.md); it takes the run's rate set from Atomic rate set: and Legacy_HHe_rates: of input.inp. What it does not share with the solver is the field, which the profiles do not carry: if a Jlya_file is set it is read (mode 1), otherwise the mode-0 parameterized estimate above is used (with F_{LyC} rebuilt from the run's H I column and the stellar EUV), and the Balmer continuum that photoionizes $n=2$ is a blackbody at Stellar Teff diluted by $(R_*/a)^2$ rather than the run's own spectrum. So a converged run with no Ly α input still yields H α /H β predictions, via the default field.

3.5 The photon grid and the spectrum that fills it

Neutral metals have ionization thresholds *far below* the hydrogen Lyman limit ($13.6 \text{ eV} \equiv 912 \text{ \AA}$): the lowest is K I at 4.34 eV ($2856 \text{ \AA}$), followed by Na I, Ca I, Mg I, Fe I, Si I, S I, and C I (Table 4). So does the He 2^3S metastable ($e_{\text{th,HeTR}} = 4.768 \text{ eV} = 2600 \text{ \AA}$) and, whenever the excited-hydrogen coupling is armed, hydrogen in $n = 2$ ($e_{\text{th,H}}/4 = 3.400 \text{ eV} = 3647 \text{ \AA}$, the head of the Balmer continuum). To photoionize any of them the photon grid has to reach their thresholds, and the spectrum has to state a flux there: the EUV/X-ray band alone ($\lambda < 912 \text{ \AA}$) leaves every neutral metal un-ionized and artificially abundant.

One spectrum type builds every band. Spectrum type: states the spectrum of the *whole* photon grid, the part below 13.6 eV and the Balmer continuum included. No band is filled from a type the input did not select.

Power-law. The power law E^{PLind} , normalized on $[E_{\text{low}}, E_{\text{mid}}]$ and $[E_{\text{mid}}, E_{\text{high}}]$ from L_{EUV} and L_X , evaluated wherever the grid reaches. Below E_{low} that is an *extrapolation* of an EUV fit, which is what the type means: real stars do not follow it into the near-ultraviolet (their flux rises steeply there). Treat power-law metal and He 2^3S photoionization as indicative.

Planck. The photospheric blackbody $\pi B_\nu(T_{\text{eff}})(R_*/a)^2$, from Stellar Teff [K] and Stellar radius [R_sun], on the whole grid. Startup refuses the type without both. It is an upper bound in the near-ultraviolet, where line blanketing and the Balmer jump depress a real F or G photosphere below the curve, and it is orders of magnitude *below* the chromospheric and coronal EUV of an active star: a run that needs both bands right needs a measured spectrum.

Load(Spectrum file:). The tabulated SED, on the whole grid. The file's own rows are the grid points, log-log interpolated where a band asks for a flux between them; no extrapolation.

Monochromatic. One energy, Photon energy [eV], and no grid at all ($N_\lambda = 1$).

The grid floor is the lowest active threshold, and a loaded table that does not reach it stops the run. The floor (photon_grid_floor_eV) is the lowest ionization threshold over every *active* absorber: the He 2^3S metastable when the triplet is included, the neutral stage of every element metals.inp gives a nonzero abundance, and H($n=2$) when Stellar Teff and Stellar radius arm the excited-hydrogen coupling. With none of them active the floor is the H I edge. The two analytic types are evaluated down to it, and a loaded table is read down to it.

A loaded table that *ends above* the band an active absorber needs stops the run at startup. There is no key to continue: a truncated table gives that absorber the photoionization rate of a truncation of the stated spectrum, not of the spectrum.

The message names the file, the lowest photon in it, the uncovered band in eV and in Å, every absorber whose threshold lies below that lowest photon (H($n=2$), the metastable, each active neutral metal), and the remedies: turn the absorber off (Include He23S? False; remove the element from metals.inp; remove Stellar Teff and Stellar radius, which is what arms the excited-hydrogen coupling), or supply an SED that reaches the stated wavelength.

How the grid is partitioned (analytic types). For Power-law and Planck the code builds the grid itself, from *bin edges*: e_v holds the *geometric center* of each bin and de_v its exact width, so the bins are contiguous and $\sum de_v$ is exactly the width of the covered band. Six bands, each with its own budget of bins, put the resolution where the absorbers are rather than spreading it over the X-ray decades:

Band	Bins	Present when
$[e_{th,H\text{I}}/4, e_{abs,low}]$	20	the excited-H coupling is armed
$[e_{abs,low}, 13.598]$	89	a metastable or low-IP metal is active
$[13.598, 24.587]$	50	always
$[24.587, 54.418]$	50	always
$[54.418, E_{mid}]$	50	always
$[E_{mid}, E_{high}]$	50	always

$e_{abs,low}$ is the lowest threshold of an absorber of the *bulk gas* (the metastable or a low-IP metal), so the first band is the one no active absorber has a cross section in; its 20 bins only have to state it, because the H($n=2$) photoionization rate is an integral over its own band and not a sum over these bins. The 89 bins of the second give it the bin ratio of the pure-H I band ($\Delta \ln E = 0.0118$), which is what the neutral-metal cross sections need: every metal photoionization rate the grid forms is then within 5×10^{-4} of the exact integral.

Every ionization threshold of an active absorber is a bin edge. A photoionization rate is the rectangle-rule sum $\sum F\sigma/E de_v$, so a bin *containing* a threshold would charge that absorber's cross section, read at the bin center, over the part of the bin lying below the threshold, where it cannot absorb at all. Each band is therefore cut at every threshold strictly inside it: H I, He I, He II, the metastable, H₂ and its dissociative- and double-ionization channel thresholds when the molecular chemistry is on, and every photo-ionizable stage of an element with a nonzero abundance, and the band's budget is shared over the resulting sub-intervals in proportion to their logarithmic widths, at least one bin each, so the budget is met exactly. A band with no threshold inside it keeps its budget in one piece.

The total is therefore $N_\lambda = 200 + N_{\lambda,TR}$, where $N_{\lambda,TR}$ counts the bins below the H I edge: 109 for a run with both sub-13.6 eV bands ($N_\lambda = 309$; He 2³S or metals plus an armed excited-H coupling), 89 without the $n = 2$ band ($N_\lambda = 289$), and 0 for a pure H/He run with no absorber below the edge ($N_\lambda = 200$). The startup report states the type, what the band below 13.6 eV was built from, the grid floor, and the integrated grid flux $\sum F \Delta E$ against the nominal $(10^{L_X} + 10^{L_{EUV}})/(4\pi a^2)$.

The energy bands set by $[E_low, E_mid, E_high]$ and the luminosities L_{EUV}/L_X still describe the *ionizing* (H/He-driving) part of the spectrum above 13.6 eV; the bands below it are in addition to them, which is why the integrated grid flux and the nominal flux need not agree.

Table 4: Neutral-metal photoionization thresholds (first ionization stage of each modeled element), as E_{th} and the corresponding wavelength $\lambda_{\text{th}} = hc/E_{\text{th}}$. A loaded SED must extend to at least the longest λ_{th} among the *active* metals, or the run stops (§3.5). The He 2^3S metastable (4.768 eV, 2600 Å) and H($n=2$) (3.400 eV, 3647 Å) floor the grid the same way. Values are the VFKY96/NIST thresholds used in the code (`species_table.f90`).

Species	E_{th} [eV]	λ_{th} [Å]	Note
K I	4.34	2856	lowest, sets the grid floor when K is present
Na I	5.14	2413	
Ca I	6.11	2028	
Mg I	7.65	1622	
Fe I	7.90	1569	
Si I	8.15	1521	
S I	10.36	1197	
C I	11.26	1101	
(H I)	13.60	912	Lyman limit (reference)
O I	13.62	910	$\simeq$ Lyman limit
NI	14.53	853	above the limit (already in the EUV band)

3.6 Diffusive separation of helium and metals

By default EXHALE is single-fluid: the composition is re-solved each step by *local* ionization equilibrium, which conserves the element ratios, so $n_{\text{He}}/n_{\text{H}}$ (and each metal/H) stays frozen at the input value at every radius. Setting `He_diffusion: True` adds a genuine transport equation for the helium *mass fraction* $X = \rho_{\text{He}}/\rho$ of the two-component H/He gas: advection at the bulk velocity plus a molecular-diffusion *settling* drift of He relative to H,

$$\Phi_{\text{He}} = -\rho(D + K_{zz})\frac{\partial X}{\partial r} - \rho X(1-X)DG, \quad G = \frac{\Delta m_{\text{eff}} m_{\text{H}} g}{kT},$$

with eddy diffusion K_{zz} (`He_Kzz`, default 0) acting on the gradient term only. The binary coefficient D is **stage-resolved**: each element is split into its ionization stages and the friction is built pair by pair, hard-sphere for neutral–neutral pairs (Banks & Kockarts 1973), a combined polarization + hard-sphere form for non-resonant ion–neutral pairs, and Coulomb for ion–ion pairs, then mixed over the stage fractions (Blanc’s law, $1/D_{\text{eff}} = \sum_{s,t} y_s y_t / D_{st}$). This matters above the ionization front, where Coulomb friction suppresses the separation by orders of magnitude relative to the neutral hard sphere and keeps the metal mixing ratios flat in the wind. Helium settles out relative to the escaping hydrogen and *He/H declines with altitude*, as in Taylor et al. (2025) and Xing et al. (2023). The equation is solved conservatively (implicit, tridiagonal) and the new split is written back into the composition conserving mass. Physics refinements: `He_ambipolar` (default on) reduces the settling in the ionized wind via the polarization field, computed from the electron-pressure gradient; `He_alphaT` adds thermal diffusion; `He_metal_diffusion` extends the same operator to each trace metal (heavier elements settle more, so they deplete faster than He).

The strength of the separation scales as (settling speed)/(wind speed): it is suppressed by fast escape and is stronger for heavier species. On a gentle escaper (HD 209458b) helium itself separates and the He 10830 line weakens; on a furious escaper (WASP-121b) helium is dragged out almost unfractionated while the heavy metals still settle

strongly. Full physics, derivation, and a two-planet before/after comparison: docs/version_compare.{tex,pdf} and md/version_compare.md; the stage-resolved coefficient is derived and measured in Update_EXHALE_stage1 §68 and §71. The flags are **default off**, so standard runs are byte-identical to before.

The enthalpy the moving elements carry. Particles that move relative to the mass-averaged velocity carry their enthalpy with them, so the energy equation of a diffusing mixture has the divergence of the interdiffusion enthalpy flux

$$q_d = \sum_s h_s J_s \quad (\text{Cook 2009, Phys. Fluids 21, 055109, eqs. 11–13}),$$

with J_s the diffusive mass flux of component s and h_s its specific enthalpy (the enthalpy and not the internal energy, because J_s is taken relative to the mass-weighted velocity). In the binary model every species moves with its element and the electrons with their ions, $J_1 = -J_{\text{He}}$, so $q_d = J_{\text{He}}(h_{\text{He}} - h_1)$, plus $\sum_X J_X(h_X - h_{\text{H}})$ for each metal with `He_metal_diffusion`. The h_s are the *sensible* enthalpies of the caloric equation of state, $\gamma/(\gamma - 1)kT$ a particle with the electrons an ion carries, plus the rovibrational energy of H_2 ; formation energy is not in them because it already rides with the moved nuclei through the formation-energy reference of the energy update (`binary_element_diffusion.f90`, `component_specific_enthalpies`). J is the operator’s whole diffusive flux, eddy term included. Its divergence is taken on the faces and volumes of the element rows, so the energy it moves telescopes over the column, and it enters both routes: the marching update subtracts $\Delta t \nabla \cdot q_d$ from the thermal energy after an accepted element step, and the stationary energy row adds $\nabla \cdot q_d$ of the state’s own element fluxes. Interdiffusion enthalpy flux: (Table 3) is **on by default whenever He_diffusion is on**, because the term belongs to the equation; the key exists to turn it off for comparison with published models that omit it. MEASURED on the molecular LHS 1140 b He/H = 1.8 state after 24 outer passes with the term (not yet certified, carrier row 1.6×10^{-5}): T lower by a median 166 K over $1.3\text{--}2 R_p$ (at most 236 K, -4.4%), He/H -3.1% , the He 2^3S column -1.3% , $\log_{10} \dot{M} + 0.002$ (md/Update_EXHALE_stage3.md section 29.3).

3.7 Lower-atmosphere connection (Tier 1–3)

Three opt-in layers connect EXHALE’s $1\text{-}\mu\text{bar}$ base to the atmosphere below (survey, implementation, examples and figures: docs/lower_atmosphere_coupling.pdf). All are **default off**.

Tier 1: analytic lower column (consistency report). Add

Lower column: 1.36 # radius of the 1-bar level [R_J], e.g. transit radius

and EXHALE integrates the isothermal- T_{eq} hypsometric column with the chemical-equilibrium $\text{H}_2/\text{H}/\text{He}$ partition (Koskinen et al. 2022) from 1 bar to $1\text{ }\mu\text{bar}$ and *reports* the derived base radius (equilibrium value + a fully-atomic bracket), the base $\text{H}_2/\text{H}/\text{He}$ fractions and μ , next to the input “Planet radius”. Nothing is overridden. A large base q_{H_2} on a genuinely cool planet signals that the molecular physics below is required.

Tier 2a: molecular base (EOS only). Molecular base: True removes the H_2 -bound particles from the base particle budget (ntot_bc) using the same equilibrium fit: a lower base pressure and heavier base μ , with the species arrays still

atomic. Crude on its own, and it is what Tier 2 turns on for itself, so the key is worth setting by hand only for a run that wants the base EOS without the chemistry.

Tier 2: molecular chemistry.

Molecular chemistry: True # H₂, H₂⁺, H₃⁺, HeH⁺ in the coupled equilibrium

adds the four molecular species to the coupled ionization equilibrium (the Koskinen et al. 2022 reaction network), the H₂ photoionization opacity and heating (Yan et al. 1998 cross section), and the optically-thin H₃⁺ infrared cooling (Miller et al. 2013 with their non-LTE factor). The output `Ion_species*.txt` gains four columns (H₂ H₂⁺ H₃⁺ HeH⁺). It also turns `Molecular` base on automatically, so that key need not be given: with an atomic `ntot_bc` under a molecular base the base temperature is inflated by the particle-count ratio ($\sim 1.8\times$ for a fully molecular base) and thermally dissociates the layer the chemistry just built, so the parser couples the two rather than leaving the inconsistent combination reachable. Requires He. `He_diffusion` is no longer refused beside it (milestone M4, 2026-08-26): the element transport of `binary_element_diffusion` closes over the molecular carriers, and the `mol_diffusion` regression case runs the combination. A `metals.inp` may be present: the trace metals are then solved in the same system as the molecular network (`System_HeH_mol_metals`), because the two blocks share the free electron density and the metals are the dominant electron donors in the shielded molecular base. H₂ photodissociation in the Lyman–Werner bands is available and opt-in: `add Stellar LW flux [erg/cm2/s]:` with the band flux at the planet. Caveats: without `Molecular carrier transport: True` (its default is on only with Oxygen chemistry) the molecular species are closed by their local equilibrium, with no advection; the caloric equation of state is the H₂ ladder of `Caloric EOS` (default), not $\gamma = 5/3$. On hot Jupiters the result is a thin molecular base under a sharp H₂ $\rightarrow$ H front with an atomic wind above: the atomic assumption becomes a result rather than an input.

Converging a molecular run takes three keys an atomic run does not need: `Solver: Newton 5.0e-2, Resid tol: 1.0e-7` and `Max steps: 150000`. The *du* descent of a molecular run is not monotonic, so the default 10^{-2} hand-off threshold can consume the whole step budget; and $\|R\|$ is set by the two or three cells just above the base, so a run stopped at the default `Resid tol: 1.0e-5` leaves the molecular layer still cooling (its energy residual there is still $0.7\times$ the local H₃⁺ cooling rate). Marching alone never reaches that layer (its H₃⁺ cooling time is $\sim 10^7$ CFL steps), so the Newton finish is not optional. The converged layer is much colder than the relaxation snapshot (~ 400 K on HD 209458 b, 190–300 K on a hot Uranus) and, without `Base IR field`, that collapse is the model's own steady state rather than a prediction; see `docs/lower_atmosphere_coupling.pdf` §7–§9 and open item (G).

Tier 2b: oxygen chemistry.

Oxygen chemistry: True # OH, H₂O, CO in the same coupled system

Molecular carrier transport: True # the default: carriers transported

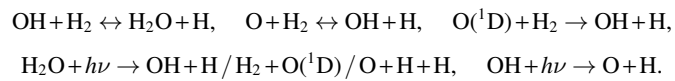
Stellar LW flux [erg/cm2/s]: 1248.5 # 912–1201 Å at the planet (LW)

Stellar Lyα flux [erg/cm2/s]: 14957.2 # 1215.67 Å, the Lyα line (B2)

Stellar FUV B3 flux [erg/cm2/s]: 1374.1 # 1231–1450 Å (B3)

Stellar FUV B4 flux [erg/cm2/s]: 46294.8 # 1451–2304 Å (B4)

With `Spectrum` type: Load the three continuum lines (LW, B3, B4) may be omitted: each band is then integrated from the spectrum file over its own interval, a trapezoid over the file's rows (the file is already at the planet), and the setup report names the source of every band. Only the $\text{Ly}\alpha$ line is always a key, because a line flux reconstructed from observations is a better number than a trapezoid over the file's rows across 1202–1230 Å. A stated value, zero included, always wins over the file. Tier 2 can build a molecular base but not *decide* how molecular it is: the H_2/H partition is either imposed through the `q_H2_base` handoff or taken from a chemical-equilibrium fit, neither of which is photochemistry. Tier 2b is the layer that computes it, by adding the oxygen carriers that the catalytic cycle runs on. Measured on a photochemical model of the HD 189733 b base, the oxygen family carries 96–99.6% of the net H_2 destruction there, and the whole of it sits in three reactions and four photolysis channels:



Every coefficient is transcribed from a named evaluation with its validity range at the code site, and the reverse of each two-body pair is generated by detailed balance against a NIST-JANAF Shomate table rather than transcribed independently: the pairs run close to cancellation on a hot base, where an inconsistent pair gives an arbitrary net rather than a small error. The audit behind the numbers is a separate audit.

$\text{O}(^1\text{D})$ is closed by a local steady state, not carried as a transported species: its only sink in this set is $\text{O}(^1\text{D}) + \text{H}_2 \rightarrow \text{OH} + \text{H}$, and that gives it a chemical lifetime of 1.6×10^{-3} s at the HD 189733 b base (shorter than every other time scale in the problem by many decades), so the steady state is the accurate treatment. Its density is reported as a diagnostic column.

CO is an oxygen reservoir, not a reactant. It locks 45–46% of the oxygen at every measured level, so a network that gave the whole oxygen abundance to the water family would over-supply the OH cycle by about a factor two. It has no chemical source in this network: with `Molecular carrier transport`: False it sits at its own $\text{CO} \leftrightarrow \text{C} + \text{O}$ chemical equilibrium, which keeps it associated below ~ 3000 K and dissociates it by ~ 4000 K, above the $\text{H}_2 \rightarrow \text{H}$ front and below the wind, so the carbon and the oxygen of the wind stay atomic. With the transport on it is a transported reservoir, and “inert” would then mean *indestructible*: measured before the ceiling below was added, the wind carried CO out to $1.67 R_p$ and 2×10^4 K, where it held 55% of the oxygen and switched off the O I and C II cooling of the whole wind. The transported CO is therefore capped at the chemical equilibrium of its own (n, T) : a one-sided constraint, inactive through the molecular layer and exact far above the turnover, which over-suppresses *quenched* CO in the narrow interval around 3000–4000 K where the dissociation is neither fast nor negligible. **Neither limit is kinetics** (the truth is quenching, and the audited set has no CO rate to give it), so a run whose *outer* wind is cool can hand the transit tool an O I column short by whatever fraction is still molecular. Every run therefore prints the molecular oxygen fraction at the top of its domain, and the number of cells the ceiling touched.

The FUV arrives as four bands with named fluxes, not as part of the photon energy grid: that grid is floored by the ionization thresholds of the run's active absorbers (§3.5) and carries their opacity, not the H_2 line system or the H_2O and OH continua, and a power-law type below 13.6 eV gives only an XUV fit extrapolated downward rather than a stellar FUV spectrum. The edges are fixed by the H_2O branching ratios, not chosen: the three-body channel $\text{H}_2\text{O} + h\nu \rightarrow \text{O} + \text{H} + \text{H}$ opens only in the interval containing $\text{Ly}\alpha$, where it is 12%. H_2O and OH absorb these bands as

continua, so the attenuation is $\exp(-\tau)$ on their star-ward columns and no self-shielding function is needed (unlike the H₂ Lyman–Werner band).

One interval, one flux, one beam. Over 912–1201 Å the same photons are absorbed by H₂ in the Lyman and Werner lines and by H₂O and OH in a continuum. That interval is therefore band LW, entered once through `Stellar LW flux`: supplying it twice would count its energy twice on the way in, and letting each absorber attenuate a private copy of the beam would count it twice on the way down. Both are closed. The H₂ rate is multiplied by $\exp(-\tau_{\text{cont}})$, the H₂O + OH continuum term of Draine & Bertoldi (1996) eq. (40) (identically 1 for a run without this option), and the H₂O and OH rates in that band are multiplied by $1 - A$, the fraction of the band the H₂ lines have *not* removed. A is the column integral of the *pump* cross section $\sigma_{\text{diss}}/p_{\text{eff}}$ of the same level-resolved table the H₂ rate comes from, so the photons the beam loses to the lines and the photons the dissociation rate spends are one number: p_{eff} is by definition the dissociations per photon removed from the beam. A cannot exceed 1, and measured over the whole table it reaches 0.69–1.00 at the top of its column axis. Until 2026-09-06 A was DB96 eq. (39), an equivalent width of the narrower 912–1110 Å band, and the two normalizations disagreed by 45 per cent. The column above a cell is not one slab: A is summed cell by cell along the path, each cell above contributing the table integral across its own column interval at its own (T, n_{H}) , the same cross section its own H₂ rate is formed from. The split is exact cell by cell, so what the three absorbers take is what the beam loses, and `output/FUV_bands.txt` reports that ledger. A run with the oxygen chemistry on, no spectrum file and `Stellar LW flux` left at 0 is warned that it gets no H₂O or OH photolysis over 912–1201 Å, where those cross sections peak; with a spectrum file that band, and B3 and B4, are integrated from it unless stated.

Three limitations to read before quoting a result. (i) The band average assumes a flat F_{λ} inside each band. Measured against a real stellar spectrum (the HD 189733 b flux at the planet), that is good to 3% for H₂O and to 18–22% for OH on both halves of LW, and to about 35% on B3, but **off by a factor 4.6–6.1 on B4**, whose cross section falls three decades across the band; a run in which B4 matters is outside the treatment. Inside LW the sensitivity is not uniform: nearly all of it sits shortward of 1110 Å, where H₂O and OH move in *opposite* directions (–60% and +89% for a 3000 K blackbody), while the 1110–1201 Å half is the best-behaved interval of the set for H₂O ($\leq 2\%$). One band average cannot show that; the two half-band numbers of `water_photolysis.f90` can. (ii) The Ly α band uses the *incident* stellar flux, attenuated only by the H₂O and OH continuum: the H I resonance scattering that actually decides how much Ly α reaches a molecular base is not applied, so that band’s rate is an upper bound. Note also that setting `Stellar Lya flux` for the H($n=2$) pumping therefore drives the photolysis as well; the startup report prints all four band fluxes so this is visible. (iii) The molecular ions H₂⁺, H₃⁺ and HeH⁺ and O(¹D) are closed by their local balance even with the transport on, and the band average applies one transmission to a whole band, so the rates are lower bounds deep in the water layer. Every run writes its own chemical, advective and diffusive time scales (`output/Oxygen_chemistry.txt`), which say where the answer is chemistry, where it is flow, and where the two are comparable.

The carriers are transported by default (`Molecular carrier transport: True`). H₂, OH, H₂O and CO are advanced by an implicit backward-Euler diffusion-advection step solved together with the same chemistry rows the local equilibrium uses, so a rate that changes in one changes in the other. The molecular coefficients come from Blanc’s law over the background carriers with the rigid-sphere pair coefficients of Banks & Kockarts (1973) that the element-diffusion operator already carries; the eddy coefficient is read from `He_Kzz` or from a lower-atmosphere profile

and from nowhere else; settling uses the same g as the hydrodynamics, with no thermal diffusion. Each cell's *element* totals are unchanged by the step (only the partition among an element's carriers moves), so element conservation holds by construction and ρ is untouched; every positivity or element limiter that fires is counted and reported in the trailer of output/Oxygen_chemistry.txt. This operator therefore transports *speciation* and not elements, which is why it cannot be run together with He_metal_diffusion: that option transports elements, and CO carries an oxygen and a carbon nucleus and so cannot follow two element factors at once. Molecular carrier transport: False restores the earlier local steady state; it isolates the chemistry for testing and is not a model of a base, because at the HD 189733 b 1 μ bar level the H₂ chemical and flow times are comparable. With $K_{zz} = 0$ and no profile the transport is pure molecular diffusion (the wrong limit for a lower atmosphere, where eddy mixing is what holds the composition well mixed below the homopause), and the run says so at startup rather than refusing.

The option removes coolants it does not replace. Where the oxygen is in H₂O and CO the [O I] 63/145/44 μ m fine-structure lines and the C I/C II line cooling correctly stop (those atoms are not there), but EXHALE carries no H₂O or CO infrared bands to put in their place, and H₃⁺ is its only molecular coolant. Measured on the hot-Uranus molecular gate, the layer just above the base cools 15 $\times$ more slowly with the option on than with it off, at comparable heating. That is open item (G) arriving through the composition instead of through the temperature, and it is why the resulting molecular-layer temperature is not a prediction. Every oxygen-chemistry run reports how much of its oxygen and carbon is molecular, where, and that those nuclei no longer cool.

The option is **default off** and refuses combinations that would give one quantity two owners: q_H2_base (it computes that partition), and He_metal_diffusion (the trace-metal transport counts each metal element over its ion stages alone, so it would transport an oxygen reservoir missing everything bound into OH, H₂O and CO (about half the element), and moving the carriers with it needs a rule for CO, which carries an oxygen *and* a carbon nucleus). He_diffusion: True on its own is accepted: the H/He element operator scales every H-bearing species by the same factor, so the molecular carriers already move with hydrogen there. A lower-atmosphere *profile* is accepted beside it: the profile owns the region below the matching level and the chemistry the region above it.

Tier 3: base.inp handoff (and VULCAN). An optional base.inp in the run directory overrides the base state before the derived constants are formed (each override is echoed; absent file = no-op):

T_base	1140.0	# [K]	-> overrides T0
r_base	1.402	# [R_J]	-> overrides "Planet radius" (1-ubar radius)
p_base	1.0e-6	# [bar]	-> the level every other value refers to
q_H2_base	0.42	#	-> H2 volume mixing ratio at the base
HeH_base	0.083	#	-> overrides the He/H number ratio
O_H_base	4.90e-4	#	-> O/H nuclei ratio, overrides metals.inp
Kzz_base	1.0e9	# [cm2/s]	-> sets He_Kzz (element + carrier transport)

Keys are matched as labels, by the same anchored rule the input.inp reader uses, and every key carries a **category** that states what the value is allowed to do to the wind. There are five, and the same table is carried by read_base.inp's header comment and by md/input_schema.md §2c:

Key	Category	What it sets, and where the value is consumed
(comments only)	provenance	Nothing. Which code, network and profile produced the file. The one comment that is parsed is <code># solution_id</code> , and only beside a profile.
T_base, r_base, p_base, q_H2_base	EOS boundary	T0, R0, p_base_bar, q_h2_base: the state of the gas at the matching level. p_base also <i>fixes</i> that level: $n_0 = p_{\text{base}} / (k_B T_0 n_{\text{tot, bc}})$, so Log10 lower boundary number density is redundant beside it and is refused if it disagrees by more than 1%. q_H2_base replaces the chemical-equilibrium fit in <code>comp_ntot_bc</code> , i.e. it sets the base <i>particle count</i> , and it is also imposed on the H ₂ partition of the inflowing ghost species, the particle count then coming from that same species state. It is refused above the attainable ceiling $0.5 / (0.5 + \text{He}/\text{H})$ rather than capped.
HeH_base, <El>_H_base	elemental reservoir	HeH, and X_<El> hence <code>melem_ab</code> : the nuclei ratio of an element to hydrogen, which the wind then transports.
(none today)	initial guess	A key here would seed a profile the solver is free to move away from. None exists yet.
Kzz_base	boundary constraint	<code>he_kzz</code> : the eddy coefficient the element-diffusion operator imposes at the base; inert with <code>He_diffusion</code> off.
q_H2O, q_CO, ...	diagnostic	Nothing; metadata written as comments by the generators. Unknown keys are ignored.

The distinction the categories draw is a real one, and q_H2_base is where it bites: it is an **EOS anchor**, not a composition pin. Nothing holds H₂ at that value, no H₂ profile is seeded from it, and the molecular network moves away from it in the first cell.

<El>_H_base is the elemental form: C_H_base, N_H_base, O_H_base, S_H_base and the same spelling for Mg Si Ca Na K Fe (any of the ten element symbols of the species table), each the El/H *nuclei* ratio at the handoff level. It overrides `metals.inp` for that element, it turns the metal system on by itself when it is the only nonzero abundance, and on a restart (`Load IC? True`) it renormalizes that element's whole loaded column onto the stated ratio, preserving the ionization split and the radial shape. An element that reached the run from `metals.inp` alone is loaded untouched. With a lower-atmosphere profile in use (§3.8) the EOS boundary, elemental reservoir and boundary constraint keys of this file are **refused** with an error `stop` naming the key and its category: the profile states all of them at the matching level, and a scalar beside it would be the same quantity said twice. Provenance comments and diagnostic keys remain allowed.

The pre-step can be invoked *by EXHALE itself*: with

```
Lower atmosphere: vulcan 1.36      # or: analytic 1.36  (arg = 1-bar radius R_J)
```

in `input.inp`, the code runs the VULCAN photochemistry (or the analytic column) at startup when no `base.inp` exists, then proceeds with the wind solve on the generated base, subroutine-style, with the first VULCAN run taking hours

and later runs reusing the cached result in `<run_dir>/vulcan_work/`. Omitting the key disables the pre-step entirely. The VULCAN/ tree (obtained by `setup_vulcan.sh`; it ships FastChem inside it) carries its own citation requirements, see §3.9. Two generators are provided in `src/utls/`:

- `run_lower.py <run_dir> --r1bar <R_J> [--guillot]`: the analytic column (isothermal T_{eq} , or the Guillot 2010 semi-grey $T(p)$); pure Python, no dependencies beyond numpy.
- `vulcan_to_base.py <out.vul> <run_dir> --mp <M_J> --r1bar <R_J>`: converts a **VULCAN** photochemical-kinetics output into the same `base.inp`, with the *photochemical* H_2/H dissociation state (on HD 189733 b photochemistry dissociates $\sim 11\times$ more H at 1 μbar than chemical equilibrium) and a hypsometric base radius using VULCAN's own $\mu(p)$ and $T(p)$.

Running VULCAN is handled automatically by the driver `src/utls/vulcan_driver.py`, so no manual VULCAN setup is required. When the pre-step is requested it (i) obtains VULCAN if needed (via `src/utls/setup_vulcan.sh`; see §3.9) and copies the EXHALE/VULCAN/ tree into `<run_dir>/vulcan_work/` (once, then reused), (ii) builds a Guillot (2010) semi-grey $T(p)/K_{\text{zz}}$ atmosphere for the planet and selects a stellar UV spectrum from the shipped spectra by the host T_{eff} ($> 6800\text{ K} \rightarrow 51\text{ Eri}$; $5300\text{--}6800\text{ K} \rightarrow \text{solar}$; $4300\text{--}5300\text{ K} \rightarrow \epsilon\text{ Eri}$; cooler $\rightarrow \text{GJ 436}$), (iii) compiles FastChem if its binary is missing, (iv) runs VULCAN to steady state, and (v) converts the result to `base.inp`. The first run takes hours; the converged `<name>-photo.vul` is cached and reused on subsequent runs (regenerate with `--force`). The code root is taken from EXHALE_ROOT when set (for relocated installs). VULCAN's networks are H/C/N/O/S: it supplies the H_2/H state and molecular mixing ratios, *not* the atomic-metal release (Na/Mg/Ca/Fe): `metals.inp` stays user-supplied.

3.8 The lower atmosphere as a profile (Lower atmosphere profile:)

The Tier-3 handoff of §3.7 states the lower atmosphere at one level, as a handful of scalars in `base.inp`. A *profile* states it over an interval of pressure instead: the lower model (photochemistry, with or without a climate solution) is solved on a column that reaches *above* the level where EXHALE places its base, and the whole overlap is handed over as one table. Point EXHALE at it with

Lower atmosphere profile: `lower_atmosphere_profile.dat`

in `input.inp`; the file lives in the run directory and the name is free. Two things ask for the interval rather than the level. The first is eddy mixing: the homopause is where K_{zz} crosses the molecular diffusion coefficient, which rises by nearly five decades between the base and $1.2R_p$, so a single `He_Kzz` cannot state where it lies (§3.6). The second is the elemental-flux closure below, in which the composition at the base stops being an input and becomes an output.

Schema. A block of # header lines of the form `# key value`, then a fixed-column table in full double precision running **deep to shallow**: the pressure column must be strictly decreasing, and at least two levels must be present. Three header keys and one schema line are required:

- `solution_id`: a sha256 fingerprint of the lower model's configuration (code and version, mechanism and thermodynamic data, stellar flux and dilution, planet mass and reference radius, elemental abundances, matching pressure, trial fluxes). It is what makes “the same lower-atmosphere solution” a checkable statement.
- `p_match_bar`: the matching pressure, i.e. where EXHALE places its base. It must lie inside the pressures the table covers.
- `p_top_bar`: the shallowest level carried. It must be smaller than `p_match_bar` (otherwise the two models never overlap) and must not lie above the shallowest tabulated level.
- `# columns::` the names of the columns, in file order. Every column is found *by name*, because the producers order their element list differently from run to run.

The remaining header keys are optional and are provenance or metadata: `source_code`, `source_version`, `mechanism`, `stellar_flux`, `p_deep_bar`, `trial_flux_H`, `trial_flux_He`, `iteration`, `reached_steady_state` and `notes`. A profile with no iteration index runs perfectly well; it is simply not drivable by the closure iteration, which the run reports as `lower_profile_iterable`. Header keys this version has no consumer for are ignored rather than refused. Nine columns are required: `p` (bar), `r` (R_J), `T` (K), `n_tot` (cm^{-3}), `rho` (g cm^{-3}), `Kzz` (cm^2s^{-1}), `q_H2`, `q_H` and `X_He`. Optional columns are the other elemental ratios `X_C`, `X_O`, `X_Mg`, ... (E/H *nuclei* ratios, summed over every carrier, for any element species_table knows), the elemental fluxes `F_H` and `F_He` (g s^{-1} , outward positive), and any number of diagnostic mixing ratios such as `q_H2O`. Columns the reader has no consumer for are **kept, not dropped**, so a file survives a round trip through the tools. A data row that does not carry as many numbers as the schema line names stops the run.

Every quantity is interpolated **linearly in** $\log p$ and is never extrapolated; a target that falls exactly on a tabulated level returns that level's value bit for bit, which is what lets a profile with a node at the matching pressure reproduce the scalar handoff exactly. K_{zz} is the one quantity taken as a profile rather than at a point: it is mapped onto the radial grid through the file's own radius column, and above the shallowest level carried each cell takes that level's value, the eddy coefficient being a lower-atmosphere property about which the file makes no statement above its top.

What it supplies, and what `base.inp` may no longer say. At the matching level the profile sets the base temperature T_0 (column `T`), the base radius R_0 (column `r`), the base pressure p_{base} (`p_match_bar`), the base H_2 fraction (column `q_H2`), the He/H ratio (column `X_He`) and each elemental reservoir it carries, which overrides `metals.inp`. It also fixes the base *level*: n_0 follows from `p_match_bar` through $n_0 k_B T_0 n_{\text{tot},\text{bc}} = p_{\text{base}}$, the same relation a `base.inp` carrying `p_base` uses, because the composition, q_{H_2} and temperature the profile hands over all refer to that one level. A `Log10 lower boundary number density` line given beside a profile is then saying the same thing twice: the two must agree to 1%, and if they do not, startup stops and prints both numbers, the n_0 each implies, $n_{\text{tot},\text{bc}}$, and the two ways to fix it. The columns `n_tot` and `rho` are carried but not imposed: the base particle count and mass follow from T_0 , He/H, q_{H_2} and the elemental ratios exactly as they do on the scalar path.

Because the profile is now the single source of those quantities, a `base.inp` sitting beside it may no longer state them. Three categories of key are **refused**, and each stops the run naming the offending key: the EOS boundary (`T_base`, `r_base`, `p_base`, `q_H2_base`), the elemental reservoirs (`HeH_base` and any `El_H_base`) and the boundary constraint

(Kzz_base). Provenance comments and diagnostic keys are still allowed, and indeed a `base.inp` whose whole content is provenance is what the adapters write. Such a file must carry a `# solution_id` comment, and it must match the profile's: a `base.inp` beside a profile with no `solution_id` line, or with a different one, stops the run, because two files describing the same gas at the same level from two different lower-atmosphere solutions are not a model of anything. `He_Kzz`: is treated differently, because it belongs to `input.inp` and not to the handoff: it is not refused, but it is *inert* while a profile is in use (the eddy coefficient comes from the Kzz column), and the run prints a warning saying so. The resolved-configuration report records what was used: `lower_profile_present`, and with a profile in use the file name, the `solution_id`, the source, `p_match_bar`, `p_top_bar`, the trial fluxes, the iteration index and whether the profile is iterable.

Producing a profile. Three tools in `src/utils/` do this, sharing one command line and one schema module (`lower_profile_schema.py`, which also computes the `solution_id` and writes the paired provenance-only `base.inp`). `photochem_to_lower_profile.py` is the production path: it solves a gas-giant photochemistry model to steady state on the column and writes the handoff, with the temperature structure either prescribed (`--tp-file`) or solved for by the radiative-convective climate model (`--climate`), in which case the tropopause, and with it the water cold trap, is part of the solution rather than an input. `vulcan_to_lower_profile.py` converts a finished VULCAN `.vul` output into the same format; it is the cross-check, not a second production path, because VULCAN carries no climate model and so can never be the route away from a prescribed $T(p)$.

Both take the same statement of the eddy coefficient. By default the Kzz column is whatever $K_{zz}(p)$ the underlying solution carried; `--kzz-const` K replaces it with one value at every level, and `--kzz-power` α , with its amplitude `--kzz-ref` K_{ref} and the pressure that amplitude is stated at `--kzz-ref-bar` P_{ref} (default 1 bar), replaces it with the saturated gravity-wave form $K_{zz}(p) = K_{\text{ref}}(p/P_{\text{ref}})^{-\alpha}$: $\alpha = 1/2$ being the Lindzen (1981) slope fitted by Parmentier et al. (2013), 0.4 that of Charnay et al. (2015). The two are mutually exclusive, `--kzz-power` without `--kzz-ref` is refused, and the power-law settings enter the `solution_id` only when they are in use, so stating neither leaves a run and its fingerprint exactly as they were. On the Photochem path a stated coefficient is the profile the chemistry is *solved* on and not only the one reported. There is no ceiling on the power law: the file's own top bounds it, and EXHALE gives every cell above the shallowest tabulated level that level's value, so the level the amplitude is anchored at decides what the wind inherits above the profile. Finally `element_flux_closure.py` drives the pair of codes to their fixed point: the trial elemental fluxes of the header are imposed on the chemistry's upper boundary, the wind's own elemental flux is measured face by face and reduced over a radial window, and the two are iterated (under-relaxed) until they agree, at which point the base composition is an output of the coupled problem instead of an input to it.

See also. `examples/17_lower_profile/` is a ready-to-run HD209458b case whose `make_example_profile.py` regenerates the synthetic column it ships (§6.3). The authoritative column-by-column schema is `md/input_schema.md` section 2d.

3.9 Obtaining VULCAN and FastChem

The lower-atmosphere pre-step uses the **VULCAN** photochemical-kinetics code, which ships the **FastChem** equilibrium-chemistry code inside it. These are third-party open-source codes and are *not* committed to the EXHALE repository (VULCAN/ is git-ignored). Fetch and prepare them with

```
src/utils/setup_vulcan.sh
```

which clones VULCAN into EXHALE/VULCAN/, applies the two modifications EXHALE needs, builds FastChem, and removes stale run products. EXHALE also runs this automatically the first time the vulcan pre-step is used and VULCAN/ is missing. Obtain updates from, and *cite*, the original sources:

- **VULCAN**: github.com/exoclimate/VULCAN (mirror github.com/shami-EEG/VULCAN). Cite Tsai et al. (2017, ApJS 228, 20) and Tsai et al. (2021, ApJ 923, 264).
- **FastChem**: github.com/NewStrangeWorlds/FastChem (shipped inside VULCAN). Cite Stock et al. (2018, MNRAS 479, 865), and Stock et al. (2022, MNRAS 517, 4070) for FastChem 2.

The modifications `setup_vulcan.sh` applies to the cloned VULCAN tree (i.e. the only changes needed if you set it up by hand):

File (in the VULCAN clone)	Modification
<code>make_chem_funs.py</code>	add <code>encoding=None</code> to the <code>np.genfromtxt(vulcan_cfg.com_file, ...)</code> call so the element-conservation check does not crash on Python 3
<code>vulcan_cfg.py</code>	set <code>use_photo = True</code> and <code>use_live_plot = False</code> (this file is the driver's template)
<code>output/*.vul</code>	delete (run products; regenerated in <code><run_dir>/vulcan_work/</code>)
<code>fastchem_vulcan/</code>	make to build the fastchem binary

On the EXHALE side (in this repository) the pre-step adds `src/utils/setup_vulcan.sh`, `vulcan_driver.py` and `vulcan_to_base.py`, and the Lower atmosphere: key in `input_read.f90/parameters.f90`.

4. Output files

All profiles are written to `output/` as plain text, one row per radial cell, in *cgs* units. Two flavors exist: the equilibrium files (`*.txt`, written during the run) and the advection-corrected files (`*_adv.txt`, written once at the end by the post-processor, which re-solves the ionization balance including advection of the wind). The `*_adv.txt` files are what the analysis tools read by default, and the equilibrium files are what they read when the solved state is asked for (EXHALE_TRANSIT_STATE=solution, §5.2) or when a stationary solve wrote no advection-corrected pair. A stationary run that alternates a wind solve with a composition relaxation also keeps the state of its last completed outer pass, the generation `output/pass_state/<state_id>/` named by `output/pass_state/current` (§4.3), a restartable pair that is never a certified state.

Caveat: the `_adv` profile at a “breathing” base. The advection-corrected (`*_adv.txt`) *temperature and ionization* would otherwise develop a spurious spike / sawtooth in the dense lower atmosphere (typically $r \lesssim 1.1\text{--}1.2R_p$ for strongly Roche-filling planets such as WASP-121b). Two compounding causes, both confined to the base, are responsible:

1. *A breathing, inflowing base (a 1-D limitation).* When the planet nearly fills its Roche lobe the 1-D wind is subsonic / only marginally launched at L_1 , so the dense lower atmosphere does not settle into a clean monotonic outflow but into a slowly recirculating state with small *negative* (inflow) velocities and a stagnation point ($v = 0$) near the wind base. This is intrinsic to the spherically-symmetric (1-D) geometry (removing it would require a 2-D/3-D model), and is not a numerical bug.
2. *A spurious hot root in the metal-cooled post-processor.* The post-processor re-solves the energy/ionization balance in each cell *including* advection, assuming an outflow (each cell takes its history from the cells below it, the recursion running outward along the flow). In the dense base the non-monotonic metal line-cooling curve gives the energy equation a second, spurious *hot* root that the Newton (MINPACK) solver can land on; through that recursion one bad cell then cascades outward, producing the sawtooth and an overshoot spike (`post_process_adv.f90`). A guard rejects solves that fall outside a band around the converged equilibrium temperature, but mild spikes *within* that band can survive.

The post-processor guards against cause 2 directly: wherever the flow is *not* a clean outflow ($v \leq 0$, the breathing inflow base) it skips the advection correction and keeps the converged equilibrium ionization and temperature (`post_process_adv.f90`). This removes the spurious spike and sawtooth at the source (it breaks the upwind cascade), while preserving the genuine few-percent `_adv` correction in the outflow above the stagnation point. Cause 1 (the breathing base itself) is intrinsic to the 1-D geometry and remains, but it is now represented there by the smooth, physical equilibrium solution. The shipped `_adv` profiles are therefore clean, and the temperature panel of Figure 6 plots the `_adv` profile directly. (Velocity and density are *identical* in the `eq` and `_adv` files: the post-processor corrects only the ionization and temperature.)

Validity range of the advection correction. The inflow condition above is one of four under which the post-processor keeps the converged equilibrium state instead of correcting it. A cell is left at equilibrium when

1. $v \leq 0$ in the cell or in the one below it (inflow): the step takes its history from the cells below, which are not upstream when the gas moves inward, and the residence time h_j/v_j is negative;
2. $\text{Da} = (h_j/v_j)\nu_{\min} > 100$, with $\nu_{\min}$ the smallest total destruction-plus-formation rate of the populations the system solves (HI; with helium also He I, He II and the He 2^3S level, whose radiative rate $A_{31} = 1.272 \times 10^{-4} \text{ s}^{-1}$ sets the floor): the gas relaxes to local ionization equilibrium many times over while it crosses the cell, so the equilibrium solution already solves the advection ODE and the correction can only add integration error;
3. $x_{\text{HII},\text{eq}} < 10^{-6}$: the residuals carry the *neutral* fraction and report the ion density as $(1 - x_{\text{HI}})n_h$, so the ion fraction is resolved only to the solver’s absolute tolerance $\sqrt{\epsilon} = 1.5 \times 10^{-8}$. In a shielded base with $x_{\text{HII},\text{eq}} \sim 10^{-13}$ the returned value is numerical noise of arbitrary sign, which the upwind cascade would then carry outward as negative ion densities;

4. the mass flux through the cell changes by too large a fraction of itself. Every equation the post-processor solves is a steady equation integrated along the recorded flow, and the flow it integrates along carries a fractional change of the *face* mass flux across the cell,

$$m_j = |R_1(j)|/s_1(j),$$

the mass row of the state divided by the largest term that row itself holds. This is the same operator the stationary certification measures the mass row by (`steady_residual.f90`, `certification.f90`), evaluated by calling those routines on the state handed in rather than by re-deriving it from center-sampled $\rho v r^2$, which is the same quantity in the continuum and a different discrete test. The terms the steady equations drop are the ones the mass divergence puts into them, each of them m_j times a term they keep, so *the correction is first order in m_j* : a cell with m_j at or below `adv_conditional_tol` = 10^{-2} is corrected, and its correction is a *conditional* one, accurate to that fraction of itself in the mass flux; a cell above it keeps the run's own temperature and the ionization equilibrium at that temperature. The value of m_j for every row is written out as the `adv_mass_row` column, so a reader has the condition of each row and not only its verdict.

The fraction above is *not* the tolerance of a certified stationary state. That is `cert_tol_mass` = 3×10^{-12} , the number the stationary certification judges the mass row of the *whole* state by, and the verdict it reaches on the input state travels in the file's own # `adv_input_certified` line. The two are different statements about one measure and the file carries both: the rows of a certified state are corrected, while a corrected row is not thereby certified. Conditions 1 and 2 are properties of the flow, 3 of the representation of the unknown and 4 of the state the equations are integrated on, so none of them depends on whether metal cooling is active. Condition 4 refuses both the temperature and the composition of the cell, condition 1 both as well (the step has no upstream history either way), and conditions 2 and 3 the composition alone. The energy solve additionally applies a *thermal* Damköhler condition of its own, $Da = (h_j/v_j)|\Gamma - \Lambda|/u_{th} > 1$: where the local radiative balance rather than the flow sets the temperature, the temperature to keep is the run's own, which is the root of that same balance with every channel the run solved. Conditions 1 to 3 trigger in the dense base; condition 4 follows the state, and it is the one that separates a converged wind from a snapshot. The counts are printed at the end of the run, and the verdict and the measure of every row are recorded in the `adv_T_status`, `adv_comp_status` and `adv_mass_row` columns of `Hydro_ioniz_adv.txt` (§4.1). Full account: `md/postprocess_advection_validity.md`.

A last condition concerns the temperature alone. The post-process carries the H/He and trace-metal gas and omits the molecular species, so its energy balance lacks the molecular chemistry heat and the H_3^+ and molecular band cooling the run solved. Wherever those omitted channels, measured on the state handed in as the difference between the run's heating and cooling and the post-process's own at the run's temperature and composition, exceed the channels it carries, the row keeps the run's temperature with `adv_T_status` = 3 (unsupported). MEASURED on the LHS 1140 b molecular states: 196 / 389 / 776 rows on 500 / 1000 / 2000 cells, up to 1.134–1.137 R_p , and an He 10830 equivalent width that moves by at most 1.2×10^{-4} of itself.

How the correction is integrated. Every equation the post-process integrates along the flow is an ODE in r , $dy/dr = F(y, r)$: the ionization balance $v dx/dr = R(x)$ and the steady internal-energy equation. Since 2026-09-27 each

is advanced cell by cell with the variable-step second-order backward differentiation formula (BDF2),

$$y_j - a_1 y_{j-1} + a_2 y_{j-2} = g h_j F(y_j, r_j), \quad h_j = r_j - r_{j-1}, \quad w = \frac{h_j}{h_{j-1}}, \quad a_1 = \frac{(1+w)^2}{1+2w}, \quad a_2 = \frac{w^2}{1+2w}, \quad g = \frac{1+w}{1+2w},$$

with every rate, heating and cooling term and the velocity of the residence time taken at r_j (post_process_adv.f90 states the derivation; the form is that of Li & Liao 2022, SIAM J. Numer. Anal. 60, 2253, read as eq. (1.4) of arXiv:2201.00527). It is second order in the cell width and L-stable, so a population that relaxes within a cell settles on its local equilibrium instead of alternating about it as it would under the trapezoid rule. Backward Euler, $y_j - y_{j-1} = h_j F(y_j, r_j)$, takes the first step of the recursion, the first step after a cell it did not integrate, a step whose width ratio exceeds the zero-stability bound $w \leq 1 + \sqrt{2}$, and the retake of a BDF2 step whose populations come out negative or whose temperature has no root. The run log counts the BDF2 and backward-Euler steps and the retakes. Until 2026-09-27 every step was backward Euler, a first-order recursion that was the main source of the first-order grid convergence of the equivalent width; MEASURED on nine certified LHS 1140 b states (500, 1000 and 2000 cells), the post-process share of the error falls from +3.0% / +1.0% to below 3×10^{-4} , and the equivalent width itself moves by -3.9% (500 cells) to -0.97% (2000 cells) at He/H = 1.6 (md/Update_EXHALE_stage3.md section 29.10). **Equivalent widths quoted from _adv files written before 2026-09-27 are those of the first-order recursion.**

4.1 Hydro_ioniz.txt / Hydro_ioniz_adv.txt

Thermodynamic profiles (7 columns, and three more in the _adv file). The dimensionless code variables are restored to cgs with the normalizations n_0 (base density), $v_0 = \sqrt{k_B T_0 / \mu}$, $p_0 = n_0 \mu v_0^2$, T_0 (base temperature), $q_0 = n_0 \mu v_0^3 / R_p$.

Col	Quantity	Unit	Note
1	r	R_p	cell-center radius (the lower face of cell 1 is the base level $r = 1$)
2	ρ (mass density)	$m_H \text{ cm}^{-3}$	mass density ($= \rho_{\text{code}} n_0$)
3	v	cm s^{-1}	radial velocity ($\times v_0$)
4	p	erg cm^{-3}	pressure ($\times p_0$)
5	T	K	temperature ($\times T_0$)
6	heating	$\text{erg cm}^{-3} \text{ s}^{-1}$	XUV photoionization heating ($\times q_0$)
7	cooling	$\text{erg cm}^{-3} \text{ s}^{-1}$	radiative cooling ($\times q_0$)
8	adv_T_status	none	_adv file only: integer, the validity of the temperature of the row
9	adv_comp_status	none	_adv file only: integer, the validity of its composition
10	adv_mass_row	none	_adv file only: m_j , the measure both fields were decided by

Column 2 is a *mass* density expressed in units of m_H per cm^3 (metals included under the eos_metals policy), not a number density; multiply by m_H to get g cm^{-3} .

Columns 8 to 10: what each row of an _adv file is. Written only in Hydro_ioniz_adv.txt; the equilibrium file has seven columns, because its rows are the run's own solution by construction. Ion_species_adv.txt carries the two status fields as its last two columns, so a tool reading the species file alone sees the same verdicts.

The advection post-process solves the steady ionization and energy equations along the recorded flow. The temperature of a row and its composition are *two* statements: a row whose temperature was corrected can carry the run's own

composition and the reverse, so each has its own field and neither is the first refusal of a single mixed integer. Both take the same five values.

Value	Name	Meaning
0	corrected	the steady equation was solved and adopted for this quantity
1	retained	the run's own value was kept: m_j above the fraction below, the local radiative balance rather than the flow sets T (thermal Damköhler > 1), or the gas enters the cell
2	failed	the cell solve did not converge, or its root was out of band
3	unsupported	the closure does not cover this cell class. Composition: the omitted molecular and oxygen carriers hold more of the particle count than the species the reconstruction carries. Temperature: the heating and cooling channels the molecule-free post-process omits (molecular chemistry heat, H_3^+ and molecular band cooling) exceed the ones it carries, so the run's own T is kept (below about $1.13 R_p$ on the LHS 1140 b molecular states)
4	not_evaluated	the post-process did not reach this row (a ghost row below the first cell at which its upwind difference can be taken)

A retained quantity is the run's own value *exactly*, not to a tolerance: the composition of such a row is the ionization equilibrium at the run's temperature, which is what the uncorrected file carries. A corrected row is a *conditional* correction, accurate to `adv_conditional_tol` of itself in the mass flux, and column 10 carries the m_j it was judged by, so a reader can weigh a corrected row rather than only count it.

The file's own # header states all of it: the schema version, the legend of both fields, an # `adv_status_counts` line with the number of rows carrying each value in each field, the # `adv_stationarity_operator` the measure comes from, the # `adv_conditional_tol` line with the fraction a corrected row is accurate to, the # `adv_input_certified` verdict of the stationary certification on the whole input state (a separate statement from the fraction), and two declarations of what the product is not: # `adv_product`, that this is a *one-way* correction on a fixed density and velocity field, so a converged temperature solve does not make the row a self-consistent solution and a profile assembled from corrected and retained rows need not satisfy any interface balance, and # `adv_model_restrictions`, that the molecular and oxygen columns are the equilibrium solution, that those species are absent from the particle and electron counts every equation here is solved with, and that the H_3^+ infrared cooling is absent from its energy balance. A file written before the schema carries no # `adv_schema` line, and the validity of its rows is UNKNOWN, not corrected.

`exhale_io.load_hydro` returns the three columns under their own names together with the header's statements, and `exhale_io.Run` exposes them as `adv_T_status`, `adv_comp_status`, `adv_mass_row` and `adv_conditional_tol` (None for an equilibrium file); `EXHALE_transit.py` prints the census of the rows its chords sample with the fraction and the largest measure among them, and copies both into the comment metadata of every saved curve. The two lowest rows are ghost rows below the first computational cell, whose temperature the energy loop never corrects whatever their status.

Header lines. The file opens with # comment lines that every loader skips: the schema tag, the # columns list, the # rows line saying which rows are ghost cells, and

```
# coupling: sec_ion=T sec_ion_step=2402 recon=WENO3 certified=T mode=init
```

which records the run state the profile was produced under: every switch that changes *while* a run converges rather than being fixed by `input.inp`. Whether the SvS85 photoelectron secondary ionization was being applied and at which step it was armed; the discrete reconstruction operator in force, which is not `rec_method` alone while the PLM $\rightarrow$ WENO3 continuation is armed; whether the state passed the stationary certification, with `cert_reason=` beside it whenever the verdict is qualified (`no_stationary_claim` for a run that ended on a cap, `certified_in_wind` for a state whose species rows were gated at $r \geq 1.20$ only); and the `Run` mode the run declared. Two fields that used to follow `sec_ion_step`, `valve` and `fluxconst`, were the switches the old base closure chose for itself and are gone with it (§2.6); a reader of an older file still finds them there, and the parser takes the fields it recognizes one at a time. This matters because the file is also the restart input: copied to `Hydro_ioniz_IC.txt` and read back under `Load IC?` True, its `sec_ion` field tells the new run to start the coupling where the old one left it, instead of re-running the staging on a state that no longer needs it (§144 of `md/Update_EXHALE_stage1.md`). A file written before this line existed simply has no such line, and such a restart starts staged exactly as it did before. The other fields are reported, never imposed: `input.inp` states them for the new run.

Provenance. Three further `#` lines say what produced the file, so that a profile found on disk in a year can be tied to an executable and an input without asking anyone:

```
# provenance: git=6d07d48afd41 tree=dirty run=2026-09-03T20:19:09
# provenance: ck_input=1260237361560544917 ck_base=-1 ck_metals=4583919659595562347
# provenance: recon=WENO3 base_bc=isothermal carrier=none restart_schema=3 resid_def=145 N=500
```

`git` and `tree` are the revision the *executable* was built from and whether the working tree carried uncommitted changes at that moment; the Makefile stamps them in and the run never calls `git`. No build time is compiled in: a timestamp in a constant would make two builds of one source different files, and the md5 of the binary is what ties a bitwise regression verdict to a revision, so the time shown is the *run* time, from `date_and_time`. The `ck_` fields are checksums of the input files as they were read (`-1` means the file was absent), and the last line records the discretization, the lower boundary model, the molecular carrier model, the version of the restart layout, the version of the residual definition the acceptance gates used, and the grid size.

The lower boundary the state was written with. Two further `#` lines carry the boundary itself, because a state read back a year later has to say which boundary it is a solution of:

```
# boundary_model characteristic_face_ps_reservoir_C_minus_contact_upwind_
ghost_fixed_point_seed_reservoir_row_face_at_level_
discrete_equilibrium_v5
# boundary_reservoir version 1 p[p0] T[T0] nhat[n0/rho0] r_level[Rp] . . .
```

(each is one line in the file). The first names the model that built the face state (§2.6); the identity changes when the model does, `v2` being the contact upwind without a stated ghost composition, `v3` adding the ghost seed and the fixed point the ghost composition is held to, `v4` (2026-09-27) putting the lower face on the base level, so that the reservoir is the face state it owns with no transport along an isentrope, and `v5` (`. . . _face_at_level_discrete_equilibrium_v5`,

2026-09-29) building the interior pressure at the face and the ghost cells from the well-balanced reconstruction's own discrete equilibrium. A run with `Well balanced: True` states v5; a run without it builds the v4 boundary unchanged and states v4. The second states the prescribed reservoir with the version of its prescription: the pressure and temperature at the base level, the particles per unit mass of the reservoir gas and the radius the three are stated at, each written at round-trip precision, so the four numbers on the line are the numbers the run held and not a rendering of them. The entropy is not among them: it is carried implicitly, as the isentrope through (p, T) at the base composition. The reservoir travels with the state because it is the only boundary input the physical column cannot reconstruct; the ghost rows the file holds are the boundary's own output and are rebuilt at the read from the column and this run's reservoir. A restart written under an older boundary model is reported, not refused on that account: the loader names both models and says that the state's residual is not the residual it was written with. A v3 or older state, however, was written on a grid whose cell centers lie lower (by half a cell on the Mixed grid), so the grid check refuses it; it reaches a v4 run only as a seed mapped with `src/utls/map_state_to_grid.py --ic` (§2.6). Both lines are # comments, so no numeric parse and no regression comparison changes.

4.2 Ion_species.txt / Ion_species_adv.txt

Ion number densities [cm^{-3}] (34 columns). Column 1 is r ; columns 2–7 are the H/He ions; columns 8–34 are the 27 metal-ion stages, in this order:

```
r ; HI HII HeI HeII HeIII HeI* ; CI CII CIII OI OII OIII NI NII NIII MgI MgII MgIII SiI SiII SiIII CaI
CaII CaIII NaI NaII KI KII SI SII FeI FeII FeIII
```

(HeI* is the metastable 2^3S triplet.) The ionization fraction of any element is the stage density divided by the sum over that element's stages. With `Molecular chemistry: True` four more columns (H_2 H_2p H_3p HeHp , cm^{-3}) are appended after the metals, and with `Oxygen chemistry: True` three more after those (OH H_2O CO , cm^{-3}); the # columns header always names the layout. In `Ion_species_adv.txt` two further columns follow, `adv_T_status` and `adv_comp_status`, the validity of the row's temperature and of its composition (§4.1), with the same header block as the hydrodynamic file, so a tool reading the species file alone sees the same verdicts; the measure they were decided by is a column of that other file. Note that with the oxygen chemistry on the OI column means *free atomic* neutral oxygen: the element total is $n(\text{OI}) + n(\text{OII}) + n(\text{OIII}) + n(\text{OH}) + n(\text{H}_2\text{O}) + n(\text{CO})$, and the same holds for carbon with CO.

4.3 pass_state/: the state of every outer pass

The state of the last completed outer pass of the stationary route (§2.7), written since 2026-09-28 so that a run stopped from outside keeps the passes it made, and published as generations since 2026-09-29 (until then it was the pair `Hydro_ioniz_last_pass.txt / Ion_species_last_pass.txt`, overwritten half by half, `md/Update_EXHALE_stage3.md` section 33). *When*: after every outer pass after which the iteration goes on, i.e. only where the route alternates a wind solve with a composition relaxation (`He_diffusion`, or molecular carriers without `Coupled carrier solve: True`). A pass that ends the iteration publishes nothing, because the run then writes its final `Hydro_ioniz.txt / Ion_species.txt`, and the last pass of the budget takes no composition update, so its state is that final state; the generation left beside a finished run is one pass older than it. *What*: the state the next pass starts from,

i.e. the accepted composition update of that pass (or the entry state of the coupled block where the route handed over to it), written by the same writers and with the same header lines as the final state files, so the two files have exactly the columns of `Hydro_ioniz.txt` and `Ion_species.txt`. Writing them changes nothing the run reads afterwards; the final state files of a run are bitwise those it wrote before the snapshot existed (MEASURED on the regression matrix, 2026-09-28, and again for the generations on `iontrans_metals` and `hp_front`, 2026-09-29).

Layout: one directory for each published pass,

```
output/pass_state/<state_id>/Hydro_ioniz.txt
output/pass_state/<state_id>/Ion_species.txt
output/pass_state/<state_id>/manifest.txt
output/pass_state/current
```

with `<state_id> = r<run identity>p<n>`. The run identity is formed once at the start of the run from the wall clock to the millisecond and the processor clock count (`yyyymmddThhmmss.sss_c<count>`), so two runs writing the same pass number never name one generation. `manifest.txt` states, one key=value a line, the `state_id`, the `run_identity`, the `pass`, the `previous_state_id` (the generation current named when this one was published), the two file names with their sizes in bytes, and the certification pair. `current` is one line, the `state_id` of the generation a restart takes.

The order of publication: (1) the three files are written and closed in `output/pass_state/.<state_id>.tmp/`; (2) that directory is renamed onto `output/pass_state/<state_id>`; (3) `output/pass_state/current.part` is written with the new `state_id` and renamed onto `output/pass_state/current`, the single rename that publishes the generation; (4) the generation current named before is kept, and the one it had replaced (its manifest's `previous_state_id`) is removed, file by file under its name. An interruption at any point therefore leaves `current` naming a complete generation, or, before the first publication in a directory, no `current`. An interruption before step 2 leaves a `.<state_id>.tmp` directory, one between steps 2 and 3 a complete generation no pointer names; neither is read, and neither is removed by a later run. EXHALE_PASS_STATE_STOP_AT (Table 1) stops a run after a chosen step, to measure this.

Headers: both halves carry the `# coupling: line, with certified=F cert_reason=pass_snapshot_p<n>` for pass *n* and, as its last field, `state_id=<state_id>`, and a line

```
# pass_snapshot pass=<n> carrier_movement_bound=<b> element_omega=<w> (the state after the
composition update of this pass; not certified)
```

A snapshot is *never* a certified state: the certification of a pass is taken before its composition update and the snapshot is the state after it, so no tool may read it as certified. `load_IC` refuses a pair whose halves state two different `state_id` values, and a pair in which one half states one and the other does not (a generation file beside a final state file); a pair in which neither states one (every final state, every file written before the field existed) is read as before. The final state files of a run carry no `state_id`.

Restarting from them: copy the pair of the generation current names to `output/Hydro_ioniz_IC.txt` / `output/Ion_species_IC.txt`

```
G=$(cat output/pass_state/current)
cp output/pass_state/$G/Hydro_ioniz.txt output/Hydro_ioniz_IC.txt
cp output/pass_state/$G/Ion_species.txt output/Ion_species_IC.txt
```

and run with `Load IC? True` as from any state file (the restart contract of §3. applies unchanged). The carrier movement bound and the element under-relaxation factor the run had reached are *not* restored by a restart; to continue at the values of the `# pass_snapshot` line, pass them as `EXHALE_CARRIER_TRUST=<b>` and `EXHALE_DIFF_OMEGA=<w>` (Table 1). MEASURED: a run killed during pass 3 left the complete pass-2 pair, which restarted and continued as the uninterrupted run did (`md/Update_EXHALE_stage3.md` section 29.12, with the pair of that time).

4.4 Cooling_breakdown.txt

Radiative cooling by channel [$\text{erg cm}^{-3} \text{ s}^{-1}$] vs. radius. Columns: 1= r , 2= T [K], 3= n_e [cm^{-3}], 4=total cooling; 5--10= the six atomic channels: recombination (the kinetic energy $k_B T \alpha(3/2 + d \ln \alpha / d \ln T)$ of the captured electrons of H II, He II, He III *and* of every recombining metal ion), collisional ionization (the ionization potential of every H, He and metal ionization), then collisional excitation split by absorber into H I (the $\text{Ly}\alpha$ -dominated H-line cooling), He I and He II, and finally free-free emission of every ion by its net charge. With the He 2^3S tracked, the He I column carries the metastable's own terms: the net $1^1\text{S} \leftrightarrow 2^3\text{S}$ exchange, which is *negative* where the superelastic $2^3\text{S} \rightarrow 1^1\text{S}$ collisions heat the gas, the triplet cascade that radiates $10830 \text{ \AA}$ and the higher triplet lines, and the $2^3\text{S} \rightarrow$ singlet excitations; 11= the H_3^+ infrared cooling (zero unless the run tracks the molecular network); 12--14= the H_2 , H_2O and CO infrared bands, which are NET rates (emission less absorption of the field from below, so a negative value is that band heating the gas) and are identically zero unless `Molecular IR bands` is on; 15--41= the 27 metal-line cooling channels (same ion order as `Ion_species`). The channels sum to column 4. Useful for identifying the dominant coolant vs. altitude (cf. Huang+2023 Fig. 10), and the H_3^+ column is the coolant of a molecular base. The `# col` header line always names the layout.

4.5 Heating_breakdown.txt

Volumetric heating by channel [$\text{erg cm}^{-3} \text{ s}^{-1}$] vs. radius, the companion of `Cooling_breakdown.txt` and written unconditionally at the end of the run. 24 columns: 1= r [R_p], 2= T [K], 3= n_e [cm^{-3}], 4=total heating; 5--10= the photoionization heating split by *absorber*: H I, He I, He II, He 2^3S , H_2 , and the sum over the metal stages (whose electrons include the Auger electrons of an inner-shell vacancy); 11= the photoelectric heating of the $\text{H}(n=2)$ population (the excited-H model); 12= the collisional de-excitation of $\text{H}(n=2)$, i.e. absorbed $\text{Ly}\alpha$ thermalized by a collision; 13= `heat_He_recomb`, the photoelectrons of the H I, He I, H_2 and metal ionizations driven by the recombination radiation absorbed on the spot (He II and He III recombinations, and the H II ground captures taken by a metal); 14= the Penning branch of $\text{He } 2^3\text{S} + \text{H}$; 15= its associative branch $\text{He } 2^3\text{S} + \text{H} \rightarrow \text{HeH}^+ + \text{e}$; 16= $\text{He } 2^3\text{S} + \text{H}_2$ Penning ionization; 17= the fragment kinetic energy of the H_2 Lyman–Werner photodissociation; 18= the collisionally de-excited Lyman–Werner fluorescence; 19= the collisional reactions of the H_2/He network; 20= the FUV photolysis of H_2O and OH; 21= the collisional reactions of the oxygen network including the $\text{O}(^1D)$ sink; 22= $\text{He}^+ + \text{CO} \rightarrow \text{C}^+ + \text{O} + \text{He}$, the charge-transfer destruction of CO; 23= the fragment kinetic energy of the CO photodissociation; 24= `heat_charge_exchange`, the energy defects of the charge-exchange reactions the run applies (metal+H, metal+He, metal+metal and $\text{He} \leftrightarrow \text{H}$), each less the excitation of a product state that leaves as radiation (an allowed decay, the photon of a radiative transfer, and, below the critical densities, the forbidden lines of an NI, S II or CI product formed out of statistical equilibrium),

negative for an endothermic reaction. A channel of physics the run does not carry is identically zero.

One state per output file. These columns are not recomputed for the dump: they are the channel array the ionization sweep filled for the state written beside them, through the one heating assembly `heating_of_composition` (`src/modules/radiation/util_ion_eq.f90`), whose total is the running sum of its channels and is the heat column of `Hydro_ioniz.txt`. Column 4 therefore reproduces that column to round-off: measured at 1.7×10^{-16} over the physical cells of the `mol_base_handoff` case (2026-09-06). The header names the layout and is built from the channel list, so a deposit added to the energy equation appears here by construction.

Rebuilding the rates on the written state instead, which earlier versions of this file did, is a different state: the rates a sweep contracts belong to the composition that *entered* it, and cannot be recovered from the composition it returned. That put a lag of one sweep between the two columns, measured at 1.2×10^{-5} in the median cell of the same case.

4.6 Lyman_Werner.txt

H₂ photodissociation diagnostic, written when a molecular run carries a `Stellar LW flux`. 8 columns: 1 r [R_p], 2 T [K], 3 $x_{H_2} = 2n_{H_2}/n_H$, 4 n_{H_2} [cm⁻³], 5 the star-ward H₂ column N_{H_2} [cm⁻²], 6 the level-resolved self-shielding factor f_{shield} , a diagnostic, $\sigma_{\text{diss}}(N_{H_2})/\sigma_{\text{diss}}$ at the bottom of the tabulated column axis, evaluated at that cell's own column, 7 the dissociation rate k_{LW} [s⁻¹], which is the cell mean of the tabulated rate between the cell's two faces and is therefore *not* f_{shield} times a thin-limit rate, and 8 the photodissociation heating [erg cm⁻³ s⁻¹]. It shows directly how deep the band penetrates: $f_{\text{shield}} \rightarrow 1$ in the thin wind above the H₂ → H front, collapsing by orders of magnitude inside the self-shielded molecular base.

4.7 Oxygen_chemistry.txt

The solved oxygen partition, written for the equilibrium state of a run with `Oxygen chemistry: True`. 18 columns: 1 r [R_p], 2 T [K], 3 $n(O)$ (*free atomic* neutral oxygen), 4 $n(O\text{ II})$, 5 $n(O\text{ III})$, 6 $n(OH)$, 7 $n(H_2O)$, 8 $n(CO)$, 9 $n(O(^1D))$ from its local steady state, 10 $x_{H_2} = 2n_{H_2}/n_H$, 11 τ_{chem} [s], 12 τ_{adv} [s], 13 the Damköhler number $\tau_{\text{adv}}/\tau_{\text{chem}}$, 14--15 the H₂O and OH photodissociation rates summed over the bands [s⁻¹], 16 $D(H_2)$ [cm² s⁻¹], 17 K_{zz} [cm² s⁻¹], and 18 $\tau_{\text{diff}} = \Delta r^2/(D_{H_2} + K_{zz})$ [s]. Columns 3--8 sum to the oxygen element total.

$\tau_{\text{chem}} = 1/(k_{O1}n_{OH} + k_{O2}n_O)$ is the H₂ lifetime against the oxygen cycle and $\tau_{\text{adv}} = r/|v|$ the radial flow time. These are a *required* output rather than a diagnostic. With `Molecular carrier transport: True` a cell whose chemistry is slow is solved rather than assumed away, and the time scales say which part of the profile the reaction set is responsible for and which part the flow is; with `Molecular carrier transport: False` they say something stronger, namely where that limit is simply wrong.

With the transport on, a trailer at the end of the file records the last transport step: `newton_steps`, the relative residual `newton_resid` it stopped at, `limited_cells` (cells where a carrier would have taken more nuclei than its element has), `worst_overshoot`, and `co_ceiling_cells`, the cells in which the transported CO was cut back to the $CO \leftrightarrow C + O$ equilibrium of their own (n, T). A limiter count that is not small is a statement about the step and is reported rather than absorbed.

Every run also closes the file with the **net H₂ loss budget at the base cell**, decomposed into eleven channels from the

same coefficients the residual uses and net of its own reverse where a channel has one (so a negative entry is a net H₂ source and a share may exceed 1). The option exists to make the oxygen cycle set the base partition, and whether it does is a property of the run: on a 2331 K base the thermal channel $\text{H}_2 + \text{M} \rightarrow \text{H} + \text{H} + \text{M}$ carries 70–97% of the net and the oxygen cycle a few percent, while on an 864 K base the oxygen family carries 96–99.6%. Read that block before reading the partition.

4.8 FUV_bands.txt

How deep each FUV band penetrates, and the band energy ledger. Written for an Oxygen chemistry run. 16 columns: 1 $r [R_p]$, 2–3 the star-ward H₂O and OH columns [cm^{-2}], 4–7 the four band optical depths (LW, B2, B3, B4), 8–11 the H₂O photodissociation rate in each band [s^{-1}], 12–15 the same for OH, and 16 the total FUV photolysis heating [$\text{erg cm}^{-3} \text{s}^{-1}$]. A header line above the data names the flux and the H₂O and OH band cross sections of each band.

The trailer of the file carries the energy ledger. For a beam attenuated as $\exp(-\tau)$ the photons absorbed over a column are exactly $N_b(1 - e^{-\tau_b})$ whatever the density profile, and the file reports the grid sum against that closed form band by band. They agree to round-off at any grid spacing, because each cell's rate is the mean over the cell and not its face value; a residual there means the rates and the columns have come apart. It also reports the absorbed energy split into deposited heat and bond energy, against the incident band flux: the absorbed energy can never exceed it, which is the property that keeps the band average from over-heating.

The last block is the shared beam of the LW band. Over 912–1201 Å the same photons are absorbed by H₂ in the Lyman–Werner lines and by H₂O and OH in a continuum, so the closed form the band's sum is checked against $N_b[1 - (1 - A)e^{-\tau}]$ with A the line-removed fraction, and the H₂ share belongs inside it: the H₂ photon count is the cell mean of the pump cross section $\sigma_{\text{diss}}/p_{\text{eff}}$, because the pumps that fluoresce back also take a photon out of the band. Since A is the column integral of that same cross section, the two sides are one normalization of one absorption and the LW row is a closure residual like B3 and B4. The block reports LW_H2_absorbed_en, LW_cont_absorbed_en, LW_incident_en and LW_line_removed_frac. The run warns if the absorbers together take more energy out of the interval than it carries. A separate warning fires where the star-ward column passes the top of the self-shielding table's column axis; above it the table is clamped at its edge cross section, so A goes on growing past 1 and is capped there, which is the one place the LW closure residual does not go to round-off.

4.9 OI_levels.txt / OI_levels_adv.txt

Populations of the three fine-structure levels of the O I $2p^4\ ^3P$ ground term, written whenever the run carries metals (no key: the presence of metals.inp, or of a lower-atmosphere profile with an oxygen reservoir, is enough). 11 columns: 1 $r [R_p]$, 2 $T [\text{K}]$, 3 n_e , 4 n_{HI} , 5 $n(\text{OI})$, 6–8 the fractional populations $f(^3P_2)$, $f(^3P_1)$, $f(^3P_0)$, and 9–11 the same three as densities [cm^{-3}], which sum to column 5 by construction.

The term is inverted, so 3P_2 is the lowest level. These are the lower levels of the O I $\lambda\lambda 1302.168, 1304.858, 1306.029$ resonance triplet, which is why they are an output field: a transit forward model that applied the total O I density to all three components would count the same atoms three times. They are not a second calculation – the file is filled by the same three-level statistical equilibrium, with the same electron and H-atom collisional rates and the

same line escape probabilities and incident infrared field, that the [O I] 63.2/145.5/44.1 μm cooling is assembled from (Cool_coeff.f90). In the dense base the solution is the Boltzmann one to round-off; it departs from it in the thin outer wind, where the levels drain radiatively toward 3P_2 .

4.10 Excited_H.txt

H($n=2$) diagnostic (written when the excited-H model is on, i.e. stellar T_{eff} and radius are set). 16 columns: 1 r , 2 T , 3 n_{HI} , 4 n_e , 5 $\bar{J}_{\text{Ly}\alpha}$, 6 n_{2s} , 7 n_{2p} , 8 Balmer proton source S_p , 9 photoelectric heating, 10 de-excitation heating, 11 $\tau_{\text{Ly}\alpha}$, 12--14 ground-state HI photoionization / collisional-ionization / recombination rates (for the proton-budget comparison), 15--16 the internal and stellar contributions to $\bar{J}_{\text{Ly}\alpha}$.

4.11 EXHALE_setup.out and the mass-loss rate

EXHALE_setup.out is a human-readable setup report (planet/star parameters, the Jeans parameter, composition, numerics) plus the final result. The **steady-state mass-loss rate** is $\dot{M} = 4\pi\rho v r^2$ evaluated near the outer boundary, corrected by the 2-D-approximation factor from 2D approximate method ($\text{Rate}/2 + \text{Mdot}/2 \Rightarrow \times \frac{1}{2}$, $\text{Mdot}/4 \Rightarrow \times \frac{1}{4}$). It is reported as $\log_{10} \dot{M} [\text{g s}^{-1}]$; divide by M_p and multiply by 1 Gyr to express it in $M_p \text{ Gyr}^{-1}$. The functions `exhale_io.mdot_log10 / mdot_Mp_per_Gyr` reproduce this.

4.12 EXHALE_resolved.out: the configuration the wind actually used

A machine-readable record written in the run directory (next to `input.inp`, not inside `output/`) by `write_setup_report.f90` once `input.inp`, `metals.inp` and any `base.inp` / lower-atmosphere-profile handoff have been resolved. Format: # comment lines, then one key value pair per line, reals at full double precision. It carries the base radius, planet mass, base temperature, He/H, orbital distance and stellar mass in effect; `base_inp_present` and `he_diffusion`; the EOS factors `mass_per_H_amu` and `ntot_bc_per_H`; one abundance_<El> line per element; and, with a profile in use (§3.8), the profile's provenance and the measured elemental fluxes. It exists so that consumers do not re-parse `input.inp` and inherit a value the handoff has since moved: `EXHALE_transit.py` reads it in preference to `input.inp` (§5.2), and `src/utls/element_budget.py` and `src/utls/element_flux_closure.py` read it as the authority on the resolved abundances.

4.13 element_flux_profile.txt

Face-by-face elemental fluxes carried by the element-diffusion operator, written to `output/` whenever a lower-atmosphere profile is in use (where they are the quantity the two models must agree on, not a diagnostic), and on demand with `EXHALE_DIFFUSION_CHECK=1`. Columns (named by the file's own # columns: line): the face index and radius, the face helium mass fraction, the diffusive and advective fluxes, then $F_{\text{He}} = 4\pi r^2(\rho X v + J)$, $F_{\text{H}} = 4\pi r^2(\rho(1-X)v - J)$ and $\dot{M}_{\text{face}} = 4\pi r^2 \rho v$ in g s^{-1} , the face Δm_{eff} , the effective and neutral diffusion coefficients with their ratio, and the dominant friction pair. Replaced at every diffusion step, so it always describes the last one.

5. Analysis tools

5.1 EXHALE_plots.py: live profile viewer

`python3 EXHALE_plots.py [--live N]` draws an 8-panel figure from output/: density, velocity, pressure, temperature, momentum $\log_{10}(\rho v r^2)$, the ion densities, and the H and He ionization fractions. With `--live N` it refreshes every N seconds (watch a run converge). If the `*_adv.txt` files exist they are overplotted (dashed) against the equilibrium profiles (solid). It also prints $\log_{10} \dot{M}$.

5.2 EXHALE_transit.py: transmission spectra

`EXHALE_transit.py` post-processes a converged run into transit transmission spectra and reports the line-center absorption depths (theoretical, instrument-convolved, and instrument+rotation-convolved) for **He 10830**, **Ly α** , **H α /H β** (from the non-LTE $n=2$ population, the level balance of Christie et al. 2013 with the solver's own rate coefficients, pumped by the Ly α field of §3.4), and the metal resonance doublets **Mg II h&k**, **Ca II H&K**, **Na I D** and the **O I 1302/1304/1306** triplet: the metals get the same full treatment as the H/He lines (all components in one window, theoretical / instrument / +rotation spectra, written under the same `tpm_<line>` names), plus the validated single-component line-center and 4 Å-band depth table; a metals-off run skips them automatically.

Which state the spectrum is built from (EXHALE_TRANSIT_STATE). The tool reads one of two state pairs, and the selection is a physical choice rather than a filename convenience. `adv`, the default, reads `Hydro_ioniz_adv.txt` and `Ion_species_adv.txt`: a post-process of a marching state, a steady advective correction cell by cell, second order in the cell width (variable-step BDF2 along the flow at the bulk velocity), with no eddy term and no element drift, refused in a cell where an assumption of the steady equations fails. It is an independent discretization of the same column and no equation of the run was solved for it. `EXHALE_TRANSIT_STATE=solution` reads `Hydro_ioniz.txt` and `Ion_species.txt`: the state the solver produced and the certification judged. With `Ionization transport` on, the composition of that state is the partition the stage transport itself produced, so an advective correction laid on top of it would be a second and different transport approximation of the same stages.

A stationary run writes no `_adv` products, and none are to be manufactured for one: the default selection then refuses by name, says why, and names the other selection instead of quietly reading it. The two files of a selection are read together, including the O I ground-term level populations below, because the temperature of one state with the composition of the other solves neither set of equations. Every saved curve carries the selection, the path, md5 and size of each file it read, and the state's own provenance, coupling, boundary model and reservoir lines.

On a marching case that carries both states the choice is small but not negligible. MEASURED on the `wasp_full` configuration (before the atomic-data corrections of 2026-09-24 to 09-26 and the solver's rates in the transit tool's $n=2$ model; not re-measured since), as the peak transit depth relative to itself: He I 10830 -0.64 per cent, H α -1.28 , H β -3.48 , Ly α unchanged at its saturated ceiling, and the optically thin metal lines $+0.1$ to $+0.8$. The metastable helium column is 12 per cent *larger* in the solved state while its line is shallower, because the advection correction moves that column outward (column-weighted mean radius 1.2941 against 1.3112 R_p) and a saturated line's disk-averaged depth

follows how far the absorbing shell reaches; $H\beta$ moves most because it is the line still on the linear part of the curve of growth.

O I 1302: the one line whose components do not share a lower level. The three components of the O I resonance triplet absorb out of the three *different* fine-structure levels of the $2p^4\ ^3P$ ground term, so the script reads their populations from the `OI_levels` file of the selected state (§4.9), `OI_levels_adv.txt` under `adv` and `OI_levels.txt` under `solution`, rather than applying the total O I density three times over, which at the near-statistical populations of these winds would triple the column. A run whose output carries no such file, like a metals-off run, simply skips the line. Because each absorption component owns its own lower-level density, the component descriptor is `(lambda0, f, A21, n_lower)` for every line, not only for this one.

The O I line is also the only one compared here against a published measurement, and the comparison is written into `tpm_OI_band_depths.txt` together with the conventions that make it legitimate: the depth is *band-integrated* over the observers' band (1300–1310 Å for Vidal-Madjar et al. 2004, 1299–1310 Å for Ben-Jaffel & Hosseini 2010), never at line center; the band weight is the stellar emission-line profile those authors measured (0.2 Å FWHM, peak ratios 1:1.5:1.17), because what was measured is a flux ratio and essentially all the flux in that band is in the three narrow chromospheric lines; the interstellar absorption is already inside those measured peak ratios and is therefore *not* applied a second time; the instrument LSF is not applied either, because a Gaussian LSF conserves the integral and cancels from a ratio over a band many LSF widths wide; and the published values, which are $(R_{\text{abs}}/R_{\star})^2$ and include the opaque planetary disk, have each paper's own continuum band subtracted before comparison. Overrides: `EXHALE_TRANSIT_RES_OI` (default 1000, HST/STIS G140L), `_OI_WEIGHT` (`star` | `flat`), `_OI_STAR_FWHM_A`, `_OI_STAR_PEAKS`, and `_OI_NSACLE`, which multiplies all three level densities at fixed ratios so the same code path can be driven up and down the curve of growth. It supports two geometries via `geometry = 'spherical'` (Phase 5a) or `'triaxial'` (Phase 5b, the 3-D Roche-equipotential reconstruction in `roche_recon.py`).

Where the system parameters come from. Every planet and star parameter is taken from the run directory, so the spectra always describe the same system as the simulation. The labels are matched exactly as `input_read.f90` does, so line order is irrelevant. Two sources are read, in this order:

`EXHALE_TRANSIT_* environment override > ./EXHALE_resolved.out > ./input.inp > built-in default,`

so an explicit override still wins for an experiment. The middle step matters whenever a `base.inp` or a lower-atmosphere profile moved the base: `EXHALE_resolved.out` (§4.12) is written by the wind solver *after* those overrides are applied, so it carries the R_p , M_p , T_0 , a , $M_{\star}$ and He/H the wind actually used, while `input.inp` alone would be stale. When the file is present the script says so at startup and takes those six from it; when it is absent it says that too and falls back on `input.inp`. A present-but-unreadable file is a hard error, never silently replaced by `input.inp`. The stellar radius and T_{eff} are not in that file and come from `input.inp` (or their environment overrides) either way.

Quantity	input.inp label	Override
R_p	Planet radius [R_J]	-
M_p	Planet mass [M_J]	-
T_{eq}	Equilibrium temperature [K]	-
a	Orbital distance [AU]	-
M_*	Parent star mass [M_sun]	-
L_{EUV}	Log10 of EUV luminosity	-
ξ (day–night)	2D approximate method	xi_override
R_*	Stellar radius [R_sun]	EXHALE_TRANSIT_RSTAR_RSUN
T_*	Stellar Teff [K]	EXHALE_TRANSIT_TSTAR
P_{spin}	from a, M_*, M_p (tidal locking)	EXHALE_TRANSIT_ROT_P

The planet spin period has no input.inp key: a close-in giant is tidally locked, so the default is the Keplerian orbital period $P_{\text{orb}} = 2\pi\sqrt{a^3/[G(M_*+M_p)]}$ built from the orbital distance and the masses already in input.inp. Only the instrument resolving powers, the wavelength windows, and the geometry switch remain script/environment settings: they describe the observation, not the system. At startup the script prints every resolved parameter together with its source (input.inp, env . . . , or built-in default), and warns when Stellar radius/Stellar Teff are missing from input.inp; read that block first when a spectrum looks wrong. A missing Stellar radius is the one worth checking before anything else: transit depths are normalized by πR_*^2 and the chord integration stops at R_*/R_p , so the built-in default radius shifts every depth, and not by a single factor. Measured on the tutorial planet (a $1.13M_\odot$ star, so $R_* \simeq 1.155R_\odot$), the built-in default returned a He 10830 depth of 22.8% where the correct radius gives 4.0%.

Outputs. Every line carries one key (He10830, Ly α , H α , H β , MgII, CaII, NaI, OI), and both products of a line are named from that key and land in the run directory: the model curve tpm_<line>.txt (always written) and, when EXHALE_TRANSIT_FIG_PREFIX is set, the figure <prefix><line>.png together with its vector .pdf. A curve file has the columns λ [Å], T_λ (theoretical), T_λ (instrument-convolved), T_λ (+rotation), so the excess absorption in percent is $(1 - T_\lambda) \times 100$; a metals-off run simply writes no metal curves. EXHALE_TRANSIT_SAVE_PREFIX decorates the curve name in the same way the figure prefix decorates the figure name; both are resolved against the run directory, so an absolute prefix writes elsewhere.

Step-by-step usage. EXHALE_transit.py reads the state pair of the selection in force from path, which is the current directory unless EXHALE_TRANSIT_PATH names the run directory: output/Hydro_ioniz_adv.txt and output/Ion_species_adv.txt by default, or output/Hydro_ioniz.txt and output/Ion_species.txt under EXHALE_TRANSIT_STATE=solution:

1. *Choose the state, and make sure its two files are there.* A stationary solve writes no _adv products, and the solution selection is how such a run is read; do not manufacture _adv files for one. A marching run writes both pairs automatically. To give an existing state the advection-corrected pair as well, copy Hydro_ioniz.txt $\rightarrow$ Hydro_ioniz_IC.txt and likewise for Ion_species, set Load IC? True and Restart intent: stationary evaluate in input.inp, and run ./EXHALE.x once (seconds): that route loads the state, derives its pressure and

temperature from the loaded conserved state, runs the ionization sweep, evaluates the certification and writes every product from that work state without entering the time integration.

2. *Set what describes the observation*, if the defaults do not: the instrument resolutions and wavelength windows (EXHALE_TRANSIT_RES_*, _HE_LMIN/LMAX/N), the azimuthal sampling of the rotation integral, and geometry for a Roche run. The star and the spin period need nothing: they come from `input.inp`.
3. *Run* `MPLBACKEND=Agg EXHALE_TRANSIT_PATH=<run_dir> python3 EXHALE_transit.py` (the Agg backend suppresses the interactive windows on headless machines). The script writes the `tpm_<line>.txt` curves into the run directory and prints the line-center absorption depths (theoretical / instrument-convolved / +rotation) and the metal-line summary; add `EXHALE_TRANSIT_FIG_PREFIX=tpm_` to save the figures as well.

Worked examples (the converged Newton solutions of WASP-121b and HD189733b, with all four H/He spectra and the metal lines): `backup/regression/tpm_wasp/` and `backup/regression/tpm_hd189/`, collected in the notebook `backup/regression/steady_solver_results.ipynb`. Figure 2 shows the four H/He spectra EXHALE_transit.py saves for the WASP-121b Newton solution of `backup/regression/tpm_wasp/` (run with `Include_He23S? True`, so the He I 10830 line is populated).

Transit-depth conventions (important). Huang et al. (2023) quote transit depths as the *effective transit radius* R_{eff}/R_* , not a percent absorption; convert from the code's line absorption fraction h via $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$. The metal NUV lines (Mg II) are quoted in a 4 Å bin, while the optical lines (Ca II, H α , Na) are line-center; line-center vs. 4 Å can differ by $\sim 3\times$ for an optically-thick line, so always compare in the same definition.

5.3 exhale_io.py: named, physical-unit loaders

`examples/exhale_io.py` loads every EXHALE output file into named arrays in physical units, so analysis code need not remember column orders (Table 5). The central object is the Run container: `load_run(outdir, inputfile)` fills it with the hydro profiles (r [R_p], n [cm^{-3}], v [cm s^{-1}] with a `v_kms` shortcut, p , T , `heat`, `cool`), the full ion-density dictionary `run.ion` (the 6 H/He plus 27 metal stages of §4.), and the parsed `input.inp` in `run.inp`. By default the `_adv` (advection-corrected) files are read; pass `adv=False` for the equilibrium snapshot. Typical use:

```
import exhale_io as aio
run = aio.load_run('output', 'input.inp')      # _adv profiles by default
print(aio.mdot_log10(run))                     # log10 Mdot [g/s], 2D factor incl.
print(aio.mdot_Mp_per_Gyr(run))                # same in M_p/Gyr
xh = run.x_ion(['HI', 'HII'])                  # stage fractions of one element
plt.plot(run.r, run.T)
plt.plot(run.r, xh['HII'])
cool = aio.load_cooling('output/Cooling_breakdown.txt')
plt.semilogy(cool['r'], cool['chan']['FeII']) # one cooling channel
```

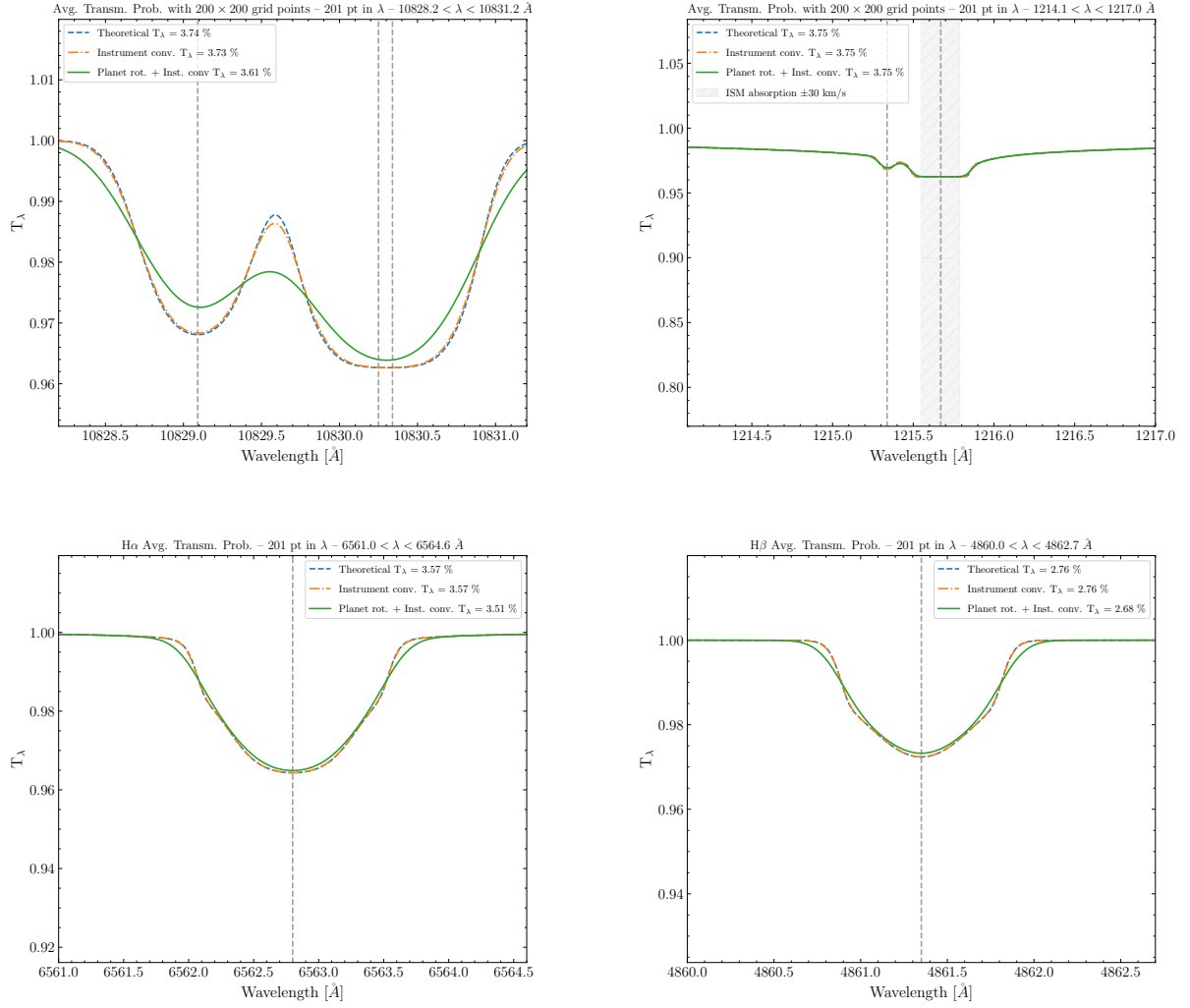


Figure 2: EXHALE_transit.py output spectra for WASP-121b (Roche, Newton solution with He 2³S on; backup/regression/tpm_wasp/): He I 10830 (top left), Ly α (top right), H α (bottom left), and H β (bottom right). Each panel shows the theoretical line, the instrument-convolved line, and the instrument+rotation-convolved line as produced by the script.

Function	Returns
<code>load_run(outdir, inputfile, adv=True)</code>	a Run: hydro profiles + run.ion dict + parsed input.
<code>Run.x_ion([stages])</code>	ionization fractions of one element, e.g. $x_{\text{ion}}(['\text{HI}', '\text{HII}'])['\text{HII}'] = n_{\text{HII}}/(n_{\text{HI}} + n_{\text{HII}})$.
<code>load_hydro(path)</code>	one <code>Hydro_ioniz(_adv).txt</code> as a dict (<code>r, n, v, p, T, heat, cool</code>).
<code>load_ions(path)</code>	(<code>r, {ion: density}</code>) from <code>Ion_species(_adv).txt</code> (all 33 stages).
<code>load_cooling(path)</code>	<code>r, T, ne, cool_total</code> + chan dict of cooling in each channel (6 H/He channels <code>rec/coll_ion/coex_HI/coex_HeI/coex_HeII/brems</code> , the H3p infrared channel, + 27 metal-line channels).
<code>load_excited_H(path)</code>	<code>Excited_H.txt</code> as a dict (<code>n2s, n2p, Jlya, tau_lya, Sproton, Hpe, . . .</code>).
<code>load_lyman_werner(path)</code>	<code>Lyman_Werner.txt</code> as a dict (<code>x_H2, NH2, f_shield, k_LW, heat_LW</code>).
<code>read_input(path)</code>	<code>input.inp</code> as a dict (<code>Rp_RJ, Mp_MJ, a_AU, HeH, LX, LEUV, . . .</code> + raw lines).
<code>mdot_log10(run, j_from_top=20)</code>	$\log_{10} \dot{M}$ [g/s]: $4\pi\rho v r^2$ near the outer boundary <i>including</i> the 2D approximate method factor (mirrors <code>EXHALE_main.f90</code> ; see §4).
<code>mdot_Mp_per_Gyr(run)</code>	the same in units of $M_p \text{ Gyr}^{-1}$.
<code>constants</code>	<code>RJ_CM</code> (and its alias <code>RJ</code>), <code>MJ</code> , <code>Msun</code> , <code>AU</code> , <code>mu</code> ($= m_{\text{H}}$), <code>GYR</code> (cgs). These are the one Python definition of each; the tools import them from here rather than writing the number down again.

Table 5: examples/exhale_io.py public interface. Run analysis scripts either from examples/ or after adding it to the Python path (`sys.path.insert(0, '<repo>/examples')`).

The notebook `examples/EXHALE_analysis.ipynb` uses these loaders to plot the profiles and the parameter-difference comparisons of §6., and `examples/make_figures.py` (which generates every figure in this manual) is a further worked example.

5.4 `exhale_to_lart.py`: LaRT Ly α input from an EXHALE run

`EXHALE_transit.py` estimates the Ly α mean intensity $\bar{J}_{\text{Ly}\alpha}$ with the simple Huang et al. (2017) prescription (§3.4). For a proper treatment of the resonant Ly α radiative transfer that sets the $n=2$ hydrogen population (and hence H α), the converged atmosphere can be post-processed with the Monte Carlo code **LaRT** (Seon & Kim 2020; Seon et al. 2022). `examples/exhale_to_lart.py` builds a complete LaRT spherical-illumination input set directly from an EXHALE run.

Given a run directory it loads the `_adv` profiles (via `exhale_io.py`; `--eq` takes the equilibrium pair instead, which is what a stationary solve leaves) and `input.inp`, and writes

- `dens_profile.txt`: $r[R_p]$, $n_{\text{HI}}[\text{cm}^{-3}]$ (the Ly α scatterer is ground-state H);
- `temp_profile.txt`: $r[R_p]$, $T[\text{K}]$;
- `velo_profile.txt`: $r[R_p]$, $v_r[\text{kms}^{-1}]$;
- `line_profile.txt`: the incident stellar Ly α profile, a double-Gaussian with peaks at $\pm m$ and Gaussian width s (defaults $m = 74$, $s = 49 \text{ kms}^{-1}$, the Huang et al. 2017 / Yan et al. 2022 values; pass `--fwhm` to read s as FWHM);
- `<name>_lya.in`: the LaRT namelist (geometry='spherical_atmosphere', source_geometry='stellar_illumination');
- `run.sh`: a multi-host `mpirun` launcher.

The length unit is set to R_p (`distance2cm = R_p` in cm), so all LaRT radii are in R_p and match the EXHALE grid; the star-planet geometry (`distance_star_to_planet = a/R_p`, `stellar_radius = R_*/R_p`) is read from `input.inp`. Because the stellar illumination breaks spherical symmetry, the scattering rate is requested as a cylindrical 2-D grid (geometry_JPa = 2): $P_\alpha(\rho, z)$ is axisymmetric about the star-planet axis, which is the form needed to build the 2-D $n_{2p}(\rho, z)$ population for the H α transit. To populate P_α , build LaRT with `make CALCPnew=1` (producing `LaRT_calcP.x`).

```
# build the LaRT input from a converged EXHALE run, then run it
python examples/exhale_to_lart.py WASP-52b/fxuv0p25_he98 \
    -o WASP-52b/LaRT_lya --name WASP-52b \
    --m 74 --s 49 --rmax 10 --ngrid 201 --nphotons 1e7 \
    --hosts lart4,lart3,lart2
cd WASP-52b/LaRT_lya && ./run.sh
```

It is also importable: `import exhale_to_lart as e2l; e2l.build(run_dir, outdir, m_kms=74, s_kms=49, geometry_jpa=2, ...)`.

6. Worked examples

6.1 A first run: a generic hot Jupiter

The directory `examples/tutorial/` contains a minimal, self-contained model of a generic HD 209458b-like hot Jupiter: a spherical domain out to $10R_p$, solar metals, EUV+X-ray power-law SED. The full input `.inp` is:

```
Planet name: demoHJ
Log10 lower boundary number density [cm^-3]: 12.70
Planet radius [R_J]: 1.38
Planet mass [M_J]: 0.69
Equilibrium temperature [K]: 1320.0
Orbital distance [AU]: 0.047
Escape radius [R_p]: 2.00
He/H number ratio: 0.083
2D approximate method: Rate/2 + Mdot/2
Parent star mass [M_sun]: 1.13
Spectrum type: Power-law
Power-law index: -1.0
Use only EUV? False
[E_low,E_mid,E_high] = [ 13.60 , 123.98 , 1.24e3 ]
Log10 of X-ray luminosity [erg/s]: 27.0
Log10 of EUV luminosity [erg/s]: 29.30
Grid type: Mixed
Numerical flux: HLLC
Reconstruction scheme: PLM+WENO3
Include He23S? True
Load IC? False
Do only PP: False
Force start: False
Domain mode: Spherical
Outer radius [R_p]: 10.0
du_th [PLM,WENO3]: 0.5 1.0e-3
Solver: Newton
Stellar radius [R_sun]: 1.155
```

The `Reconstruction scheme: PLM+WENO3` line enables the two-stage reconstruction (with the two `du_th` values as the PLM and WENO3 thresholds), and the `Solver: Newton` line adds the steady-state Newton finish (§2.7), so the run converges to the true steady state without manual restarts. With `metals.inp` present (solar abundances, §3.), run `./EXHALE.x` in that directory. Figure 3 shows the converged profiles read back with `exhale_io`: the wind heats to $\sim 10^4$ K, the velocity rises monotonically outward (transonic outflow), and hydrogen ionizes across a front near $1.2\text{--}1.4R_p$.

Tutorial run: generic hot Jupiter (spherical, H/He + metals)

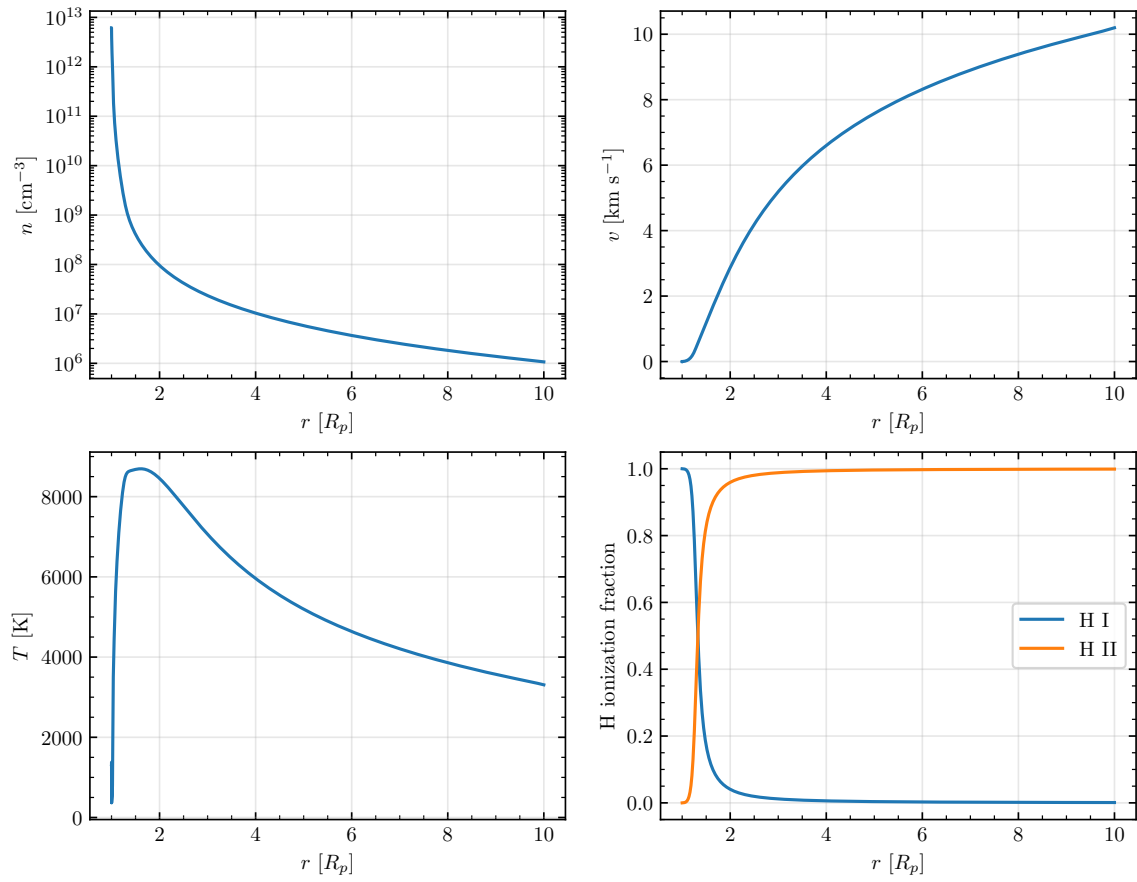


Figure 3: First-run output of the generic hot-Jupiter tutorial: density, velocity, temperature, and the H ionization fraction vs. radius.

6.2 Turning physics on and off

The same model is re-run with single-line changes to illustrate each option: *metals off* by removing `metals.inp`; *Roche/RLOF* by removing the `Domain mode/Outer radius` lines (the default truncates at L1); the *atomic-helium limit* by setting `Include He23S? False` (the triplet is on by default, so this is the change that removes it); the *H α /Balmer* physics by adding `Stellar Teff/Stellar radius`; and the *transonic IC* by adding `Transonic IC: True` for a near-unbound (deep-RLOF) base; and *diffusive He/H separation* by adding `He_diffusion: True` (extend it to the metals with `He_metal_diffusion: True`; §3.6). The comparison studies below quantify the effect of the first two.

6.3 Ready-made example configurations (examples/)

For every solver/physics combination there is a ready-to-run folder under `examples/` (paths relative to the EXHALE root). Folders 01–11 are the *same* planet (HD189733b) so that the effect of each option can be isolated; 12 and 14–17 use HD209458b (where the Wind-AE warm start, diffusion, molecular chemistry with and without metals, and the lower-atmosphere profile handoff are validated), and 13 spans four planets. Each folder is self-contained: `cd` into it and run the repo-root binary (`././EXHALE.x`); outputs land in the folder’s own `output/`. The “Added / changed lines” column of Table 6 is relative to each example’s natural base: the legacy `01_legacy_marching/input.inp` for the solver examples 02–04 (and Wind-AE 11/12), and the recommended-solver `03_newton/input.inp` (two-stage PLM+WENO3 + Solver: Newton) for the physics examples 05–10, which all inherit that solver and add only their signature option on top. A `diff` against the appropriate base shows exactly what an option adds; all configurations have been smoke-tested. The metastable helium triplet is on by default in the code, but the ladder holds `Include He23S? False` on purpose: in both bases and in every rung whose signature option is something else (02, 04, 05, 09–12), so that atomic helium is the baseline and 06_he23s remains a one-line difference. Outside the ladder (tutorial/, 13–16, the planet folders) the triplet is on. 11_windae_ic (HD189733b) exercises the self-consistent-BC continuation: its Wind-AE ramp converges and writes the IC, though EXHALE’s own time-integration of HD189733b is limited by a separate base-breathing instability (§2.10). 12_windae_ic_hd209 is the seed-adjacent HD209458b case. 14_diffusion and 15_molecular turn on the He/metal diffusive separation (§3.6) and the full molecular network (§3.7) respectively, on HD209458b, and 16_molecular_metals adds a `metals.inp` to the latter. 17_lower_profile replaces the scalar `base.inp` handoff by the profile of §3.8.

Table 6: Ready-made example configurations under `examples/` (01–11 are HD189733b; 12, 14–17 are HD209458b; 13 spans four planets). The Wind-AE IC ramp converges for both of the planets that use it (HD 189733 b (11) and HD 209458 b (12)), and 11, being far from the seed, additionally exercises the self-consistent-BC continuation and the molecular-layer turn-off. Metals on/off is a runtime switch: any folder becomes metals-on by copying a `metals.inp` into it, molecular folders included: that is what `16_molecular_metals` is.

examples/...	Demonstrates	Added / changed lines
01_legacy_marching	classic ATES v2 baseline: PLM marching, H/He only, Roche domain, <i>du</i> -based stop	(none, baseline)
02_two_stage	two-stage PLM→WENO3 marching (§2.4)	Reconstruction scheme: PLM+WENO3, <code>du_th [PLM,WENO3]: 0.5 1.0e-3</code>
03_newton	recommended default: two-stage warm-up + Newton finish (§2.7)	+ Solver: Newton
04_newton_from_state	Newton re-convergence of an existing state (no marching; needs output/*_IC.txt, see the folder README)	Load IC? True, Resid tol: 1.0e-5; run via the EXHALE_PTC hook
05_metals	trace-metal cooling, solar C/N/O/Mg/Ca/Na/Fe (§3.)	<code>metals.inp</code> present
06_he23s	He 2^3S metastable level (HeI 10830); the one rung that leaves the ladder's atomic-helium baseline	Include He23S? True
07_balmer_lya	non-LTE $H(n=2)$ + $Ly\alpha$ pumping ($H\alpha/H\beta$, §3.4)	+ Stellar T_{eff} /radius, Deexc heat, J_{lya} escape-prob, Stellar $Ly\alpha$ flux/halfwidth/boost
08_full	everything on (= the HD189733b/planet folder + Newton); feeds EXHALE_transit.py	07 + <code>metals.inp</code>
09_spherical	spherical domain instead of the default Roche/L1 truncation (§1.3)	Domain mode: Spherical, Outer radius <code>[R_p]: 10.0</code>
10_warm_seed_ic	warm-seed (hot Parker overlay) initial condition	Hot Parker IC: 10000

Table 6, continued

examples/...	Demonstrates	Added / changed lines
11_windae_ic	in-process Wind-AE IC for HD189733b (§2.10): far from the seed, so the self-consistent-BC continuation (base-BC re-convergence + molecular-layer turn-off) is exercised; the Wind-AE ramp converges and writes the IC. EXHALE's own HD189733b base-breathing instability (separate from the IC) then limits the warm start	+ IC mode: windae, Solver: Newton; symlinks inputdata/ for the shipped seed
12_windae_ic_hd209	in-process Wind-AE warm-start IC that <i>works</i> : HD 209458b (not HD189733b) is close to the shipped seed, so the ramp converges and EXHALE warm-starts cleanly (spherical $10R_p$)	HD209458b params + Domain mode: Spherical, IC mode: windae, Solver: Newton; symlinks inputdata/
13_lower_atmosphere	lower-atmosphere connection for four planets (HD209458b, HD189733b, WASP-121b, WASP-52b; §3.7): an analytic $1\mu\text{bar}$ base column plus a base.inp hand-off (isothermal or Guillot $T(p)$ generator); see the folder README	Lower column: <R_1bar>; driver-generated base_iso.inp / base_guillot.inp
14_diffusion	diffusive separation of He and metals (HD209458b; §3.6): the He/H ratio declines with altitude and each trace metal settles independently, re-shaping He I 10830	HD209458b params + Include He23S? True, He_diffusion: True, He_metal_diffusion: True, He_Kzz: 1.0e9, He_alphaT: 0.0; metals.inp present

Table 6, continued

examples/...	Demonstrates	Added / changed lines
15_molecular	full molecular chemistry (HD209458b; §3.7): $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ in the coupled ionization equilibrium; a sharp $\text{H}_2 \rightarrow \text{H}$ front forms above a thin molecular base, the wind above it essentially atomic. Newton-converged from a cold start, re-run 2026-08-15 with the current code ($\ R\ = 6.105 \times 10^{-6}$ in 89 outer iterations): the front sits at $r = 1.0097 R_p$ and the molecular layer below it collapses to ~ 400 K	HD209458b params + Molecular chemistry: True, Solver: Newton 5.0e-2, Resid tol: 1.0e-7, Max steps: 150000 (no metals.inp here)
16_molecular_metals	molecular chemistry <i>and</i> trace metals solved in one system (HD209458b): the $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ network and the metal ionization stages share the free electron density, which the metals dominate in the shielded molecular base. Converges only from a warm restart, and the seed matters: seed it with a 15 converged by <i>the same binary</i> (a state left by an older binary is a different seed and fails, info = 2), then expect info = 0 at Resid tol: 1.0e-7 and $\log_{10} \dot{M} = 9.65-9.67$. The cold path is closed: with metals on, the JFNK line search bottoms near $\ R\ \sim 10^{-3}$ (worst cell $r = 1.021$, momentum) and falls back to marching at every Low-Mach damping coefficient tried: off, 5×10^{-3} , 10^{-2} , 2×10^{-2} (measured 2026-08-15)	as shipped, 15_molecular + metals.inp (solar C/N/O) without 15's three convergence keys. For the warm restart put the seed run's Hydro_ioniz.txt/Ion_species.txt in output/*_IC.txt and use 15's three keys + Load IC? True + Low-Mach damping: 5.0e-3 (run 1.0e-2 as well and compare, see the folder README)

Table 6, continued

examples/...	Demonstrates	Added / changed lines
17_lower_profile	lower-atmosphere <i>profile</i> handoff (HD209458b; §3.8): the lower atmosphere handed over as a table over an interval of pressure instead of the scalars of <code>base.inp</code> . The base state (T_0 , R_0 , p_{base} , q_{H_2}), the elemental reservoirs (He/H and solar C/N/O) and $K_{\text{zz}}(r)$ all come from the one file; <code>make_example_profile.py</code> regenerates it	14_diffusion + Lower atmosphere profile: <code>lower_atmosphere_profile.dat</code> , <i>minus</i> <code>He_Kzz</code> (the profile carries K_{zz}) and <i>minus</i> <code>metals.inp</code> (the profile carries C/N/O)

7. Comparison studies

7.1 Effect of metal-line cooling (metals on vs. off)

Trace metals add strong line cooling (and some photoionization heating and free electrons). Figure 4 compares the tutorial planet with and without `metals.inp`: metal-line cooling lowers the peak temperature by ~ 1800 K (acting as a thermostat that prevents the H/He-only over-heating), while the H ionization structure is barely changed. The effect is far stronger for hotter, more irradiated planets (e.g. WASP-121b, where metal cooling is essential to match the observed temperatures); the detailed cooling by channel budget is in Figure 5.

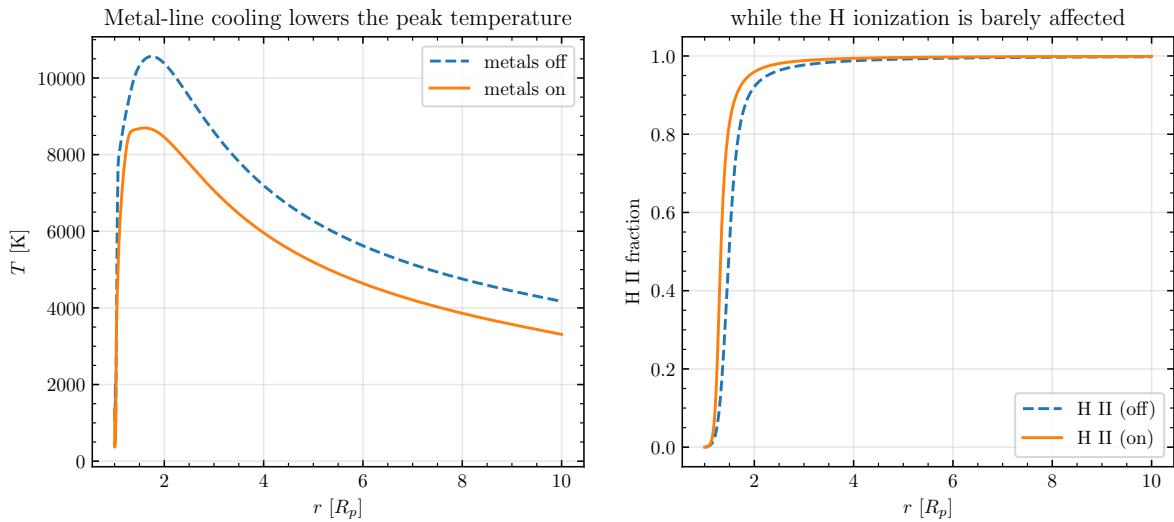


Figure 4: Metals on vs. off for the tutorial planet: metal-line cooling lowers the peak temperature (left) while the H ionization fraction is nearly unchanged (right).

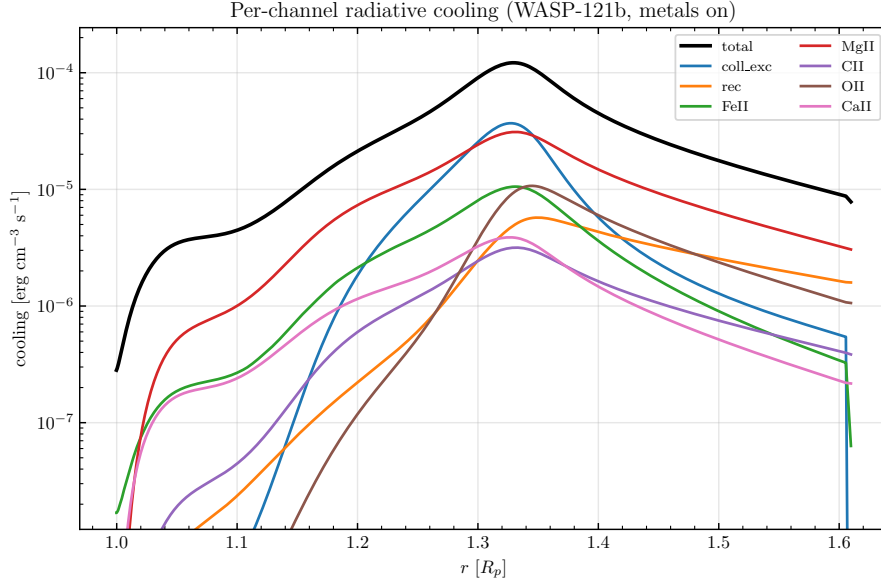


Figure 5: Radiative cooling by channel vs. radius (WASP-121b, metals on; from `Cooling_breakdown.txt`). Below $\sim 1.4 R_p$ the metal lines (e.g. Fe II, Mg II, C/O) compete with or dominate the H/He collisional channels.

7.2 Spherical vs. Roche (RLOF) and the mass-loss rate

Removing the `Domain` mode: `Spherical` line switches on the full Roche tidal potential, truncated at the L1 radius. Figure 6 compares a spherical model (Case A) and a Roche model (Case B) of WASP-121b: the tidal potential lowers the effective gravity toward the star, so the Roche wind accelerates far faster and the mass-loss rate rises by an order of magnitude ($\dot{M}$ from ~ 0.05 to $\sim 0.34 M_p \text{ Gyr}^{-1}$ here), the “Roche-lobe overflow” enhancement.

7.3 Transmission spectra

`EXHALE_transit.py` turns a converged model into transit absorption. The metal resonance lines are the key NUV/optical diagnostics; Figure 7 compares the modeled effective transit radii R_{eff}/R_* (spherical Case A) against Huang et al. (2023). Recall the conventions of §5.2: Mg II is a 4 Å bin, Ca II/Na are line-center, and $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$.

8. Atomic and molecular data

By default EXHALE is an *atomic/ionic* code: the modeled species are H, He (including the He 2^3S metastable triplet), and ten trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) across their first two–three ionization stages. The optional molecular network (`Molecular chemistry: True`, §3.7) adds H_2 , H_2^+ , H_3^+ and HeH^+ to the coupled equilibrium, with H_2 photoionization opacity (Yan et al. 1998) and H_3^+ infrared cooling (Miller et al. 2013); `Oxygen chemistry: True` adds OH, H_2O and CO (§3.7), and `Molecular IR bands: True` the H_2 line spectrum (Roueff et al. 2019) and the H_2O and CO bands (HITEMP) as infrared coolants. There is no H^- , and the only molecular opacities are those of H_2 (its ionizing continuum and the Lyman–Werner lines) and, with the oxygen chemistry, the FUV photodissociation of H_2O , OH and CO. Table 7 collects the provenance of the atomic rates and cross sections, grouped by physical process.

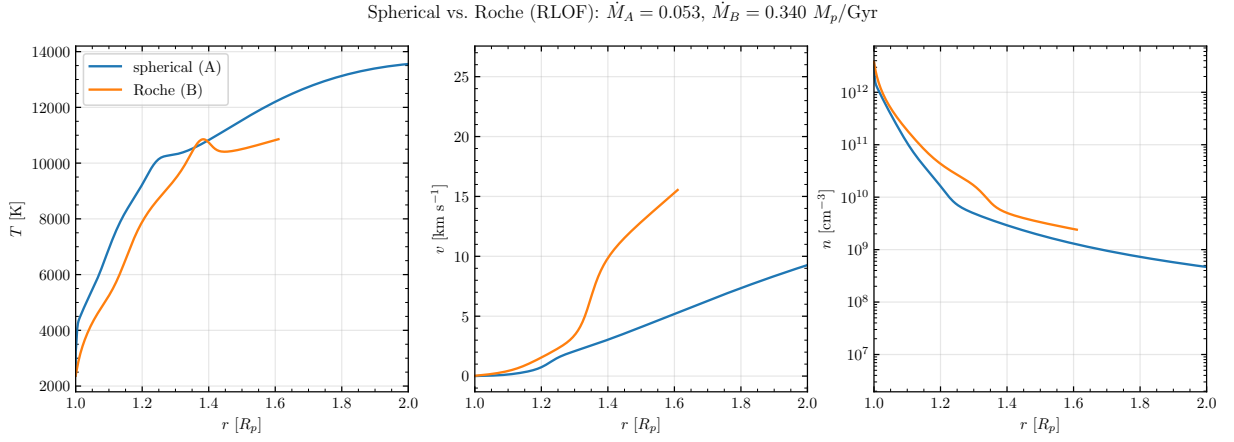


Figure 6: Spherical (Case A) vs. Roche (Case B) WASP-121b: temperature, velocity, and density. The Roche wind reaches L1 ($\approx 1.6R_p$) and accelerates much faster; $\dot{M}$ is annotated. The temperature panel is the `_adv` profile; its base is smooth because the post-processor falls back to the equilibrium solution where the flow is not a clean outflow (see the caveat in §4.). The Roche velocity (middle) dips slightly negative near the base: the genuine, physical “breathing” inflow of a near-Roche-filling planet.

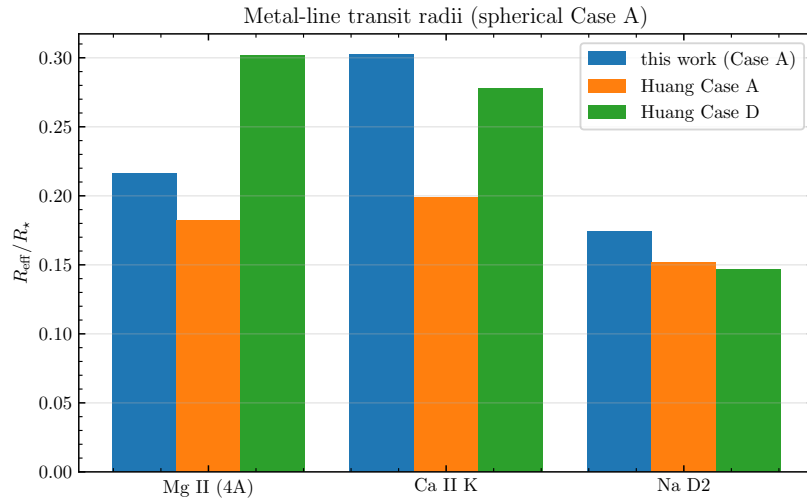


Figure 7: Metal-line transit radii R_{eff}/R_* for spherical Case A, compared with Huang et al. (2023) Case A and Case D.

Of the H/He data of the original ATES (Caldiroli et al. 2021, 2022) the code keeps the hydrogenic cross sections of H I and He II and, as comparison options, the legacy fits (ATES_photoionization_rate, Legacy_HHe_rates); the default recombination, collisional and cooling coefficients of H and He were replaced after the atomic-data audits of 2026-09-24 to 09-26 (md/Update_EXHALE_stage3.md sections 17, 19 and 22), and the metal, excited-hydrogen, and charge-exchange data were added in this work.

Table 7: Atomic data sources, by process. Source files: cross_sec.f90 (cross sections); Cool_coeff.f90 (collisional rates, recombination, cooling); charge_exchange.f90; excited_hydrogen.f90; EXHALE_transit.py (transmission line data).

Process	Species	Source
Elemental abundances		
Solar reference	10 metals (metals.inp template)	Asplund et al. (2009)
Helium	He/H ratio	input parameter (He/H number ratio)
Photoionization cross sections		
Bound-free $\sigma(E)$	H I, He II	hydrogenic (Kramers) analytic form on the measured thresholds (orig. ATES)
	He I	Verner, Ferland, Korista & Yakovlev (1996) default; legacy ATES 2-term fit via ATES_photoionization_rate
	He I 2 ³ S (metastable)	two VFKY96 (Verner) wings + power-law bridge, fitted to TOPbase/Opacity-Project (extends the orig. Norcross 1971 broken power law past 60 eV)
	C, N, O, Mg, Si, Ca, Na, S, Fe (+ ions)	Verner, Ferland, Korista & Yakovlev (1996) outer-shell fits below the first inner-shell edge, Verner & Yakovlev (1995) above it; the inner subshells (Verner & Yakovlev 1995, their phfit2) are in the sum, so σ is the whole photoabsorption of the ion
	K I	Verner & Yakovlev (1995) at every energy, outer shell and subshells

Table 7, continued

Process	Species	Source
Inner-shell relaxation	the 17 photo-ionizable metal ions	a vacancy below the outer shell relaxes by Auger decay (one or more further electrons, the ion jumping two or more stages) or radiatively, with the probabilities of Kaastra & Mewe (1993, A&AS 97, 443) Table 2 and the fluorescence of their Table 3, the vacancy energy closed exactly on the code's thresholds ($E_{\text{th},s} = I_i + \text{further ionization energies} + \text{Auger energy} + \text{fluorescence}$); the photoelectron and the Auger electrons are the electron energy, the fluorescence leaves the cell. A neutral of C, N, O, Mg, Si, Ca, Fe that loses two or more electrons goes to its third stage in the balance; a jump beyond the carried stages is counted in the top carried stage
Opacity model (runtime)	all (via opacity.inp)	A=analytic above (default); C=constant above threshold, $\sigma = \text{OPA_CONST_X}$ times the analytic value at the edge; P=that constant base times the pressure-broadening factor (Robinson & Catling); T=tabulated files
Collisional (electron-impact) ionization		
Rate $C(T)$	H I, He I, He II	Voronov (1997, ADNDT 65, 1) Table I default; legacy ATES fits (Abel+1997 / Hui & Gnedin 1997) via Legacy_HHe_rates
	C–Fe, all stages	Voronov (1997) Table I, each row with Voronov's own fitted ΔE ; the fits are stated for 1 eV–20 keV (3 eV He II, 2 eV N II, O II) and are evaluated below that range with their exact e^{-U} threshold factor and no accuracy claim
	He 2^3S	Black (1981, MNRAS 197, 553) eq. 12, $8.38 \times 10^{-10} T^{1/2} e^{-55338/T}$
Recombination (radiative RR + dielectronic DR)		

Table 7, continued

Process	Species	Source
Ground capture α_1	H I, He I, He II	the Milne relation on the ground-state photoionization cross section the transfer uses, so the capture and the absorption of its photon are one process (0.998 of Hummer 1994 for H I, 1.006–1.011 of Hummer & Storey 1998 for He I, as the code comment records them)
Case B	H II, He II, He III	Badnell (2023) RR (+ Badnell DR for He II) minus that α_1 default; legacy Hui & Gnedin (1997) fits via Legacy_HHe_rates; Koskinen et al. (2022) Table 1 R1, R2 via Atomic rate set
Case A where thin (def. on, H_rec_escape, He_rec_coupling)	H II, He II, He III	the balance coefficient is $\alpha_B + y\alpha_1$, y the fraction of the ground-capture photons the recombined species does not re-absorb in the cell (case A in a thin cell, case B in a thick one); False = case B
Recombination radiation absorbed on the spot (def. on, He_rec_coupling)	He II $\rightarrow$ He I, He III $\rightarrow$ He II	He II: the case-B captures leave through their own exits (the 584 Å line, the 19.8 eV 2^3S line, the 2^1S two-photon continuum of Drake, Victor & Dalgarno 1969), shared by the network's rates as in Draine (2011, sect. 14.3.2, 15.5) and Osterbrock & Ferland (2006, sect. 2.4); He III: the ground capture, the direct capture into $n = 2$ (Seaton 1959), He II Ly α and the $2s$ two-photon continuum (Nussbaumer & Schmutz 1984; $A(2s) = 526.823 \text{ s}^{-1}$ of Drake 1986), with the $2s$ share of Pengelly (1964) and the $2s$ – $2p$ ion mixing of Pengelly & Seaton (1964). Every photon is shared among H I, He I, He II, H ₂ and the metal ions by their opacity in the cell; what leaves the cell is dropped

Table 7, continued

Process	Species	Source
He I 584 Å resonance line	He I $2^1P \rightarrow 1^1S$	scattered by He I in the cell and converted into the 2^1S two-photon continuum with probability 1/912.0 per scattering (NIST A values), a flight ending in continuum absorption, line scattering or escape, after Wood, Mathis & Ercolano (2004, sect. 4.1); in a cell thick in line and continuum the continuum share tends to their eq. (17) with every continuum absorber in place of H I. Doppler mean cross section, no frequency redistribution, and no electron-collisional $2^1P \rightarrow 2^1S$ transfer (validity stated at the code)
Metastable	He II $\rightarrow 2^3S$, He II $\rightarrow$ excited singlets	case B split into triplet and singlet captures with the triplet share of Hummer & Storey (1998) Table 5 (0.771 at 10^4 K), so ground + singlets + triplets = the Badnell α_A
Electron collisions	He $1^1S \leftrightarrow 2^3S$, $2^3S \rightarrow 2^1S$, 2^1P	$8.629132 \times 10^{-6} \Upsilon e^{-E/kT} / (g\sqrt{T})$ with the Bray et al. (2000) effective collision strengths (their table inside $10^{3.75}$ – $10^{5.75}$ K, the CHIANTI 11.0.2 he_1 temperature dependence joined continuously outside it) and NIST level energies; the $2^3S \rightarrow 1^1S$ de-excitation q_{31g} is the detailed-balance reverse of the excitation
	He $1^1S \rightarrow n^3L$, $2^3S \rightarrow n^1L$ ($n \geq 3$)	the excitations into the higher triplets cascade into 2^3S and are added to q_{13} (0.019, 0.107, 0.331 of it at 5×10^3 , 10^4 , 2×10^4 K); the excitations of 2^3S into the singlets above 2^1P destroy it and are added to q_{31b} ; both are coronal sums over the Bray et al. (2000) terms $n \leq 5$
Penning ionization	He $2^3S + H$	García Muñoz (2025) Maxwell average of the Movre & Meyer (1997) cross sections (always on); the total quenches the metastable, 0.9 of it is Penning ($He + H^+ + e^-$) and 0.1 associative ($HeH^+ + e^-$)

Table 7, continued

Process	Species	Source
	He 2 ³ S + H ₂	García Muñoz (2025) network rows (Cohen & Lane 1977 cross sections; 500–10 000 K), split 0.9 Penning (H ₂ ⁺) : 0.1 associative (H + HeH ⁺). Active only when a molecular run also tracks the triplet
RR + DR	C, N, O, Mg, Si, Na, K, S, Ca II (+ stages)	Badnell (2006, ApJS 167, 334) RR + Badnell adf48 DR (TAMOC clist_K)
RR only	Ca I (Ca II+e)	Shull & Van Steenberg (1982); no DR
RR + DR	Fe I, Fe II	Huang et al. (2023) analytic fit; the Badnell DR project does not cover these isoelectronic sequences (see the note below the table)
Charge exchange (charge transfer)		
metal + H/H ⁺ (default)	group A: the 23 rows of Huang et al. (2023, ApJ 951, 123) Table 4, 19 carried	the reaction list of that table, each rate from the original source where it could be read (Kingdon & Ferland 1996 and the calculations behind it: Allan et al. 1988, Butler & Dalgarno 1979, 1980, Neufeld & Dalgarno 1987, Gargaud et al. 1982; Kimura et al. 1996; Stancil et al. 1998, 1999; Zhao et al. 2005; Rutherford & Vroom 1972; Dutta et al. 2001; Watanabe et al. 2002). The direction the source calculated or measured is carried as published; the other is computed from it by detailed balance with NIST internal partition functions. Four reverses (Mg ⁺ , Fe ⁺ , Si ⁺ , Na ⁺ + H) are not carried, their forward ending in an excited or radiative product
density-dependent reverses	N + H ⁺ , S ⁺ + H, C + S ⁺	for the three forwards whose sources name their product levels, the reverse is the thermal one times the product populations of a five-level statistical equilibrium of N I, S II and C I at the cell's (T, n_e) (CHIANTI 11.0.2 A values and electron collision strengths); it equals the thermal reverse at the collisional limit

Table 7, continued

Process	Species	Source
He + H/H ⁺ (default on, He_H_charge_exchange)	group B and He ²⁺ + H	He ⁺ + H: radiative (Stancil, Lepp & Dalgarno 1998, Table 1 row 19, $1.20 \times 10^{-15}(T/300)^{0.25}$, from Zygelman et al. 1989) + non-radiative (Loreau, Ryabchenko & Vaeck 2014, by reciprocity, with $\sigma_r \propto \sqrt{(x-x_{th})/(15-x_{th})}$ between the threshold $x_{th} = 13.655$ and 15 eV/u where their calculation starts); He + H ⁺ : detailed balance of the non-radiative channel; He ²⁺ + H $\rightarrow$ He ⁺ + H ⁺ + $h\nu$: West, Lane & Cohen (1982), Maxwellian average of their cross section, no reverse. Applied by dedicated routines in every helium-bearing system, <i>not</i> gated by <code>cx_full</code>
full network (<code>cx_full</code>)	groups C, D: metal+He/He ⁺ , metal+metal	Table 4 reaction list with the originals where read (Kimura et al. 1993; Zhao et al. 2004; Satta et al. 2013; Prasad & Huntress 1980; Rutherford & Vroom 1972; Chenel et al. 2010; Kwolek et al. 2019; Lavvas et al. 2014), reverses by detailed balance; C ⁺ , O ⁺ , Si ⁺ + He and Fe ⁺ + O, Fe ⁺ + N, Na ⁺ + Ca are not carried; C + Si ⁺ as printed
O ²⁺ , N ²⁺ + H (default on, <code>cx_O2p_H</code> , <code>cx_N2p_H</code>)	group E, not in Table 4	Barragán et al. (2006) Tables 1 and 2, interpolated over 10 ² –10 ⁵ K; no reverse; not gated by <code>cx_full</code>
heat (channel <code>heat_charge_exchange</code>)	every applied reaction	the energy defect less the excitation of a product state that leaves as radiation (product states from the sources read; Fe ⁺ products in the near-resonant z^4F/z^4D terms)
Radiative cooling		
recombination	H II, He II, He III, metal ions	the kinetic energy of the captured electrons, $kT\alpha(3/2 + d\ln\alpha/d\ln T)$, with the coefficient α the ionization balance uses (for H II, He II and He III the net coefficient $\alpha_B + y\alpha_1$, at the frozen escape weight; for the metals their RR+DR coefficient); checked against Hummer (1994)

Table 7, continued

Process	Species	Source
coll. excitation	H I	coronal sum over the CHIANTI 11.0.2 h_1 levels (Anderson et al. 2000 collision strengths), observed energies, the $1s \rightarrow 2s, 2p$ rates the H($n=2$) model uses
	He I	coronal sum over the 28 terms $n \leq 5$ of Bray et al. (2000); with the He 2^3S tracked, the $1^1S \rightarrow 2^3S$ channel is charged with the network's own excitation and its detailed-balance reverse (a net that turns negative where the metastable is overpopulated), plus the triplet cascade cooling of 2^3S (Bray terms $n \leq 5$) and the thresholds of its excitations into the singlets
	He II	coronal sum over the CHIANTI 11.0.2 he_2 levels (Ballance 2003)
coll. ionization, free-free	H, He, metals	the measured ionization potential per event (He 2^3S and the metal ions included); free-free bremsstrahlung with the van Hoof et al. (2014) thermal Gaunt factor at $\gamma^2 = Z^2 \text{Ry}/kT$, Ry the infinite-mass Rydberg (their eq. 21 and sect. 2; $\text{Ry}/k = 157887.5 \text{ K}$), weighted by ion net charge
CHIANTI descaling	every CHIANTI-based coefficient	the Burgess–Tully scaled collision strengths descaled with the natural cubic spline of CHIANTI's <code>descale_scups.pro</code> , as CHIANTI prescribes
H ₃ ⁺ infrared (molecular runs)	H ₃ ⁺	Miller et al. (2013) LTE emission per molecule (Table 5 $z(T)$ fits) times their Table 6 non-LTE departure factor $s(T, n_{\text{H}_2})$, interpolated as a power law in $n(\text{H}_2)$ between the columns and linearly in T ; optically thin. The dominant coolant of a molecular base, where every atomic channel above is exponentially suppressed

Table 7, continued

Process	Species	Source
metal line cooling	C I/II, N I/II, O I/II	the ground term in statistical equilibrium (row below) plus the channels that leave it from (T, n_e) statistical-equilibrium tables of CHIANTI 11.0.2 (default); legacy AIOLOS fits via <code>cno_cool 0</code> (no N cooling)
	Mg II	closed-form fit (0.2%) to the coronal sum over the 32 bound levels of the CHIANTI 11.0.2 model with the observed level energies
	Mg I, Ca II	(T, n_e) statistical-equilibrium tables of CHIANTI 11.0.2 carrying every channel of the ion
	Na ID	Van Regemorter with effective $\bar{g} = 0.2$ (not in CHIANTI)
	Fe I	NIST f -values + Van Regemorter, lower levels Boltzmann-populated: the high-density (LTE) limit, an upper limit at most 2.0 times the low-density value
	Fe II	(T, n_e) multilevel statistical-equilibrium table (CHIANTI)
	all (T, n_e) tables	above their top density the last slope in $\log n_e$ is carried on, clipped to $[-1, 0]$
ground-term fine structure (statistical equilibrium)	C I $2p^2\ ^3P$, C II $2p^2\ ^3P$, N II $2p^2\ ^3P$, O I $2p^4\ ^3P$	the split ground term is solved in exact statistical equilibrium at the local (n_e, n_{HI}) instead of the low-density coronal limit, which overstates these lines by 2.9–8.3 decades at the base. Electron collision strengths from CHIANTI v11.0.2; H-atom rates from Yan & Babb (2023) for C I/N II, Abrahamsen, Krems & Dalgarno (2007) for O I, Barinovs et al. (2005) for C II. The complete set: the other coolants have a single ground level, are built from permitted lines (Fe I), or already use a statistical-equilibrium table (Fe II)

Table 7, continued

Process	Species	Source
line trapping	the eight ground-term fine-structure lines	line-center escape probability from the column above the cell, Hollenbach & McKee (1979) / de Jong, Boland & Dalgarno (1980) plane-parallel Doppler form renormalized to $\beta(0) = 1$; applied as $A_{ul} \rightarrow \beta A_{ul}$ inside the statistical-equilibrium solution, the only placement correct in both the low- and high-density limits. Other lines, including Mg II h&k, are treated as optically thin
validity floor of the coronal fits	all CHIANTI-derived coefficients	the coronal part is cut off smoothly below 10^3 K (the lower end of the fitted range), continuous in value and dT -slope and bit-identical above it; only the explicitly solved ground-term fine-structure terms survive below. The roll-off is $\exp(-x^2)$ with $x = (T_{\text{floor}}/T - 1)/w$ and $w = 0.1$ by default, settable as <code>Coronal cutoff width</code> . With the C/N/O ground-term fine-structure floors saturated no coronal cooling survives below the floor for the guard to remove, and the base-cell balance temperature of all four paper planets is identical over $w = 0.02$ – 1.2 . Not applied to the legacy AIOLOS branch (<code>cno_cool 0</code>)
Photoelectric heating		
secondary ionization (def. on, Secondary_ionization)	H I, He I, H ₂	Shull & van Steenberg (1985) $f_{\text{ion}}(x)$ fits with the Dalgarno, Yan & Liu (1999) energy dependence, and Dalgarno et al.'s heating fraction $f_{\text{heat}}(x, E_0)$, H ₂ target channel and molecular heat returns; photoelectrons with $E_0 > 30$ eV drive secondary H I / He I / H ₂ ionizations, heating scaled to f_{heat} ; the He 2 ³ S photoelectron and the metal photoelectrons and Auger electrons feed the same heating and secondary-source budget
Excited hydrogen H($n=2$) and Lyα		

Table 7, continued

Process	Species	Source
$2s/2p$ rate equilibrium	H($n=2$)	the level balance of Christie, Arras & Li (2013, eqs. 12–13); $A(2p \rightarrow 1s) = 6.2649 \times 10^8 \text{ s}^{-1}$ (NIST), $A(2s \rightarrow 1s) = 8.22461 \text{ s}^{-1}$ (Drake 1986, eq. 27); $1s \leftrightarrow 2s, 2p$ collisions from the CHI-ANTI 11.0.2 h_1 strengths and their detailed-balance reverses; $2s$ – $2p$ mixing by electrons, Seaton (1955) eq. (55), and by H^+ , He^+ and He^{2+} , Pengelly & Seaton (1964) with their cut-off; in a molecular run the H_2^+ and HeH^+ dissociative recombinations as an $n=2$ source
recombination cascade to $n=2$	H	the case-B coefficient of the balance, split with the $2s$ fraction of Pengelly (1964, Table I)
Balmer continuum	H($n=2$)	$\sigma_2 = 1.4 \times 10^{-17} \text{ cm}^2$ at the $n = 2$ edge (Seaton 1959, 1.378×10^{-17} with his Gaunt factor), falling as E^{-3} , over the run’s own spectrum from $e_{\text{th,HI}}/4$ to $e_{\text{th,HI}}$
$\text{Ly}\alpha$ pumping field (mode 0)	H	Huang et al. (2017) Eq. 6
Transmission line data (EXHALE_transit.py)		
λ , f -value, A_{ki}	He 10830, Ly α , H α , H β , Mg II, Ca II H&K, Na I D, O I 1302/1304/1306	NIST ASD
line profile	all	Voigt (Faddeeva, <code>scipy.special.wofz</code>)

Note on Fe. Iron recombination uses the Huang et al. (2023) analytic RR+DR fit while every other metal uses the Badnell tabulation, and this is not a deferred item: no Badnell fit exists for either stage. Fe I is formed by recombining Fe II (Mn-like, 25 electrons) and Fe II by Fe III (Cr-like, 24), whereas the Badnell dielectronic-recombination project reaches the phosphorus isoelectronic sequence (15 electrons) as of its paper XVI (2022). Ca I has the same problem one sequence lower (K-like, 19 electrons) and likewise keeps a non-Badnell rate. **Note on completeness.** The metal grid carries the stages needed for the XUV-driven ionization balance and the observed NUV/optical lines: three stages of C, N, O, Mg, Si, Ca, Fe and two of Na, K, S. Twelve of the 27 carry a line-cooling channel (C I, C II, N I, N II, O I, O II, Mg I, Mg II, Ca II, Na I, Fe I, Fe II); the others (the doubly ionized stages, Si I, Si II, Ca I, Na II, K I, K II, S I, S II) cool only through recombination, collisional ionization and free–free emission, where each applies. Higher stages and additional elements are not modeled.